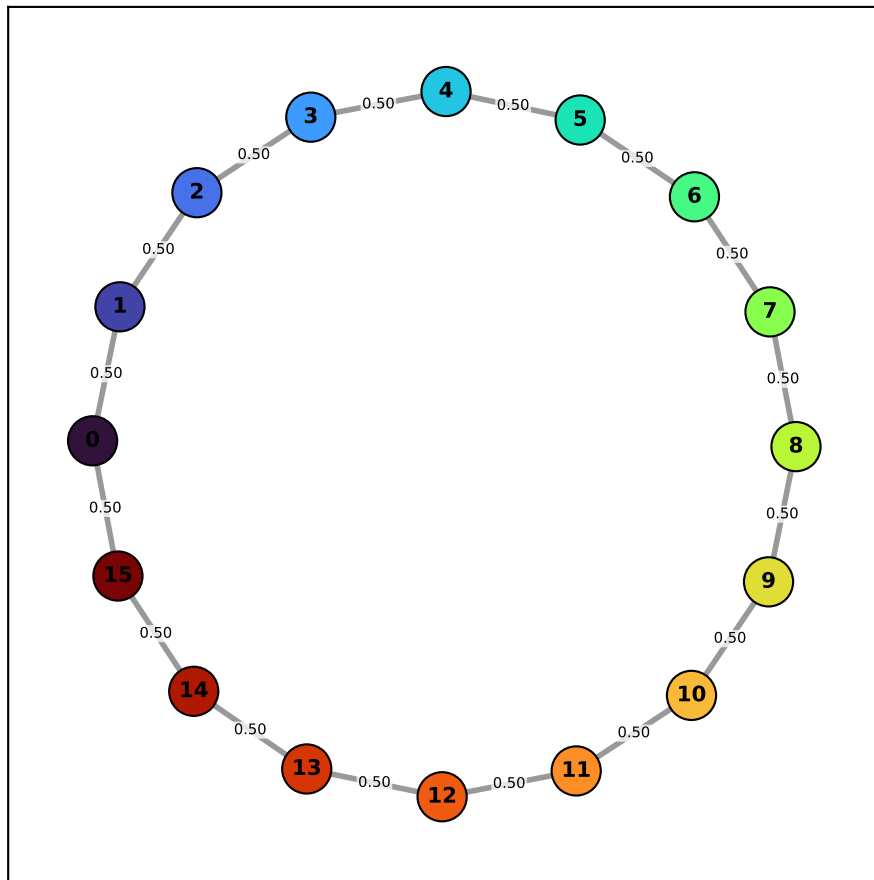
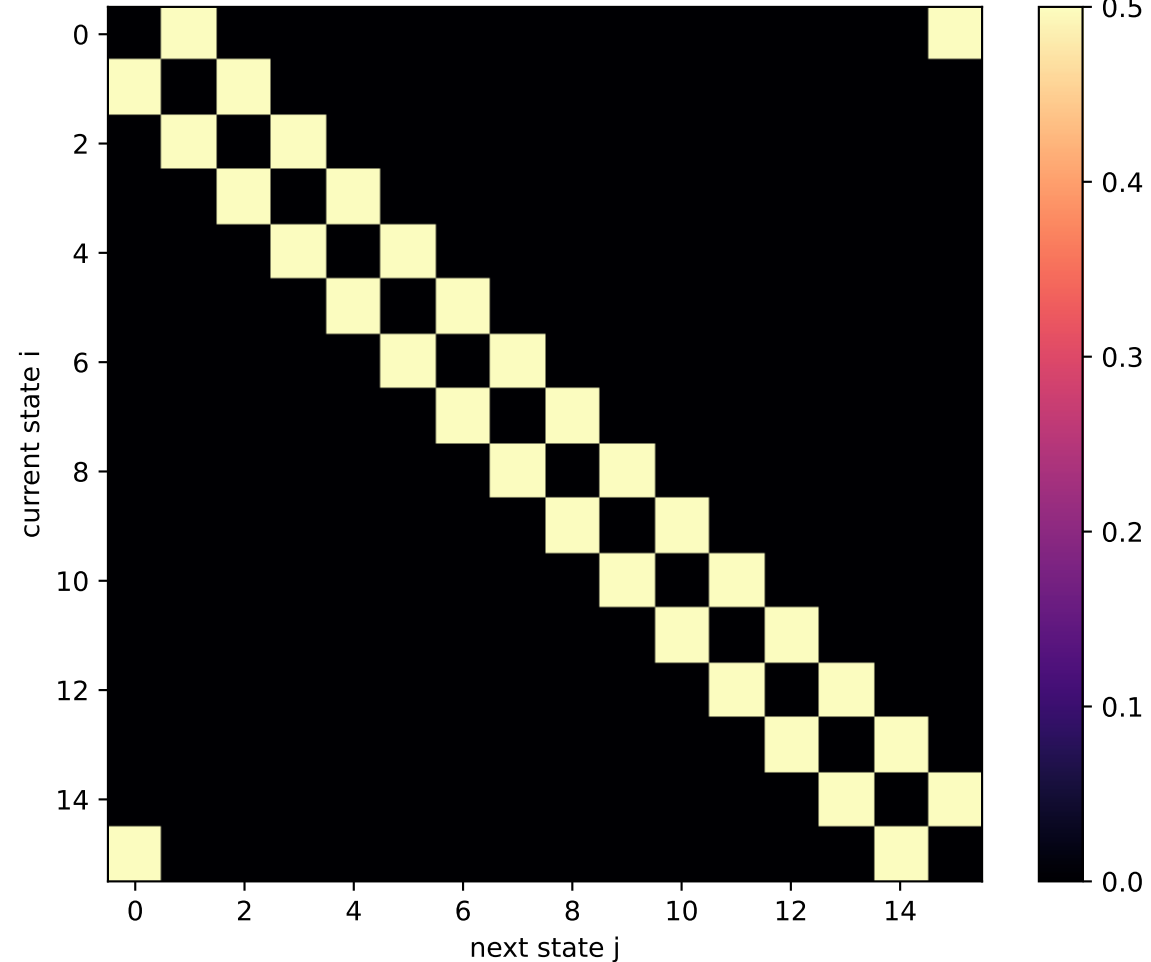


Markov family: ring (uniform random walks; edge width/label = transition probability 1/deg)

ring: the Markov chain
16 states, 16 transitions, mean degree 2.0

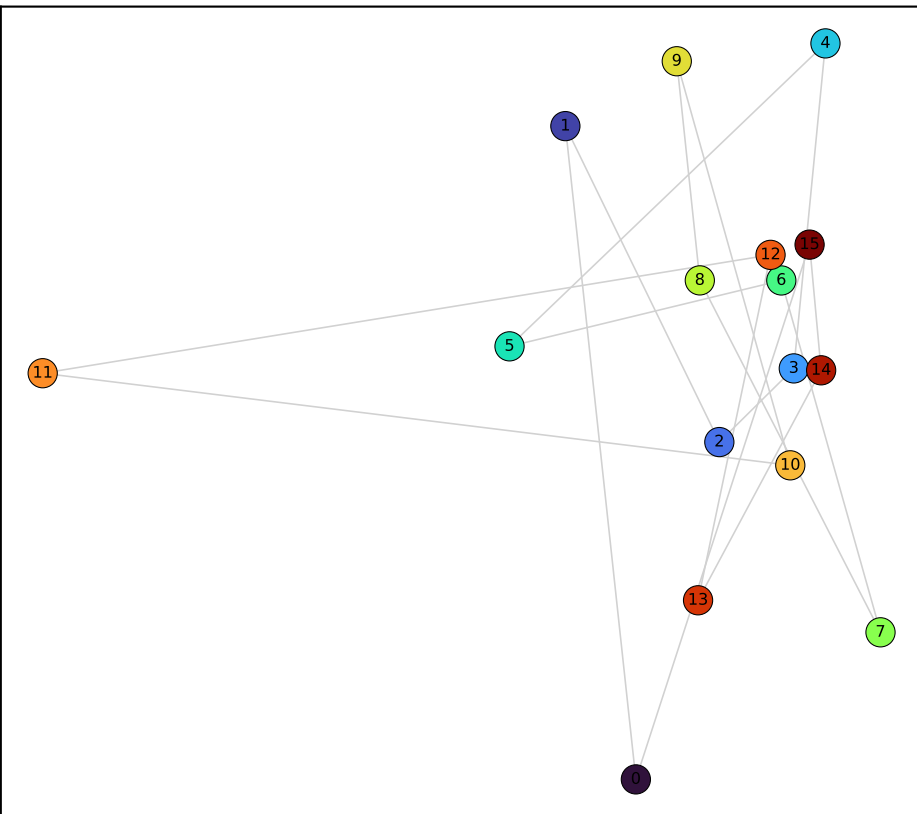


transition matrix
 $P(\text{next}=j \mid \text{current}=i) = 1/\text{deg}(i)$

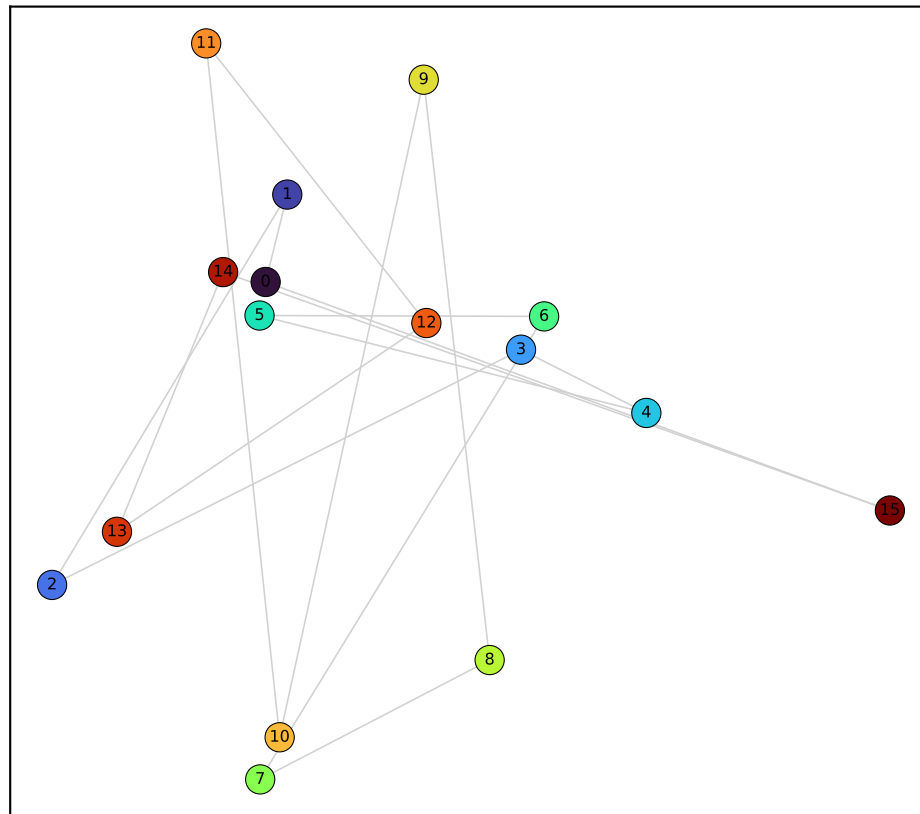


ring — node-mean PCA, depth 0% (edges = graph adjacency)

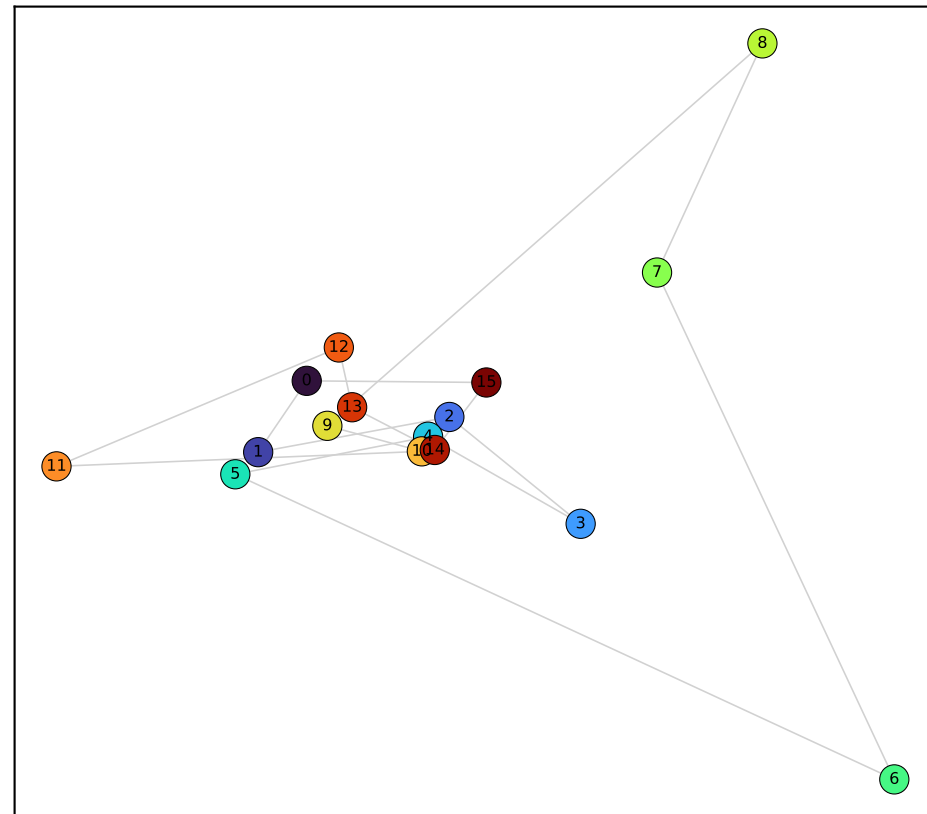
Llama layer 0/31 (PC var 10.8%)



Gemma layer 0/41 (PC var 8.8%)

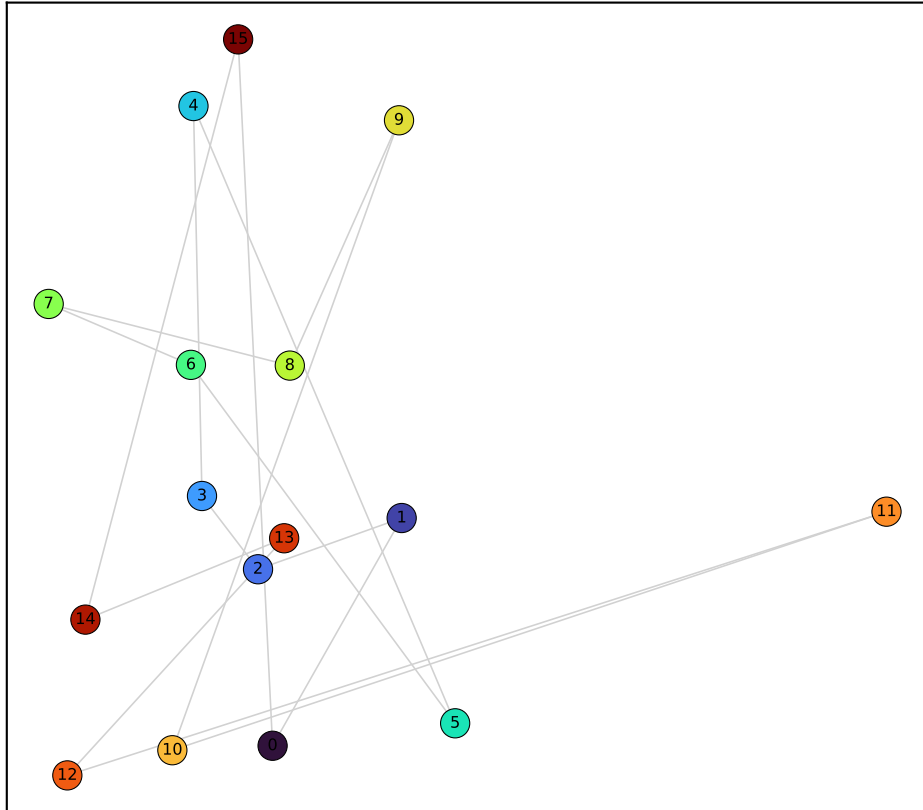


Qwen layer 0/35 (PC var 11.10%)

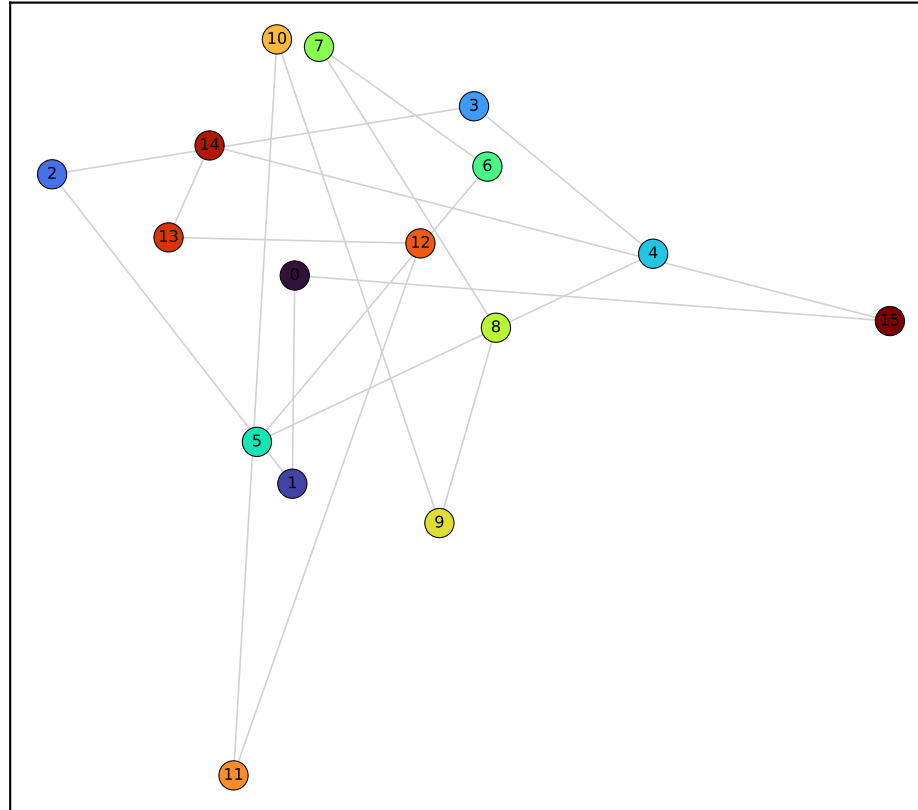


ring — node-mean PCA, depth 2% (edges = graph adjacency)

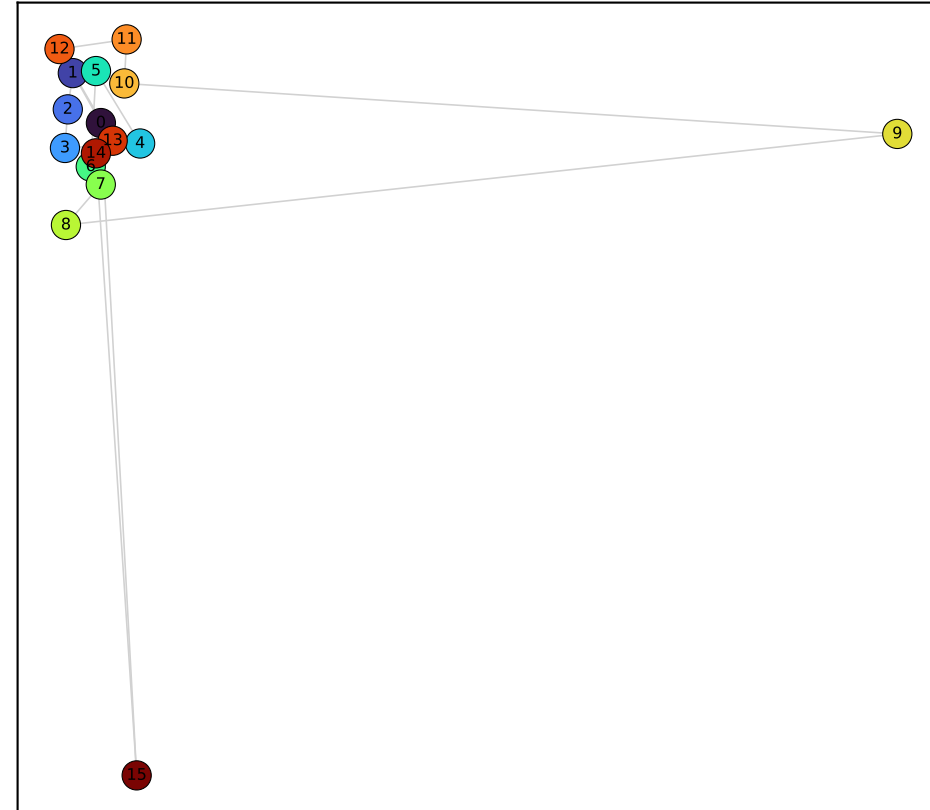
Llama layer 1/31 (PC var 9.8%)



Gemma layer 1/41 (PC var 9.8%)

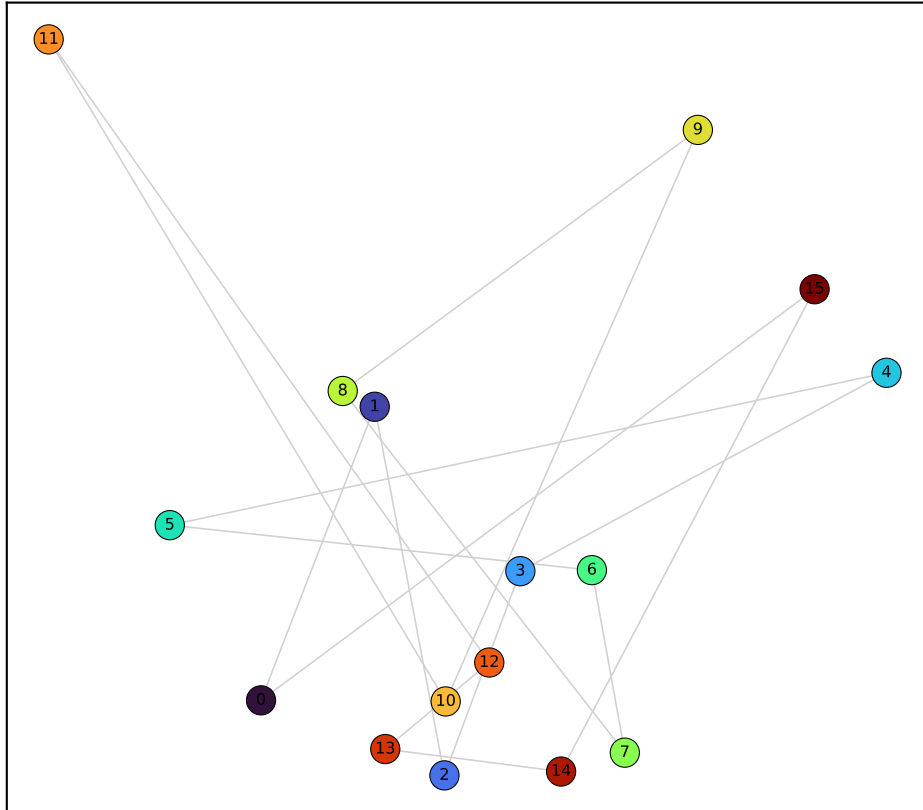


Qwen layer 1/35 (PC var 18.10%)

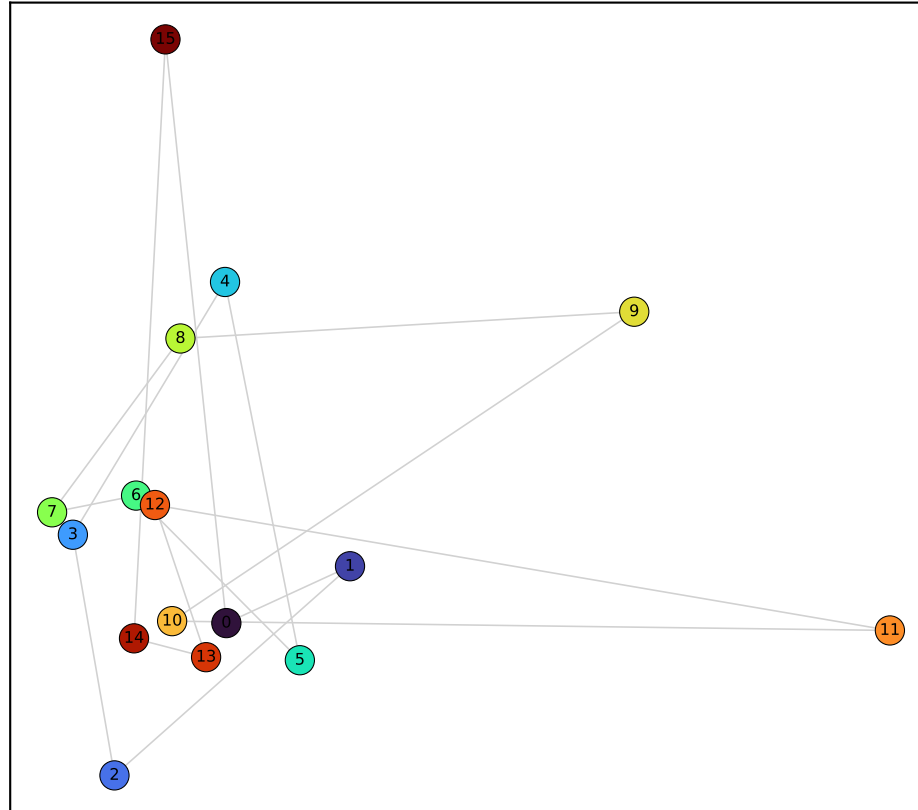


ring — node-mean PCA, depth 5% (edges = graph adjacency)

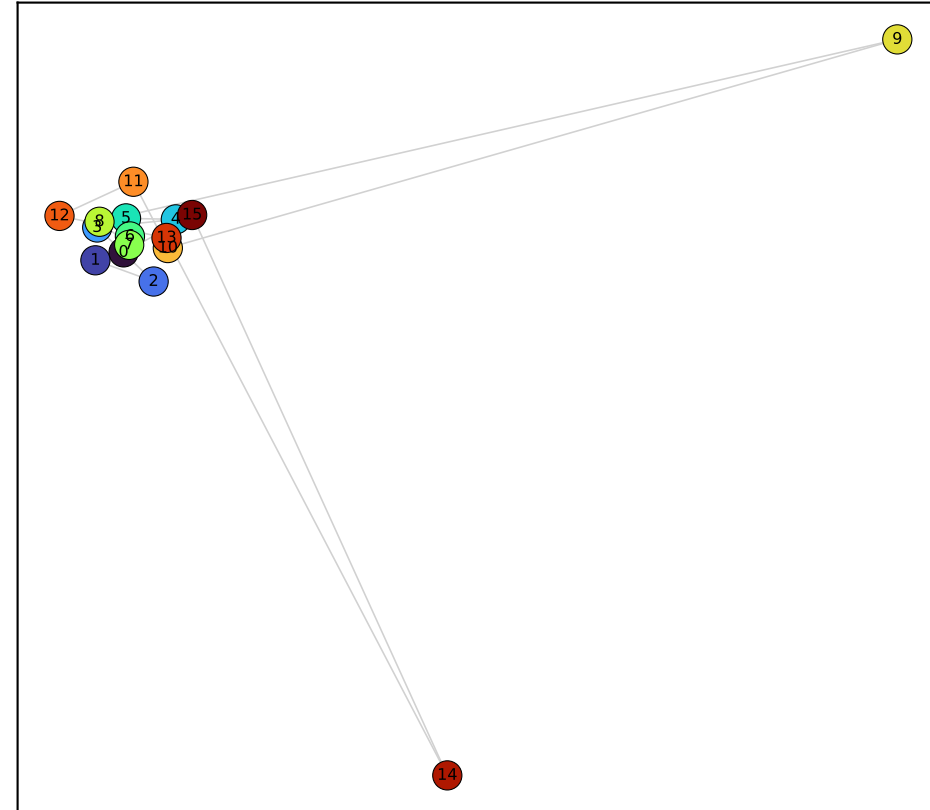
Llama layer 2/31 (PC var 9.8%)



Gemma layer 2/41 (PC var 9.9%)

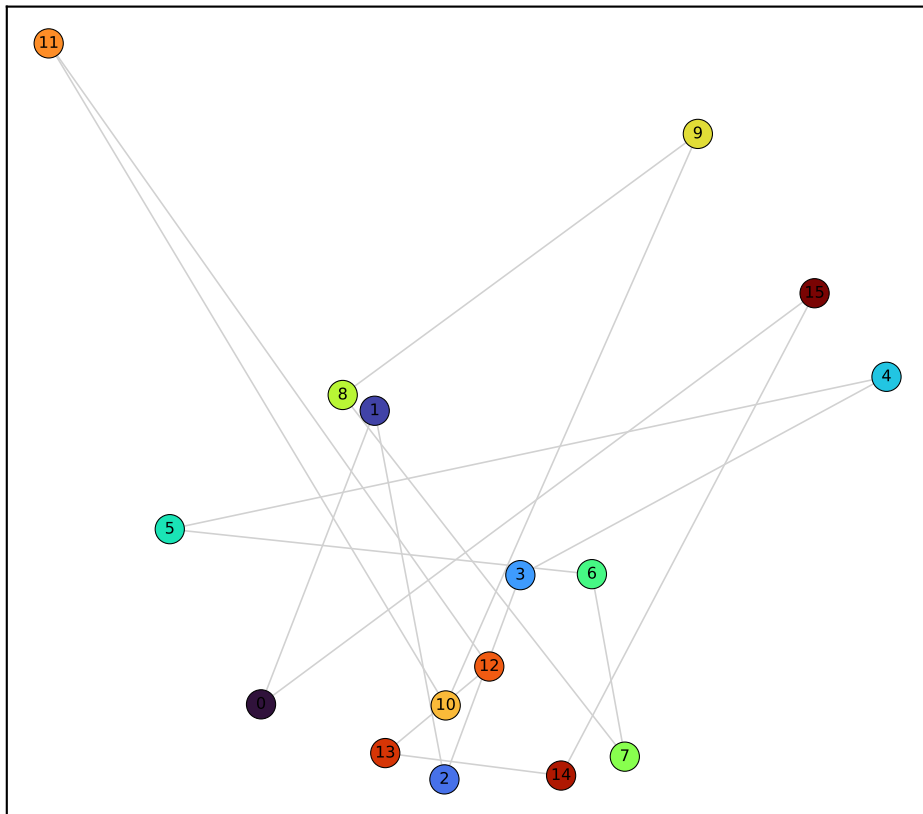


Qwen layer 2/35 (PC var 16.15%)

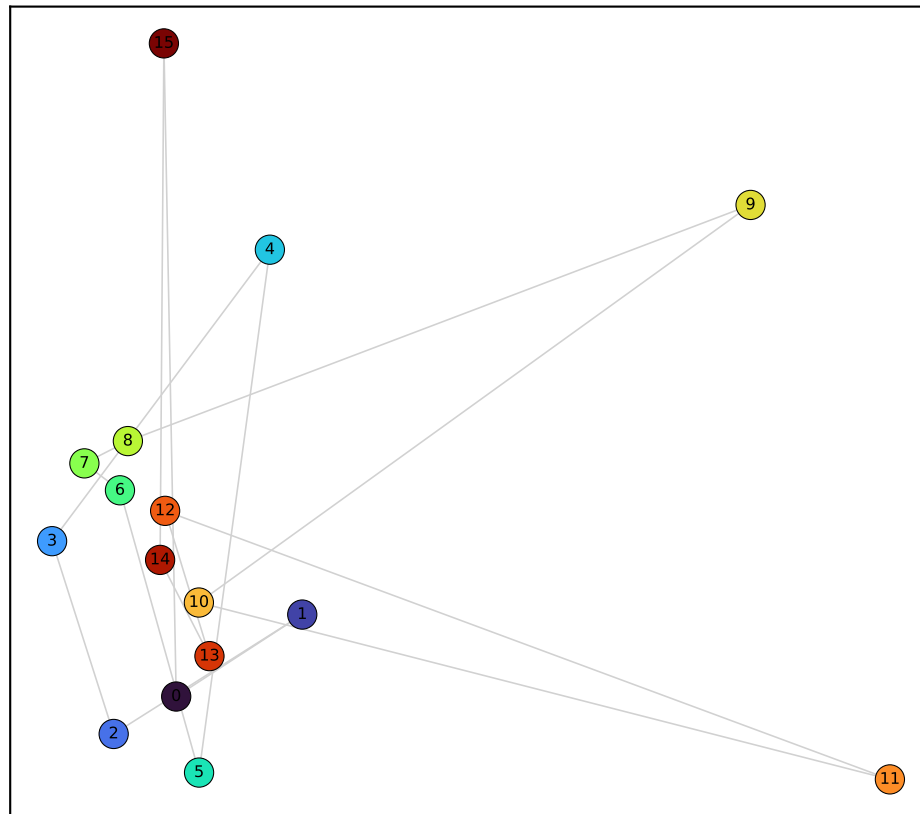


ring — node-mean PCA, depth 7% (edges = graph adjacency)

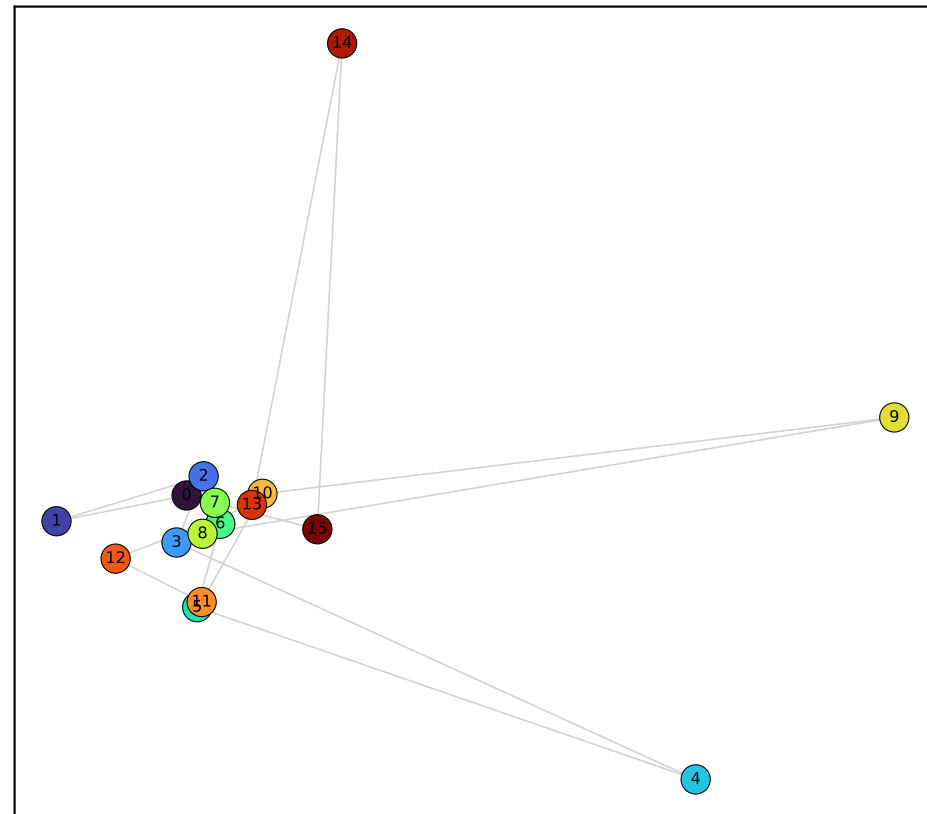
Llama layer 2/31 (PC var 9.8%)



Gemma layer 3/41 (PC var 10.9%)

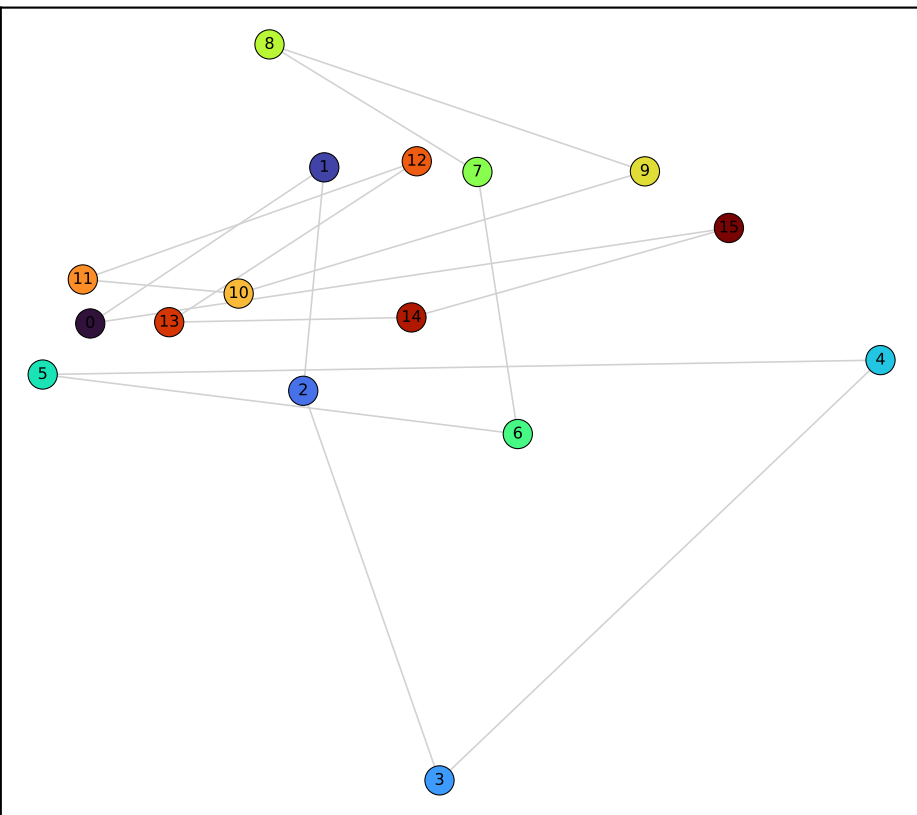


Qwen layer 3/35 (PC var 12.11%)

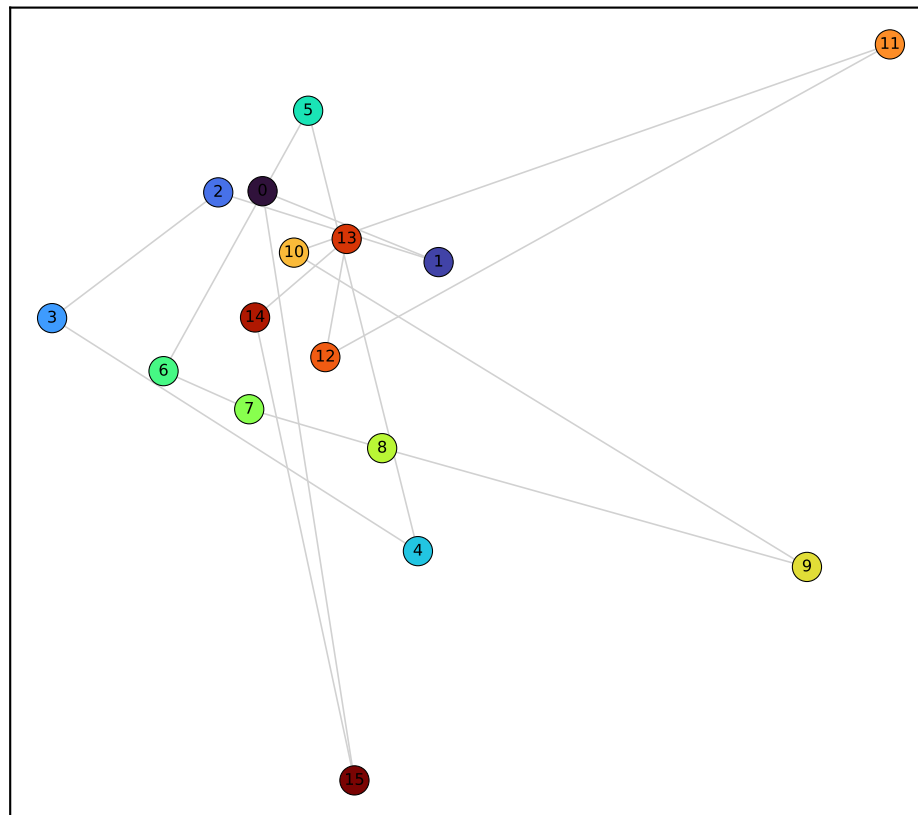


ring — node-mean PCA, depth 10% (edges = graph adjacency)

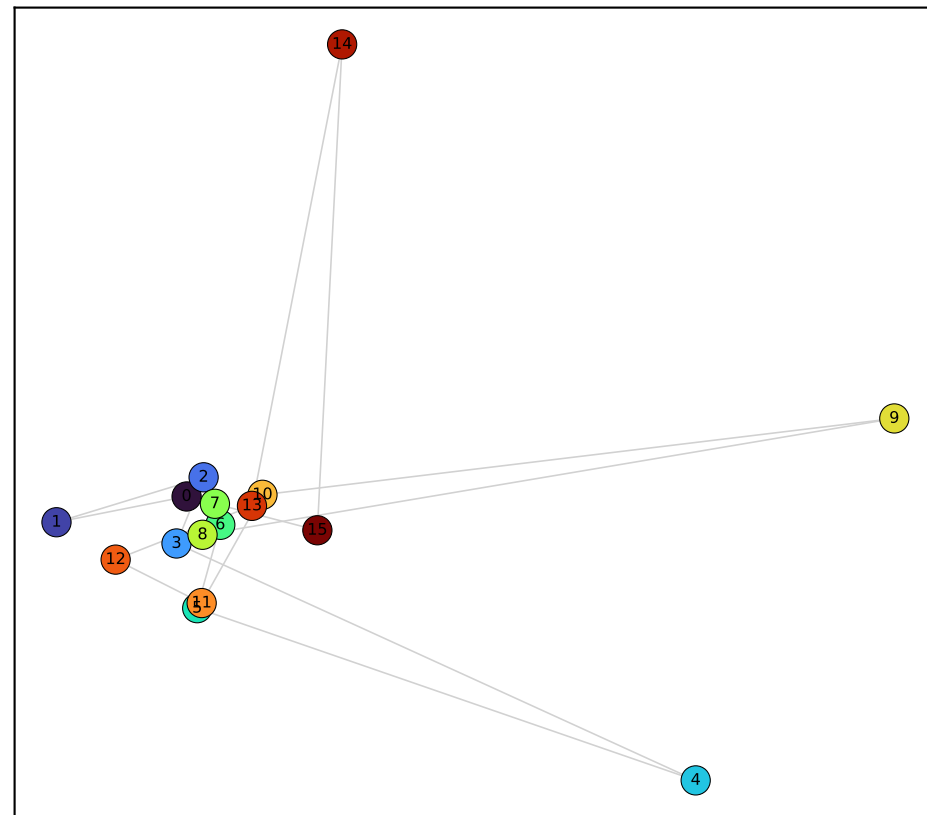
Llama layer 3/31 (PC var 9.8%)



Gemma layer 4/41 (PC var 10.9%)

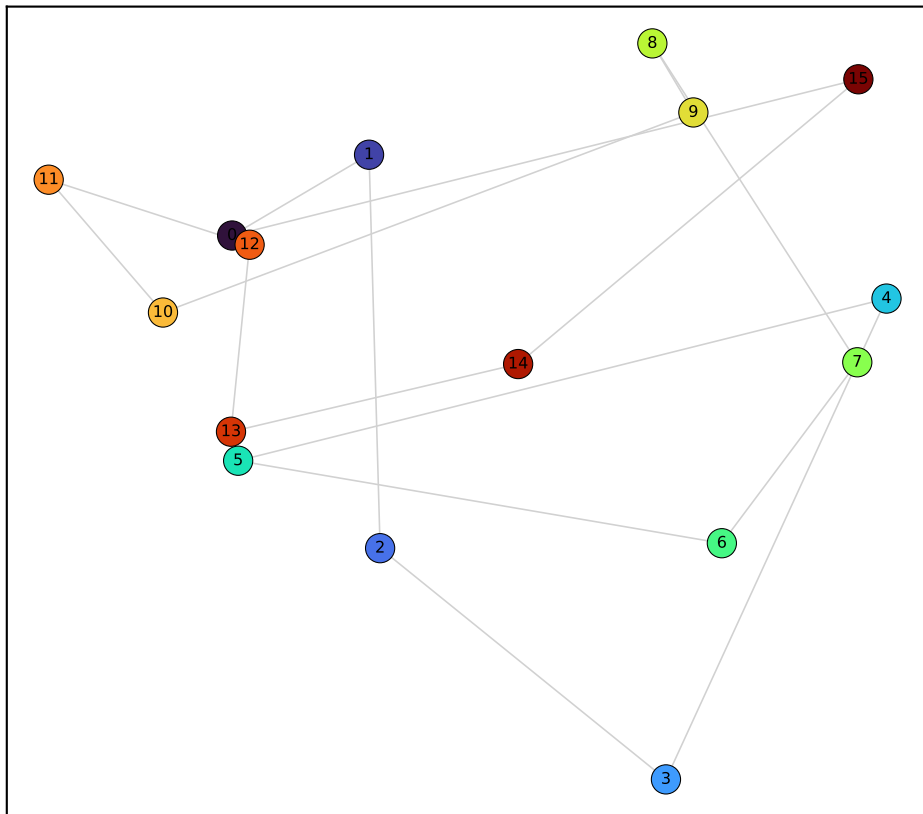


Qwen layer 3/35 (PC var 12.11%)

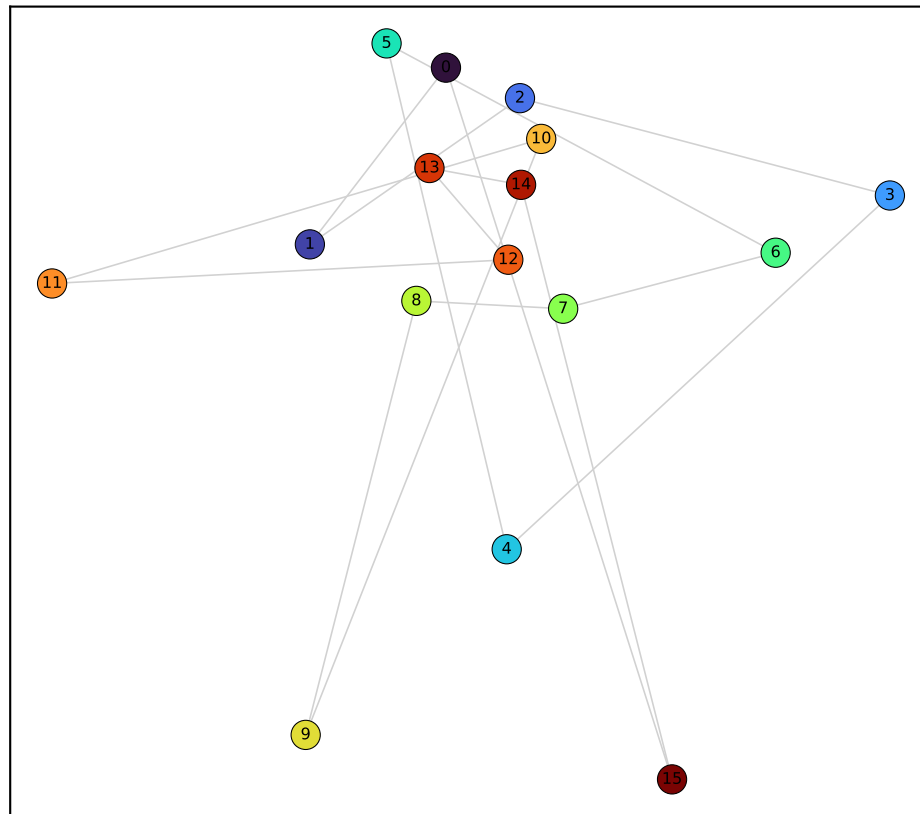


ring — node-mean PCA, depth 12% (edges = graph adjacency)

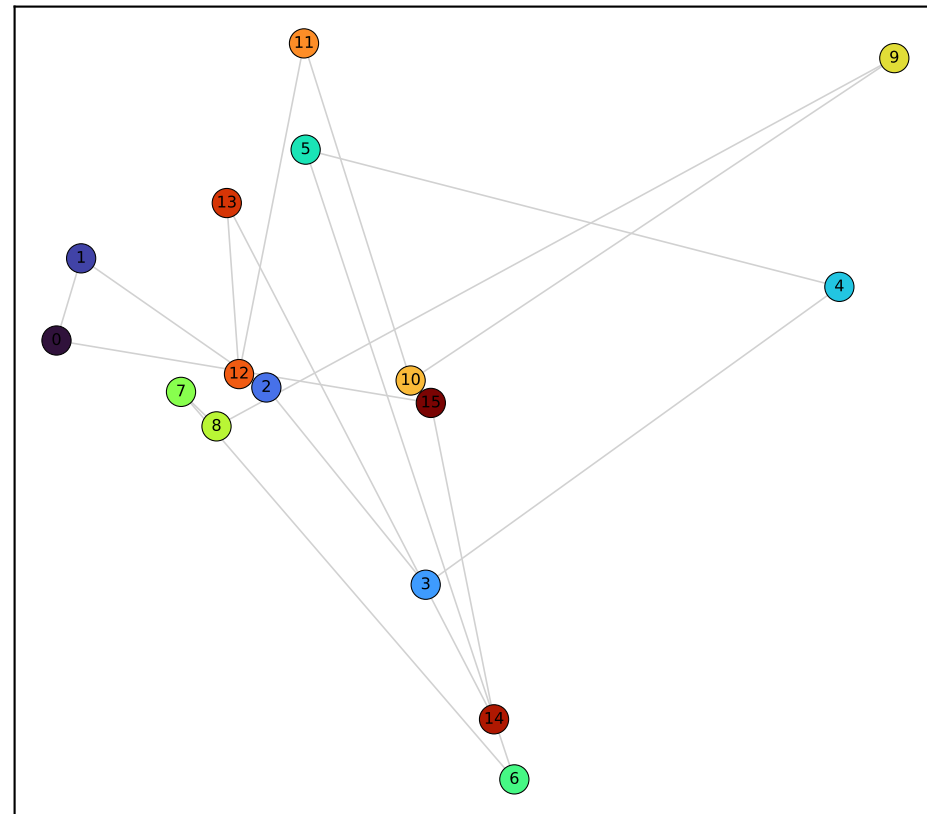
Llama layer 4/31 (PC var 10.9%)



Gemma layer 5/41 (PC var 10.10%)

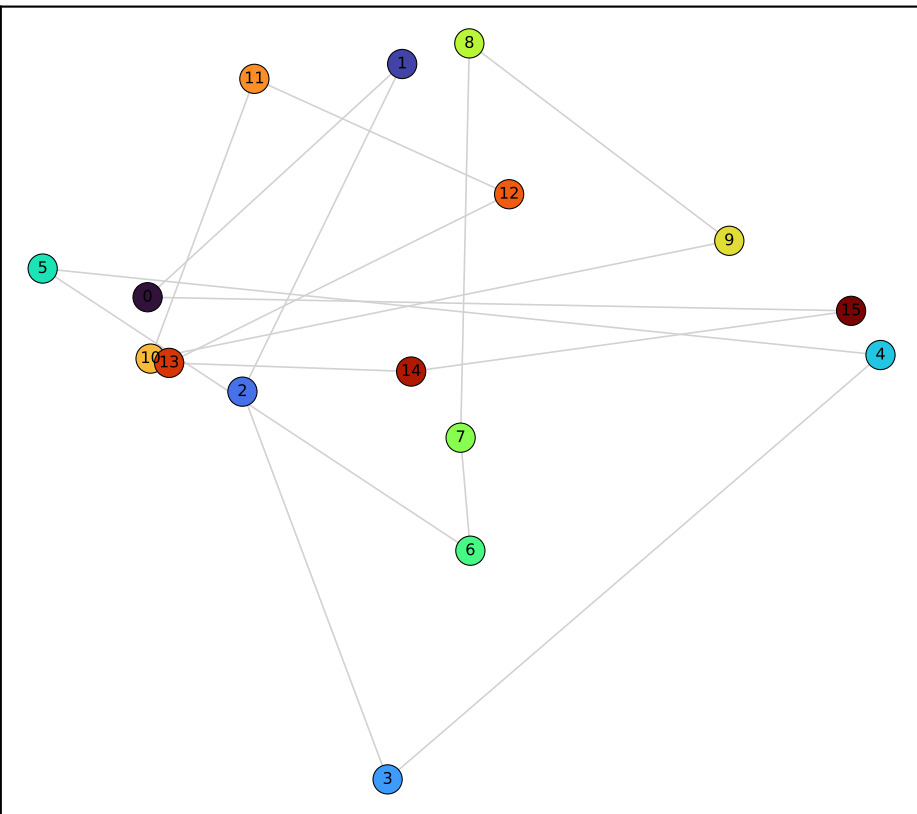


Qwen layer 4/35 (PC var 10.9%)

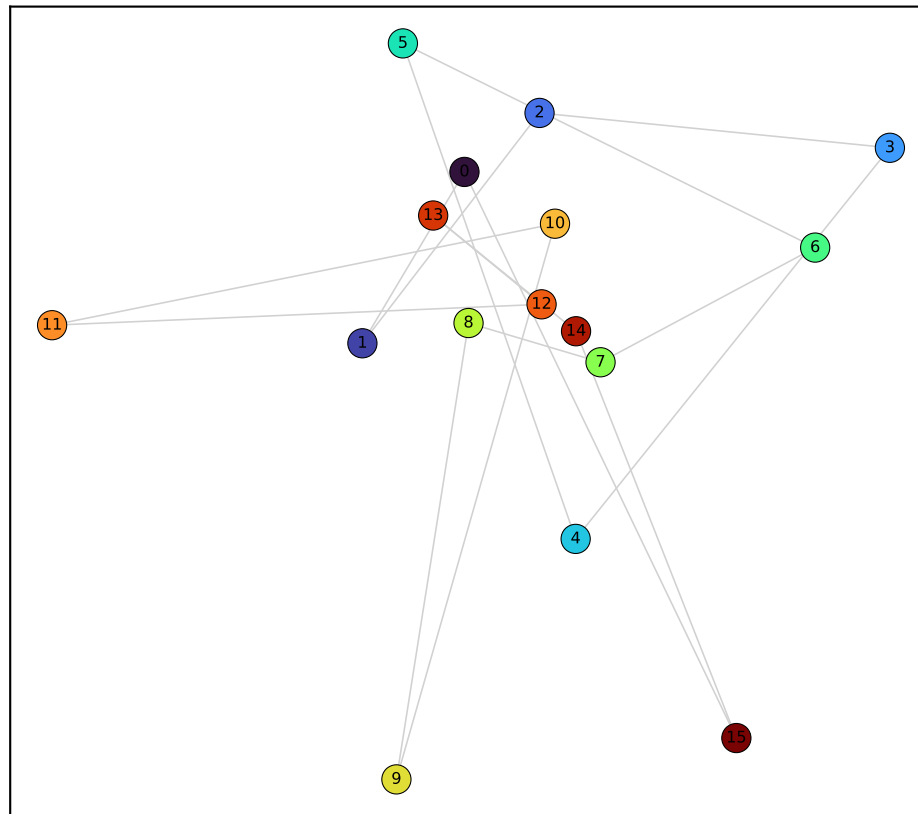


ring — node-mean PCA, depth 15% (edges = graph adjacency)

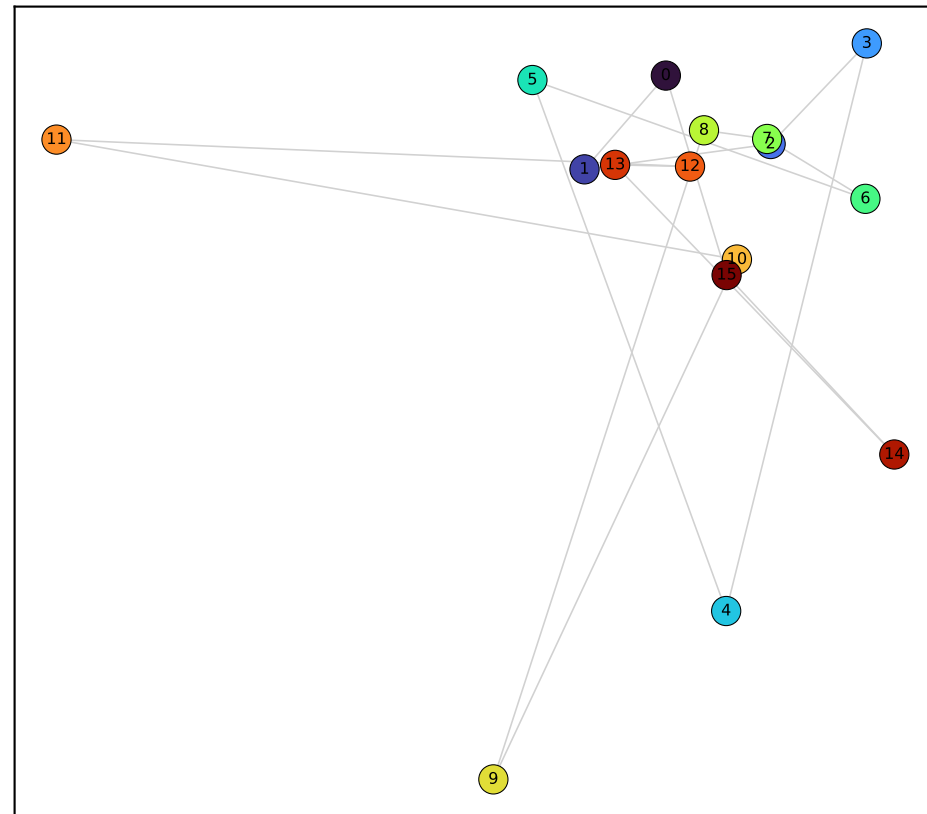
Llama layer 5/31 (PC var 11·10%)



Gemma layer 6/41 (PC var 10·10%)

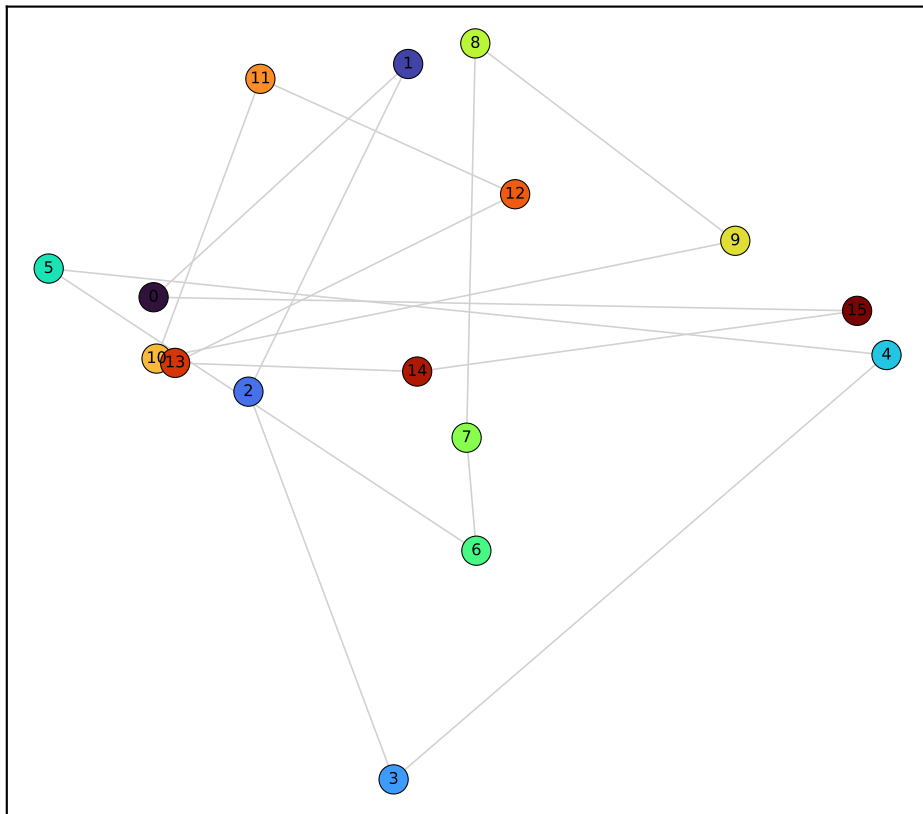


Qwen layer 5/35 (PC var 11·10%)

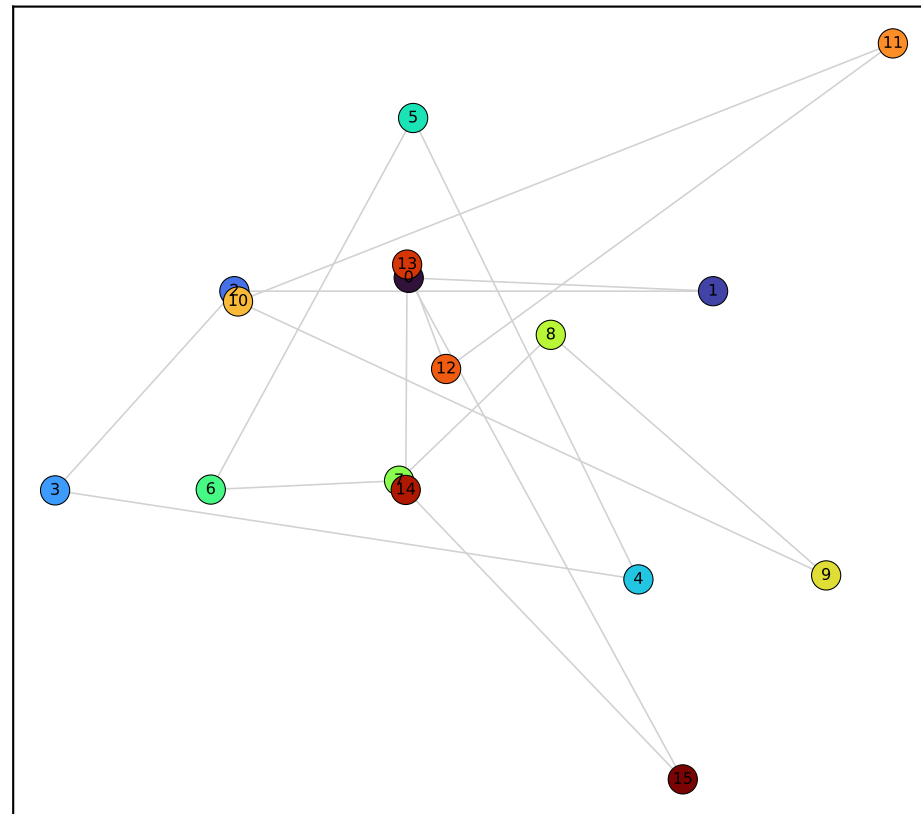


ring — node-mean PCA, depth 17% (edges = graph adjacency)

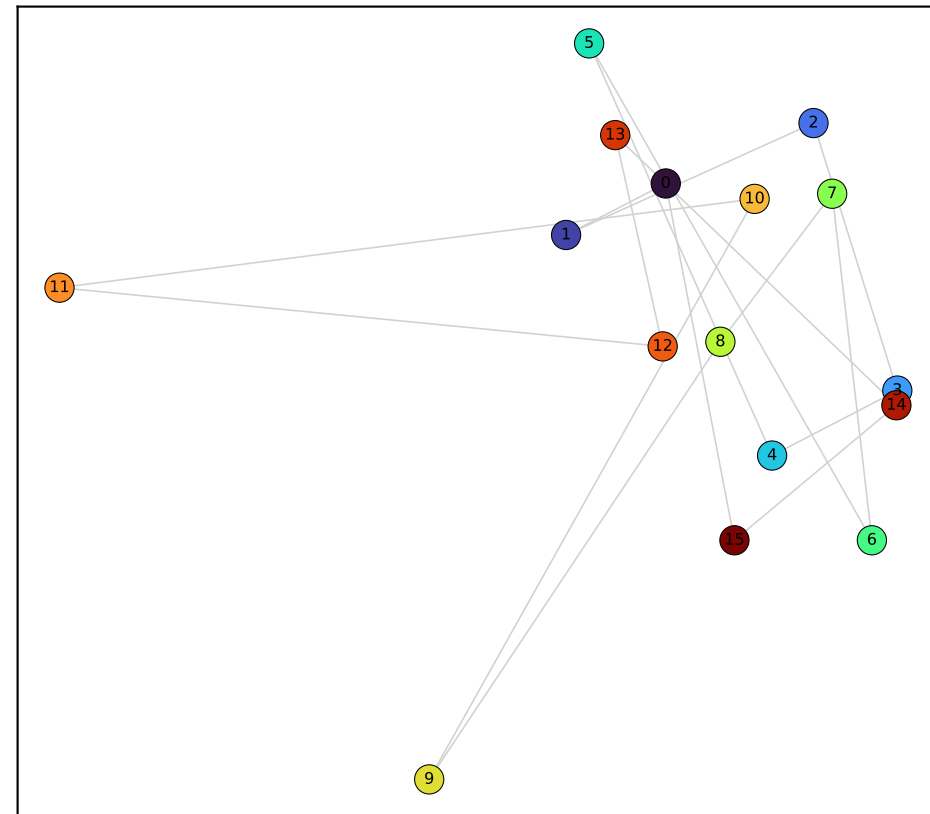
Llama layer 5/31 (PC var 11·10%)



Gemma layer 7/41 (PC var 10·10%)

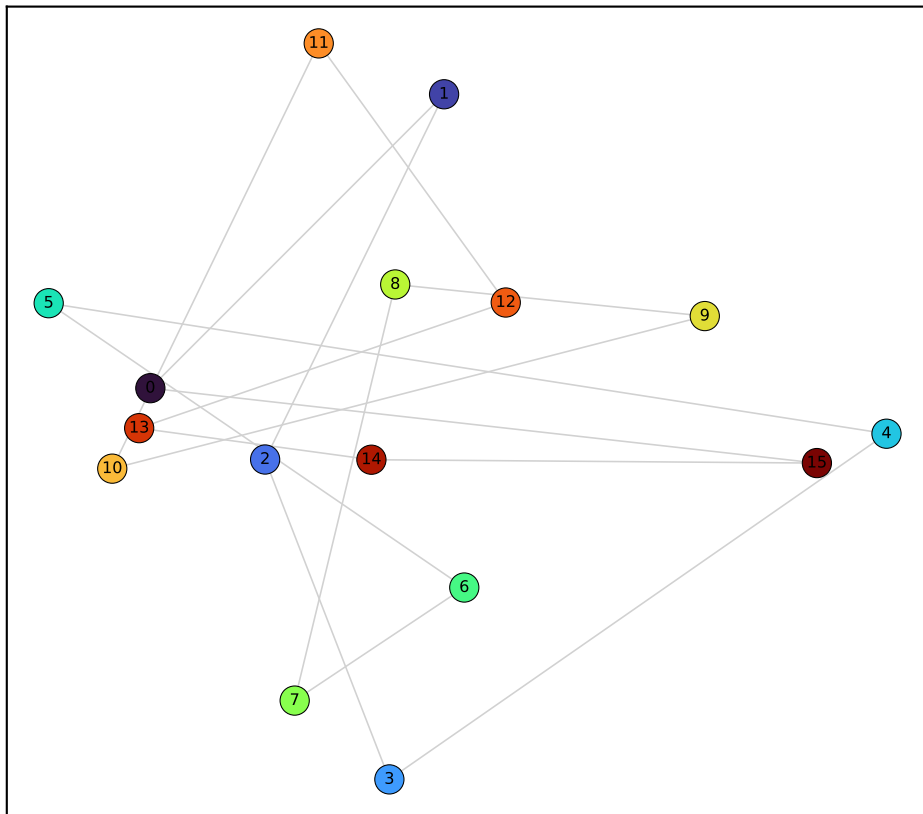


Qwen layer 6/35 (PC var 11·9%)

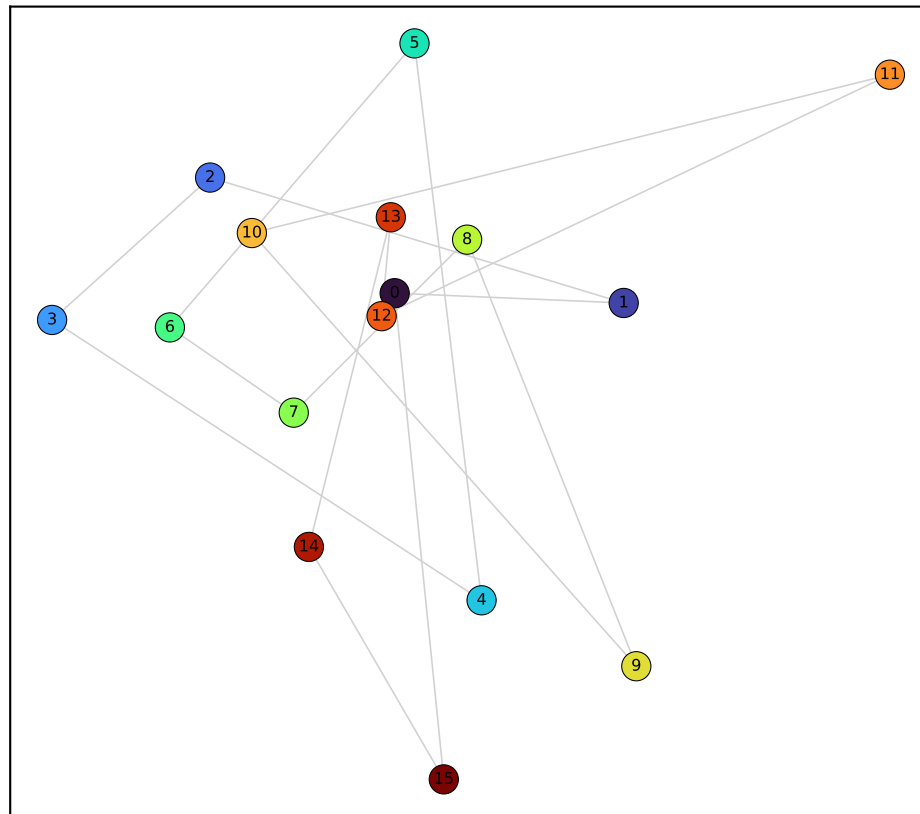


ring — node-mean PCA, depth 20% (edges = graph adjacency)

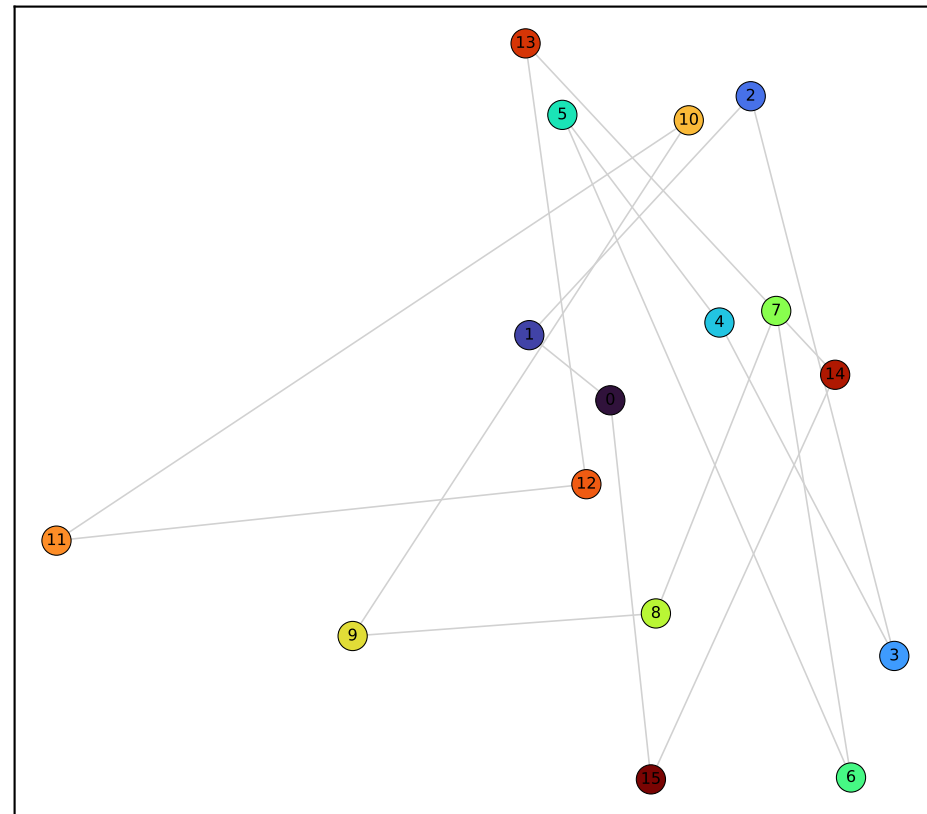
Llama layer 6/31 (PC var 13·11%)



Gemma layer 8/41 (PC var 10·10%)

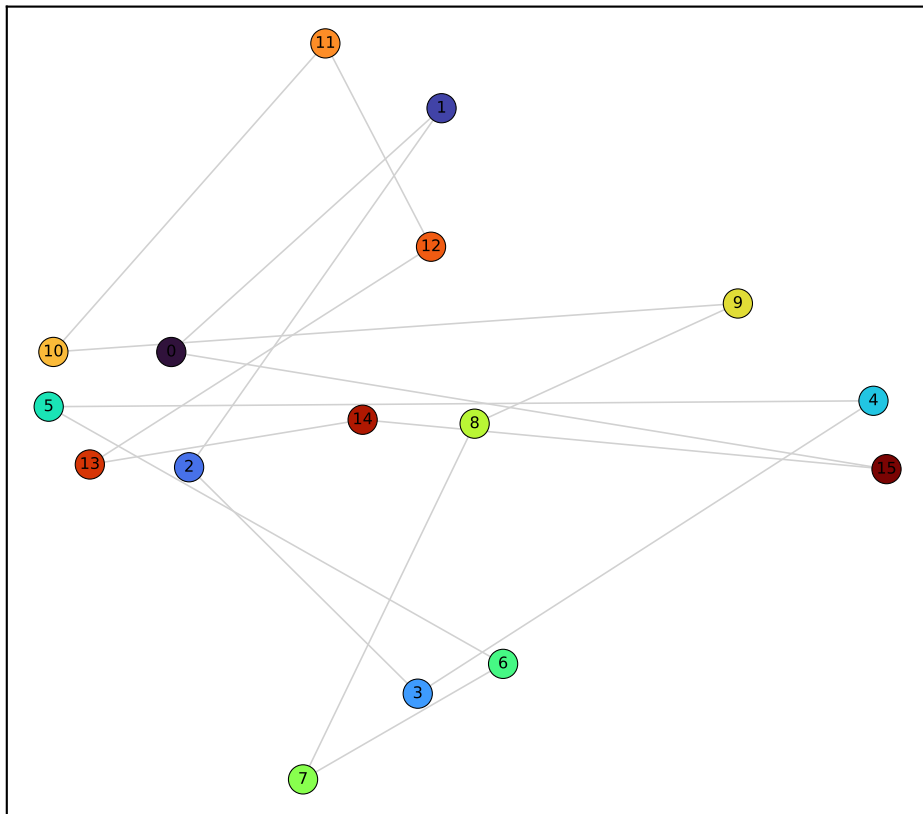


Qwen layer 7/35 (PC var 11·9%)

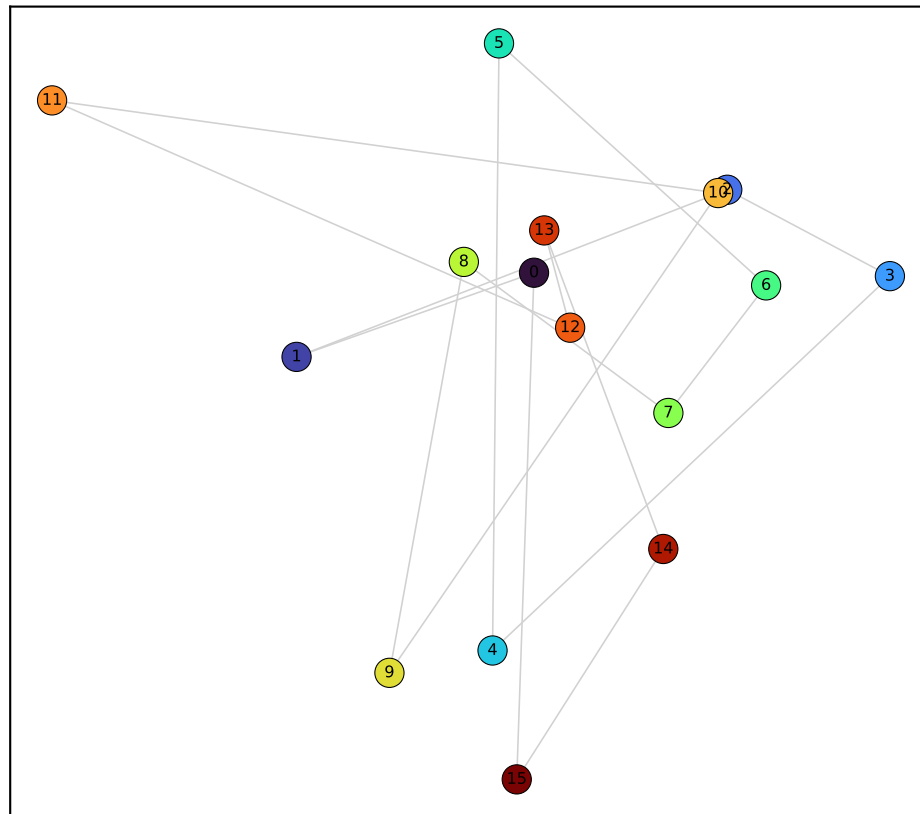


ring — node-mean PCA, depth 22% (edges = graph adjacency)

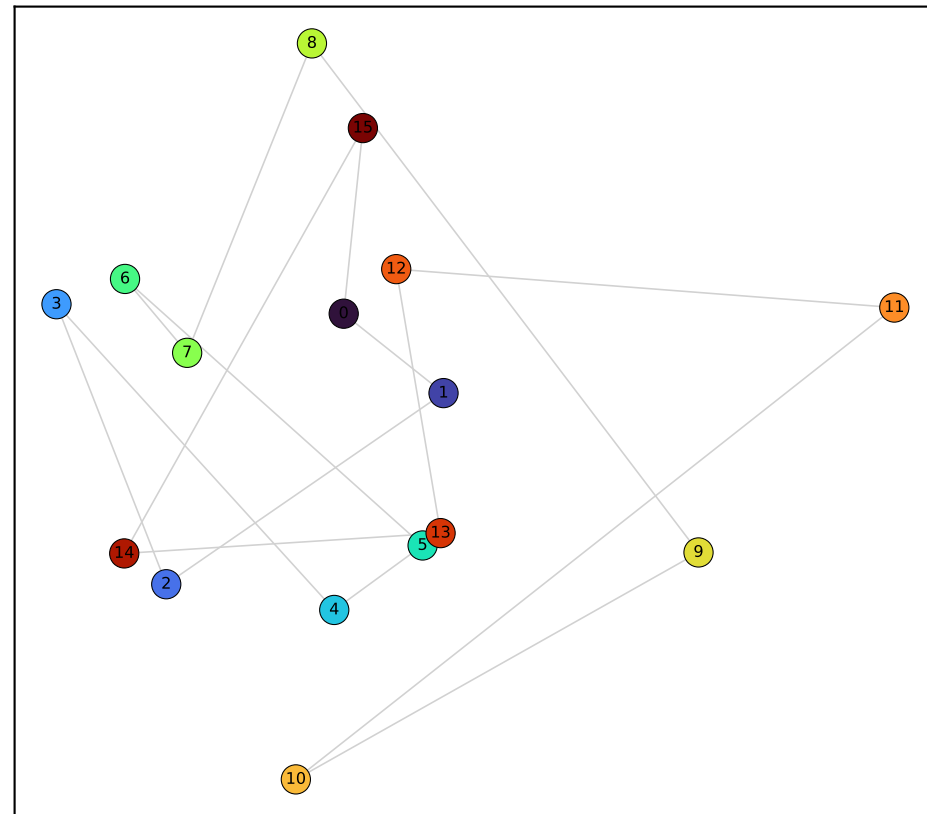
Llama layer 7/31 (PC var 12·11%)



Gemma layer 9/41 (PC var 11·10%)

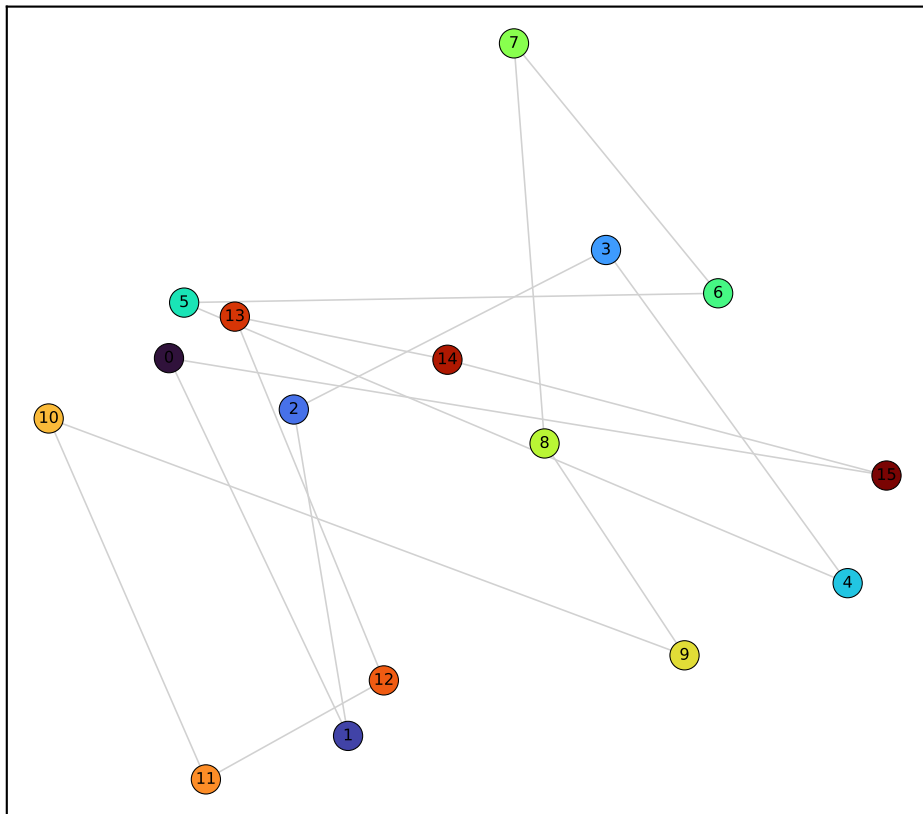


Qwen layer 8/35 (PC var 10·9%)

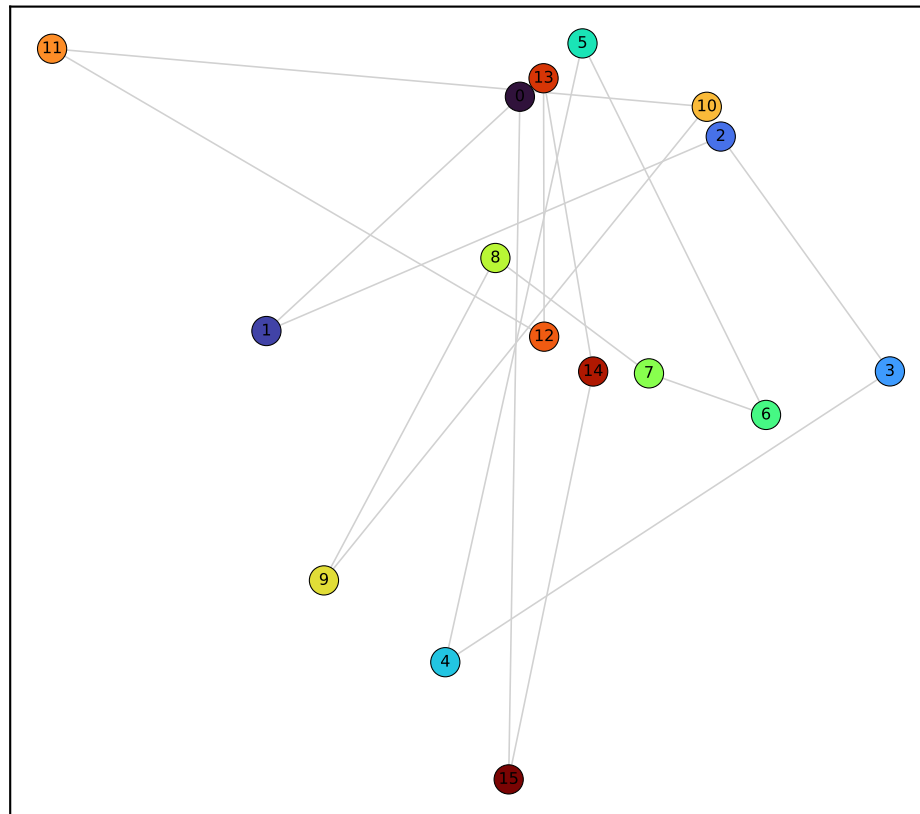


ring — node-mean PCA, depth 27% (edges = graph adjacency)

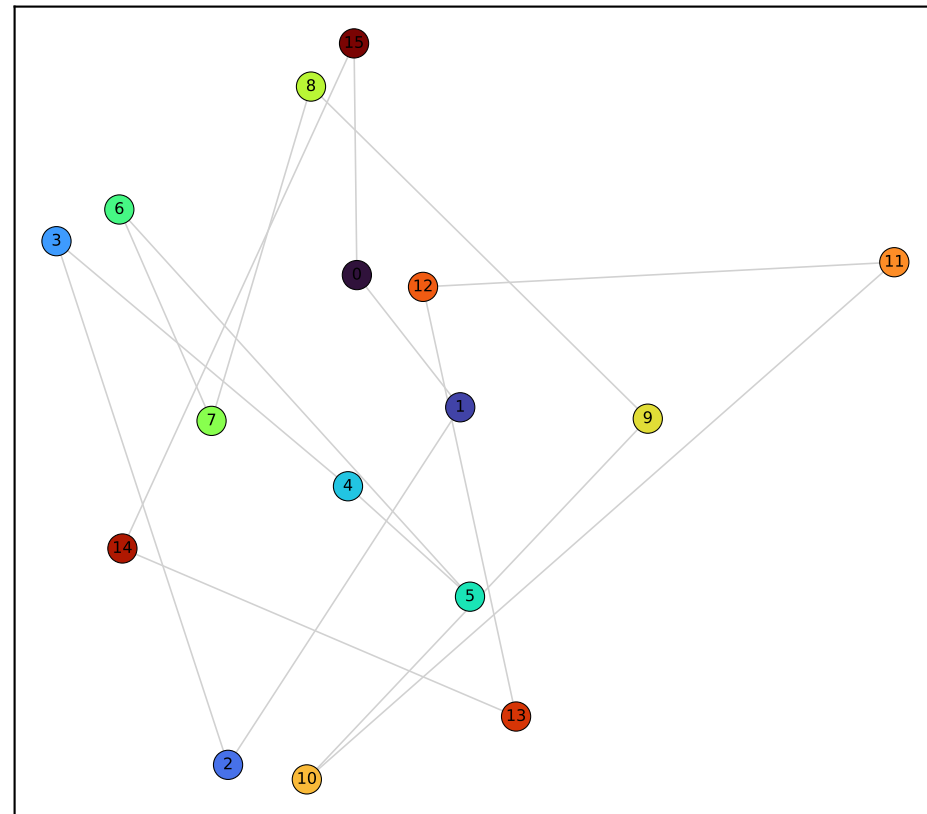
Llama layer 8/31 (PC var 12·12%)



Gemma layer 11/41 (PC var 11·10%)

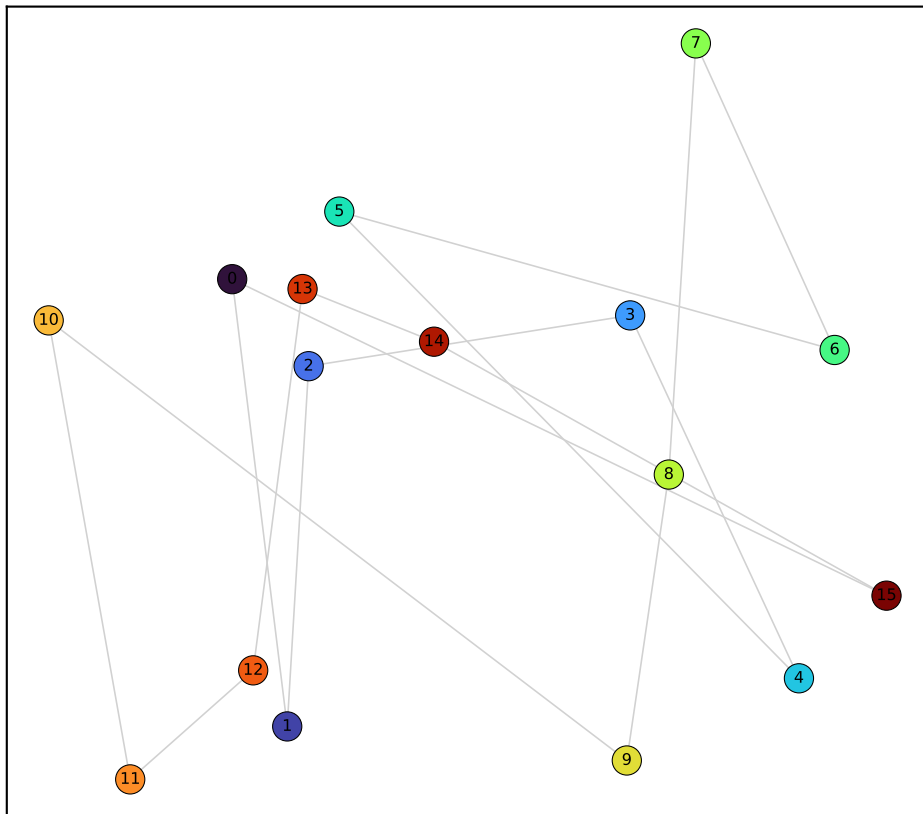


Qwen layer 9/35 (PC var 11·10%)

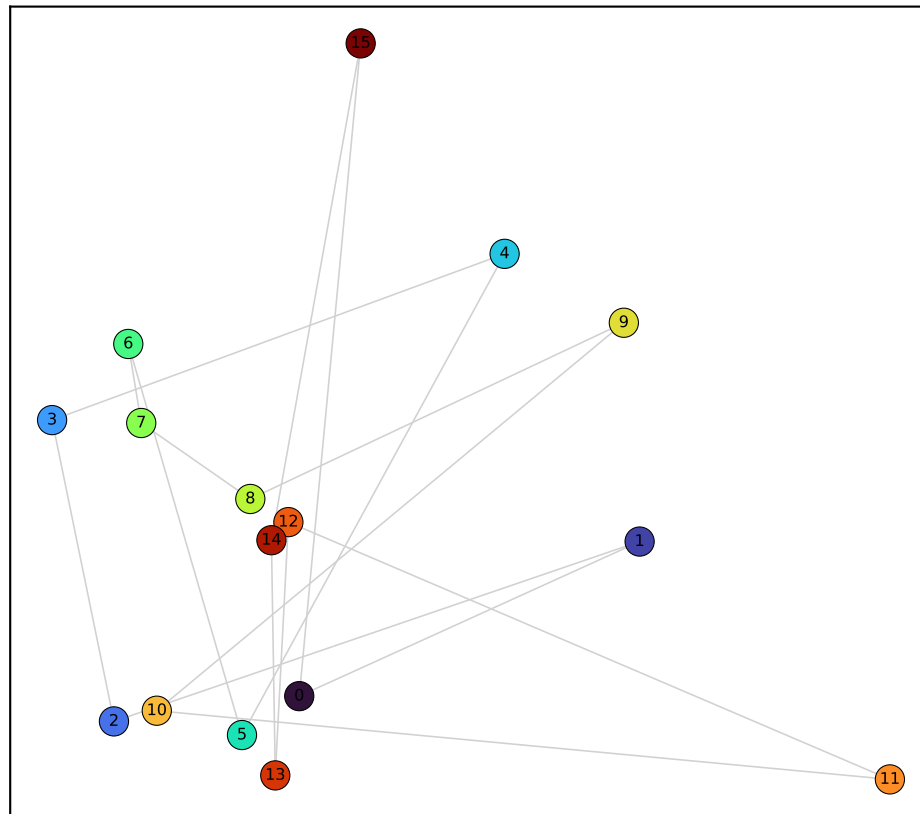


ring — node-mean PCA, depth 29% (edges = graph adjacency)

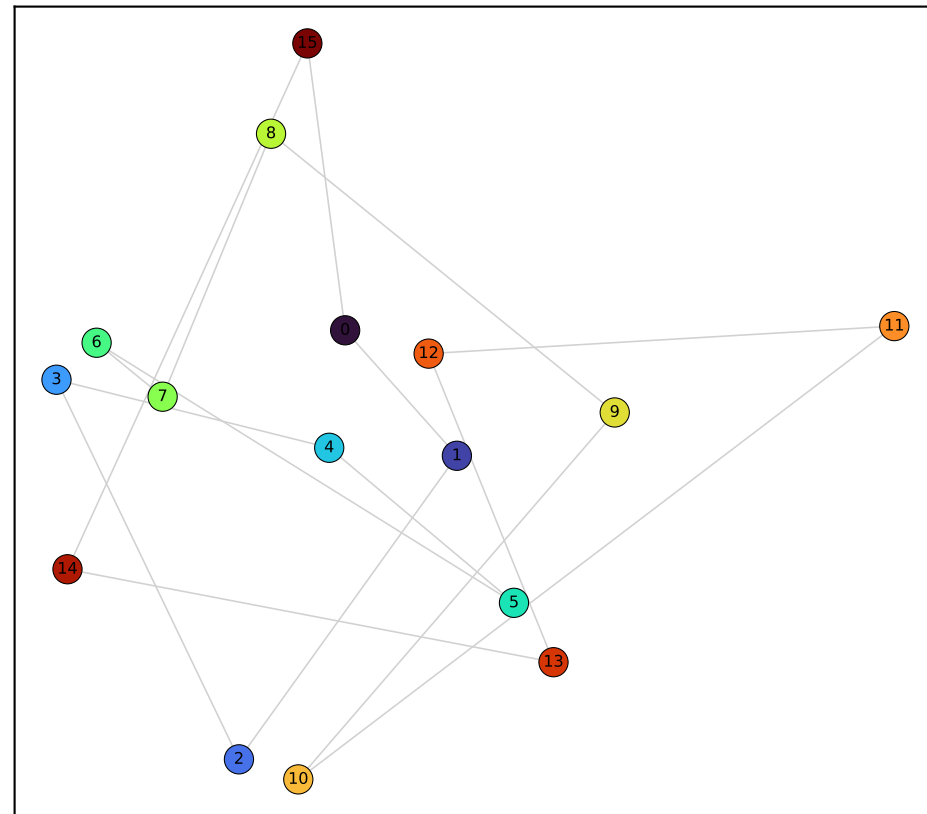
Llama layer 9/31 (PC var 12·12%)



Gemma layer 12/41 (PC var 11·11%)

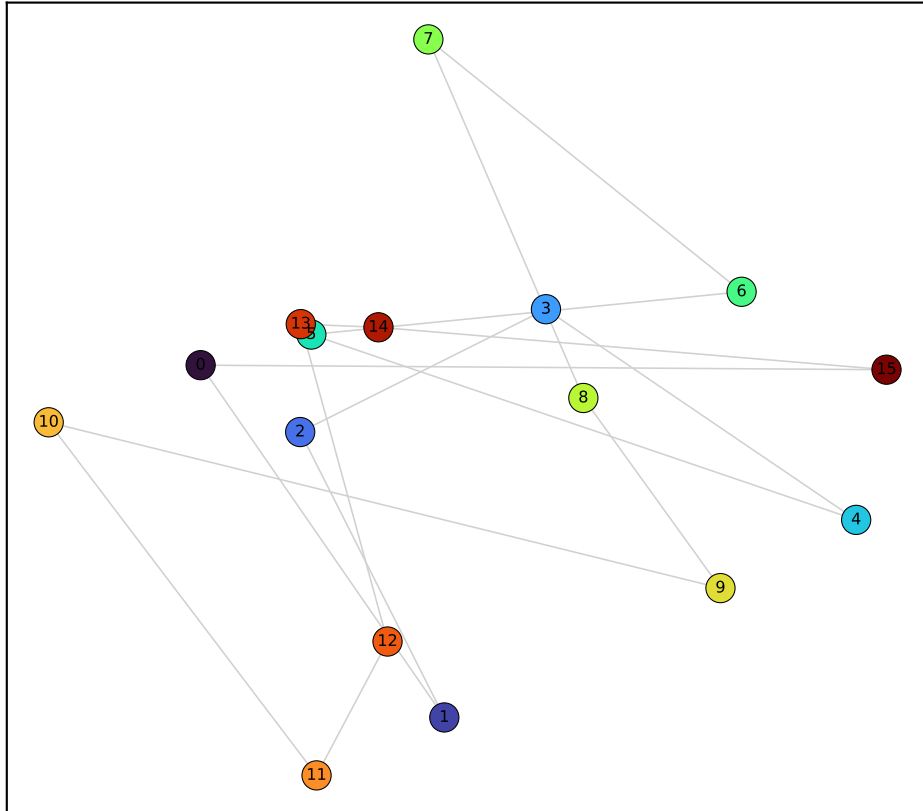


Qwen layer 10/35 (PC var 11·10%)

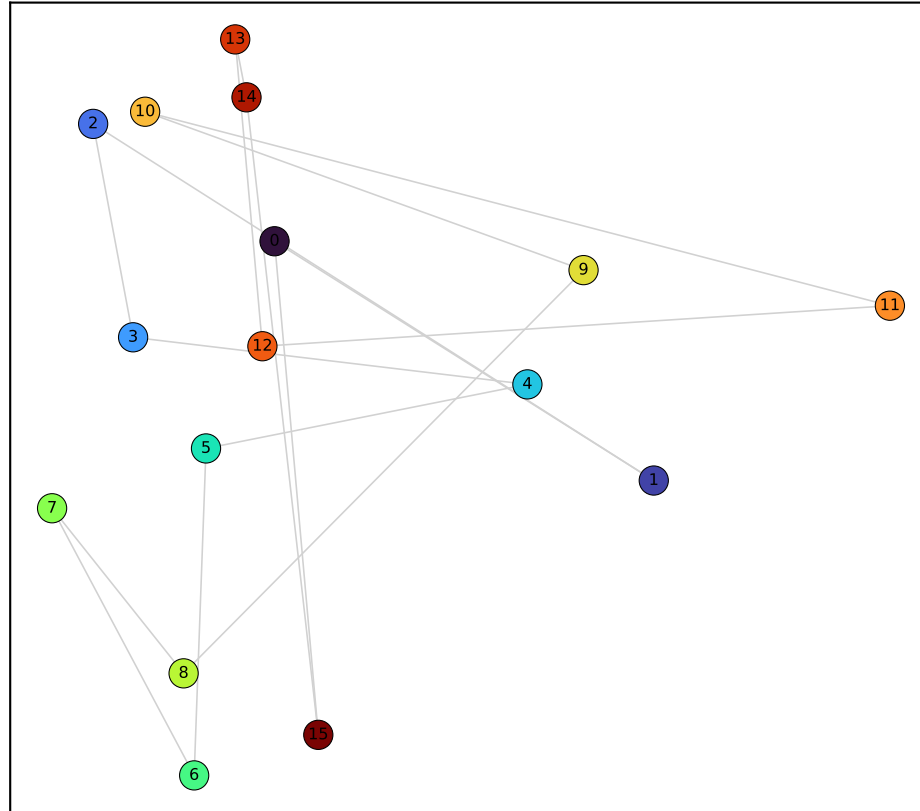


ring — node-mean PCA, depth 32% (edges = graph adjacency)

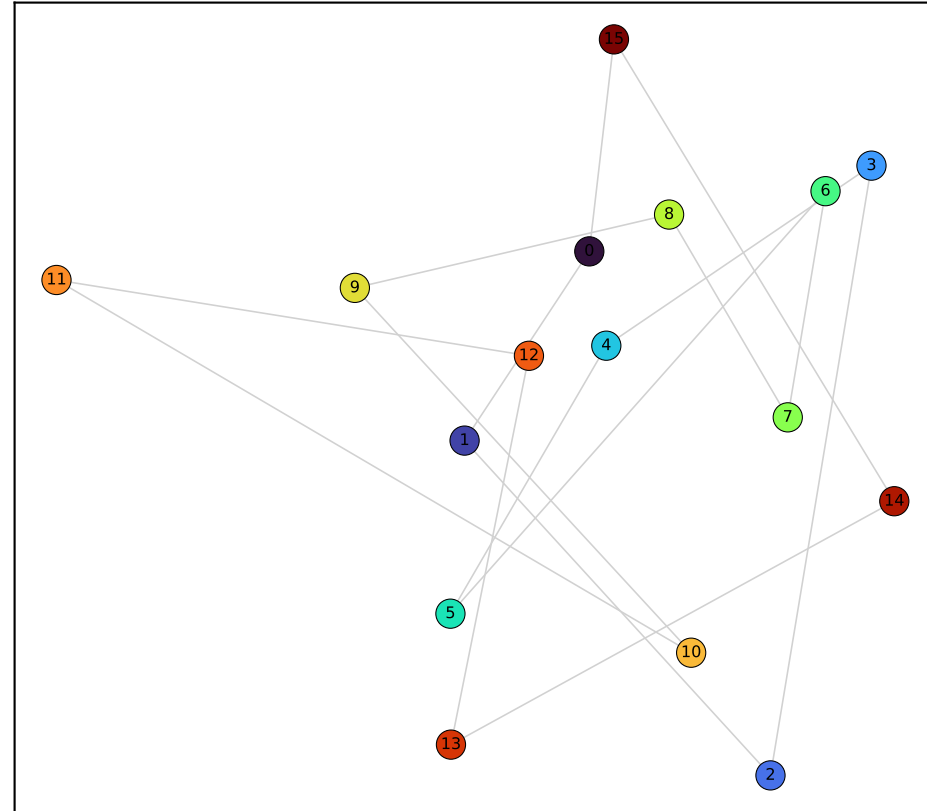
Llama layer 10/31 (PC var 13·12%)



Gemma layer 13/41 (PC var 12·11%)

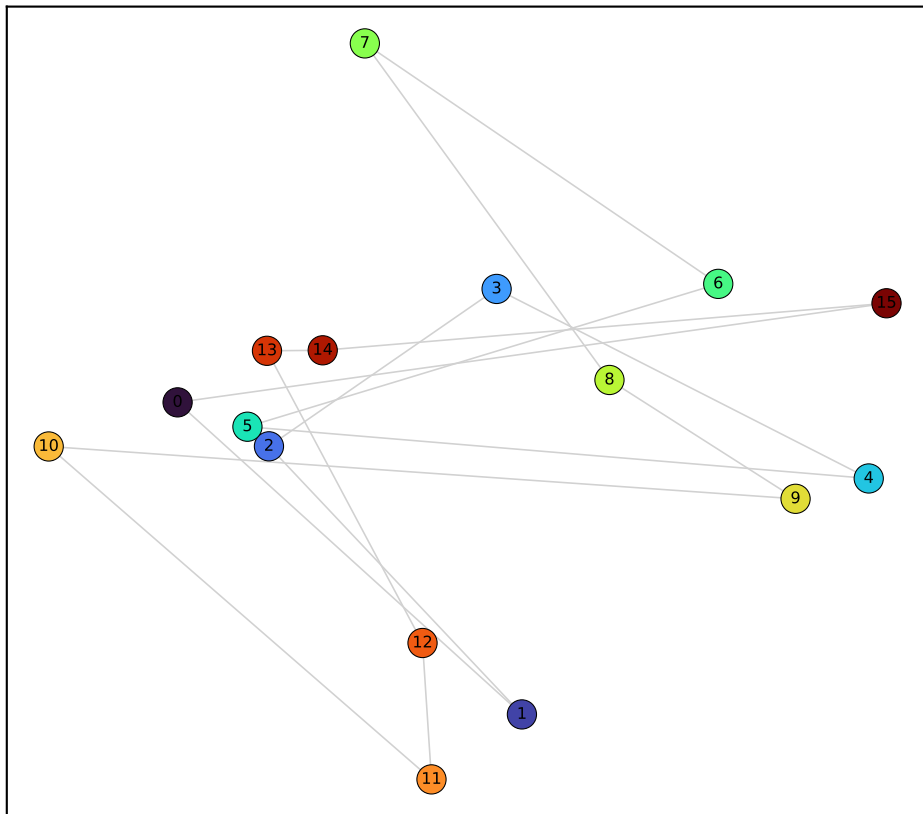


Qwen layer 11/35 (PC var 11·10%)

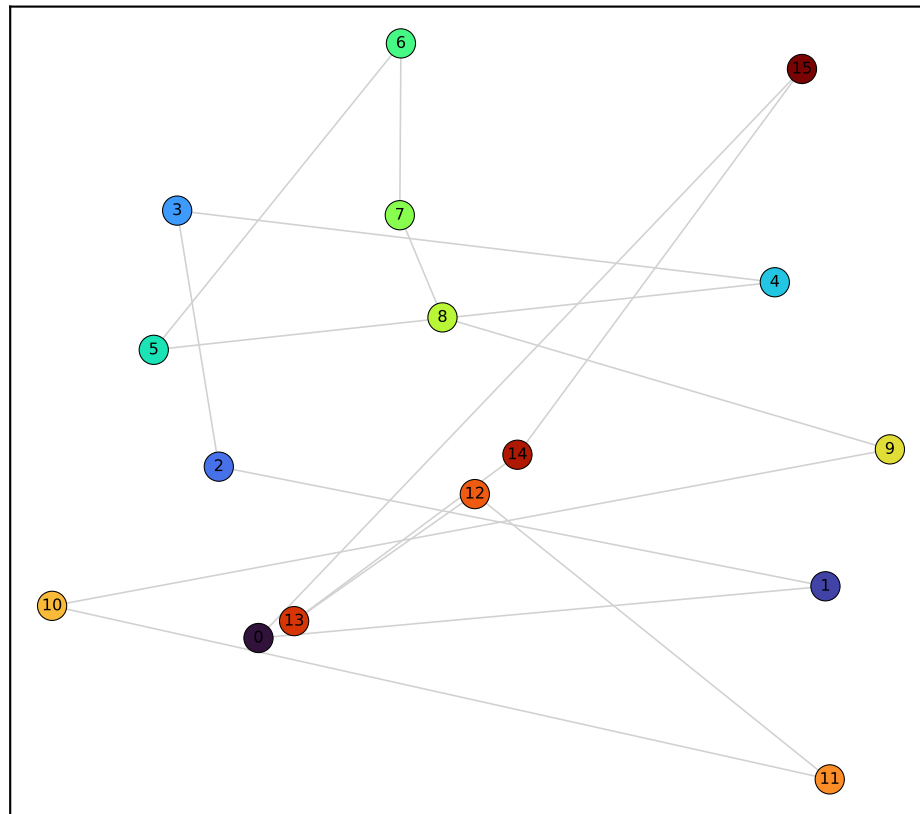


ring — node-mean PCA, depth 34% (edges = graph adjacency)

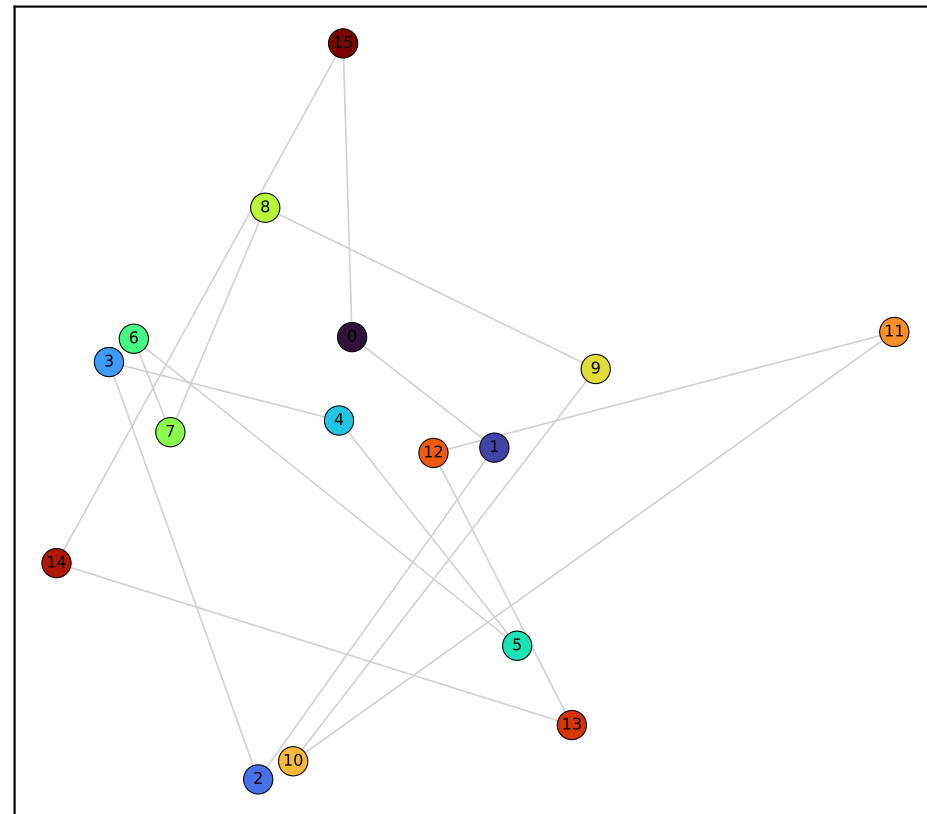
Llama layer 11/31 (PC var 14·11%)



Gemma layer 14/41 (PC var 12·11%)

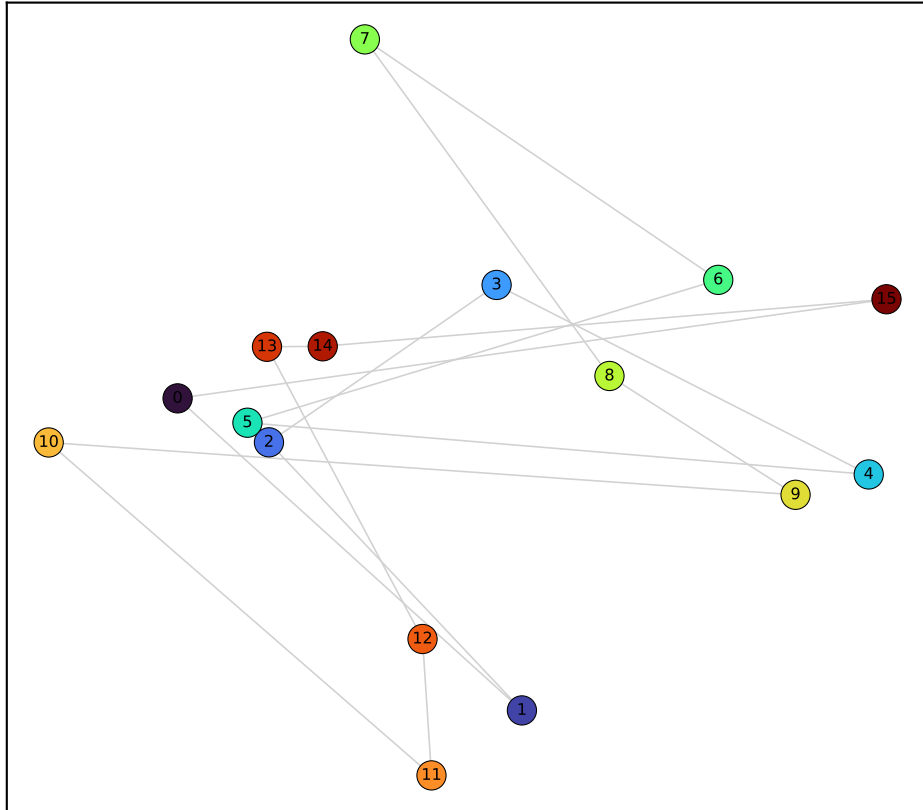


Qwen layer 12/35 (PC var 11·10%)

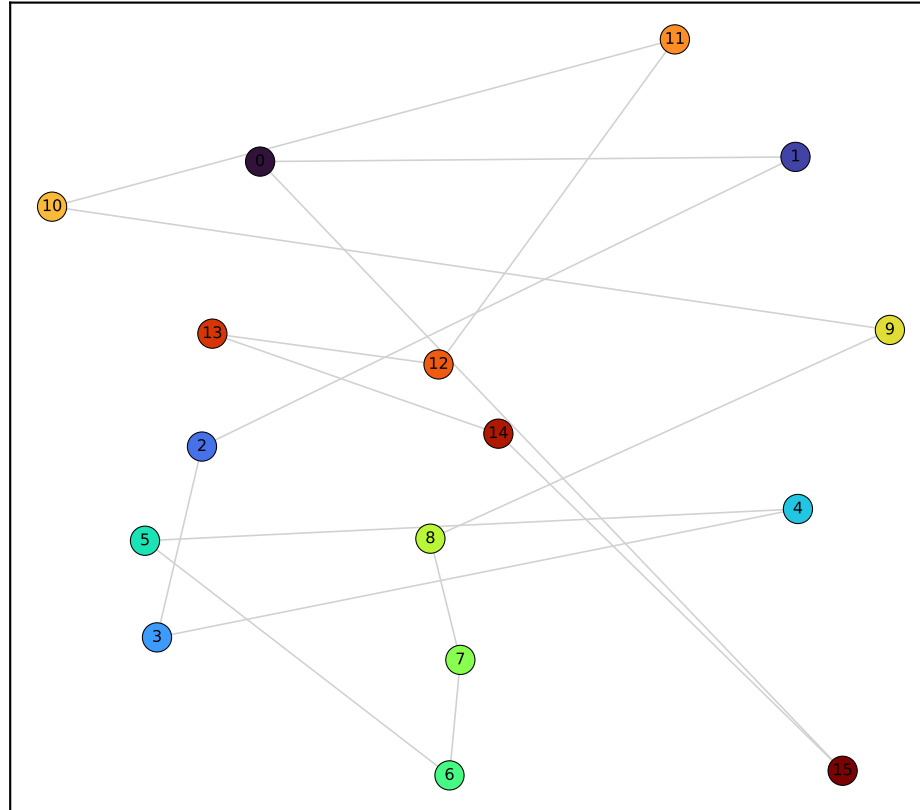


ring — node-mean PCA, depth 37% (edges = graph adjacency)

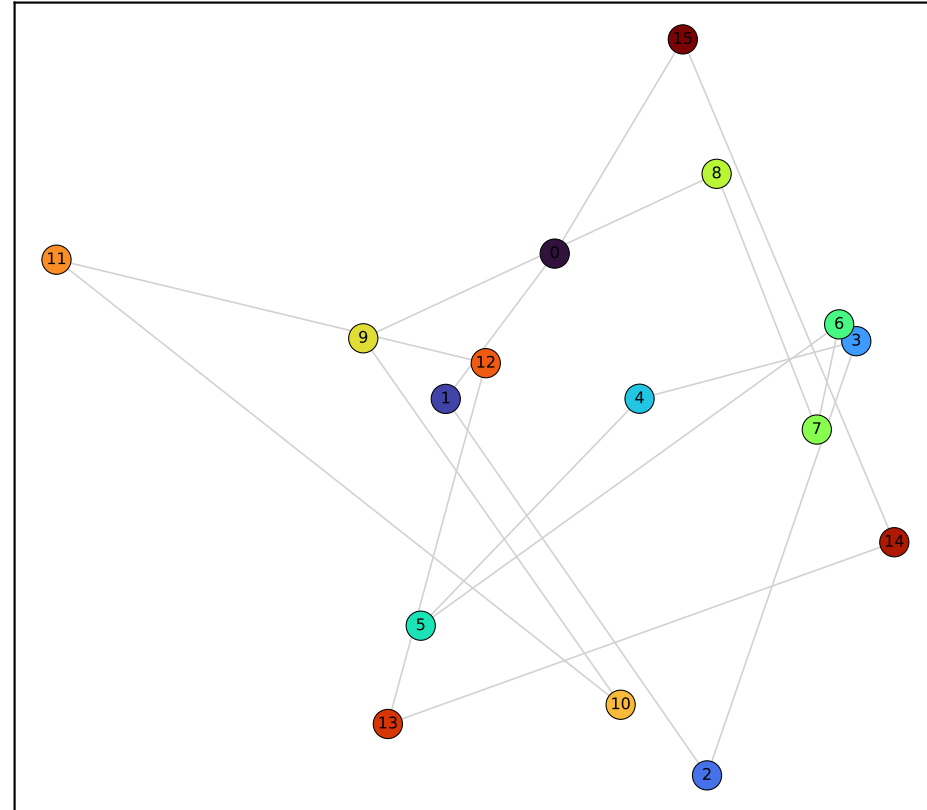
Llama layer 11/31 (PC var 14·11%)



Gemma layer 15/41 (PC var 12·11%)

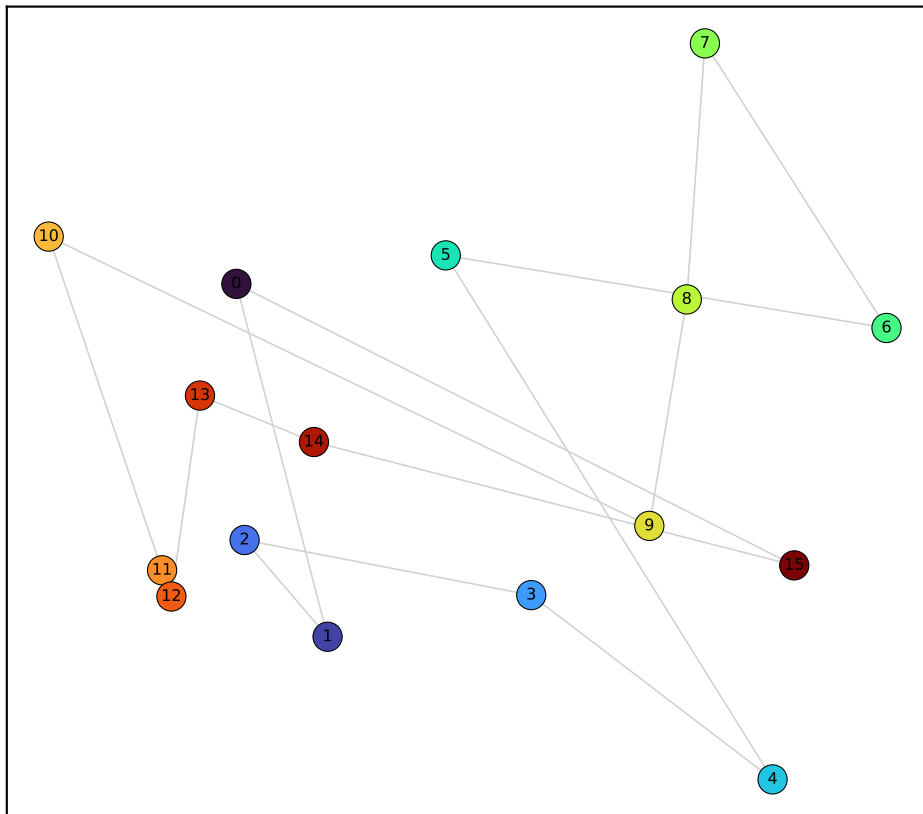


Qwen layer 13/35 (PC var 11·10%)

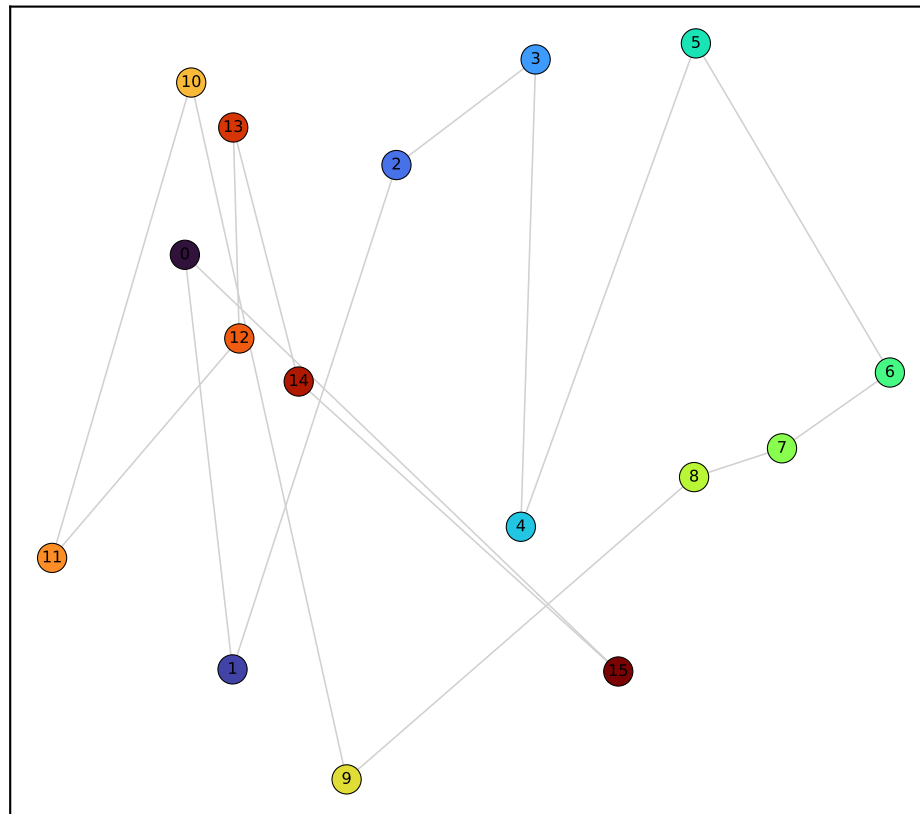


ring — node-mean PCA, depth 39% (edges = graph adjacency)

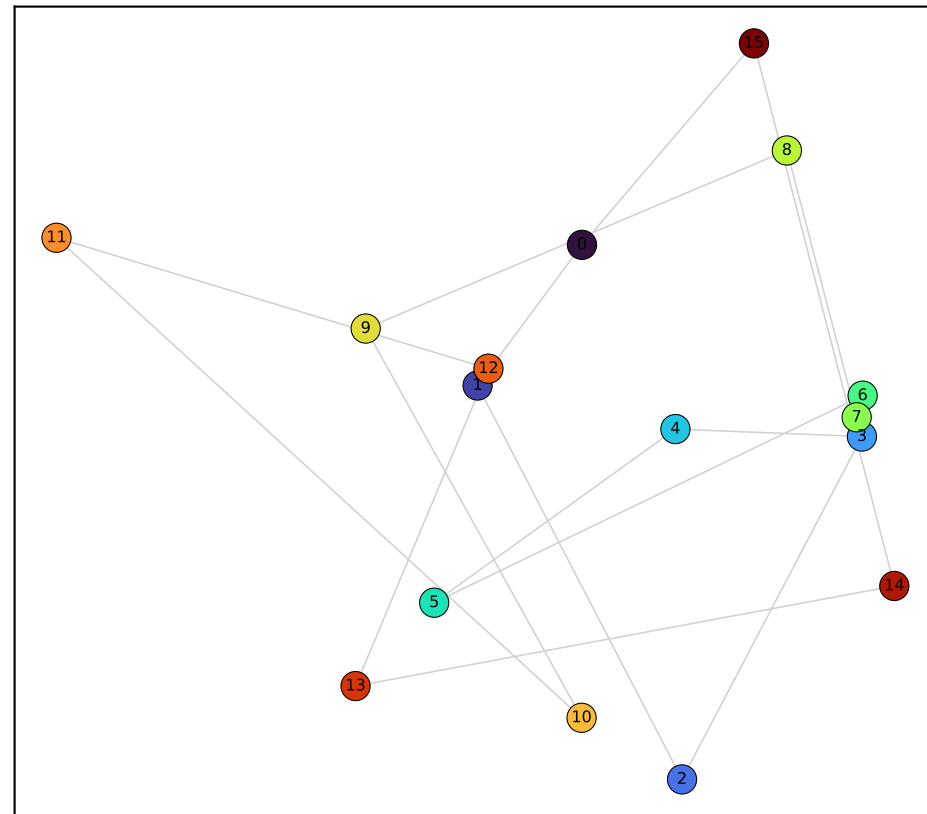
Llama layer 12/31 (PC var 13·12%)



Gemma layer 16/41 (PC var 13·11%)

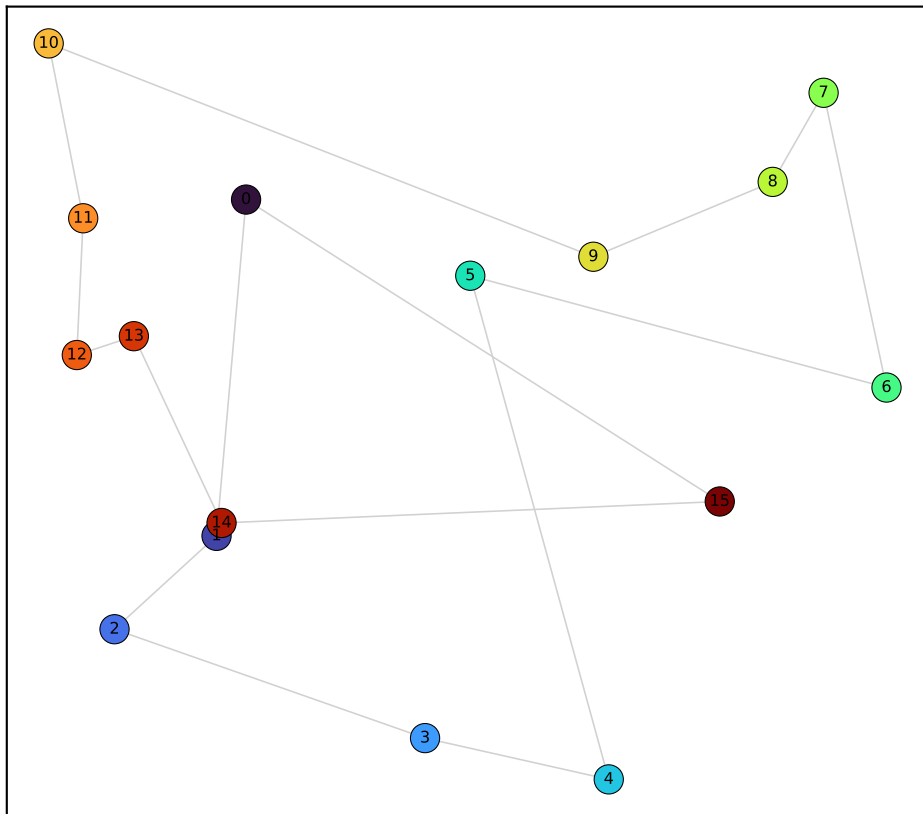


Qwen layer 14/35 (PC var 11·10%)

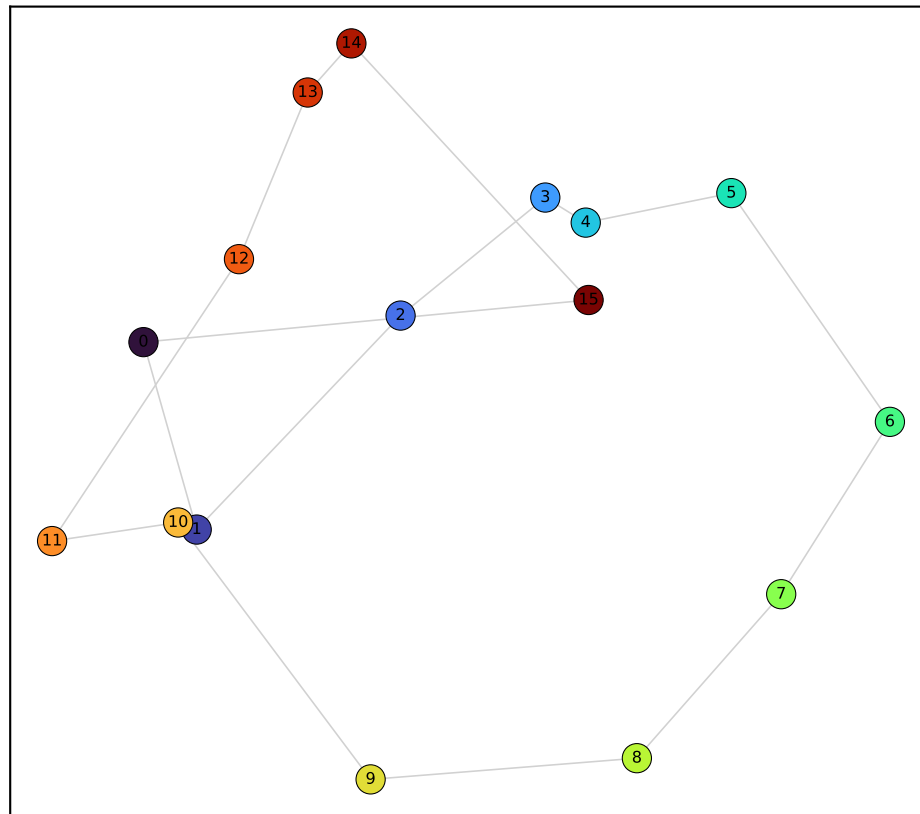


ring — node-mean PCA, depth 41% (edges = graph adjacency)

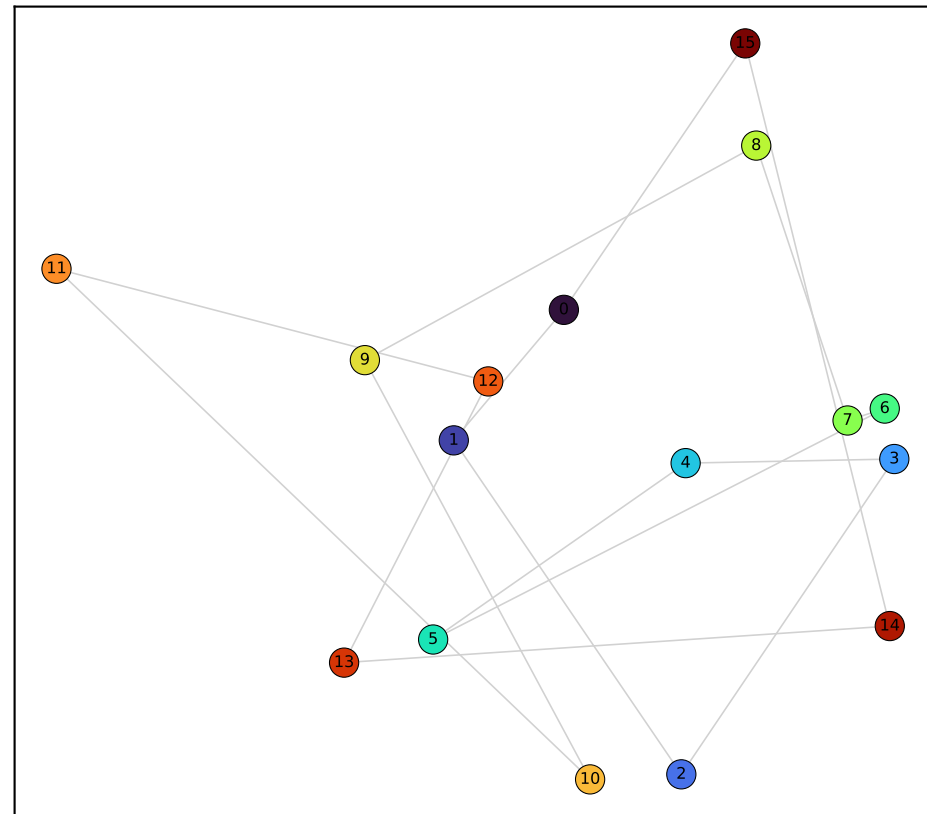
Llama layer 13/31 (PC var 13·11%)



Gemma layer 17/41 (PC var 14·12%)

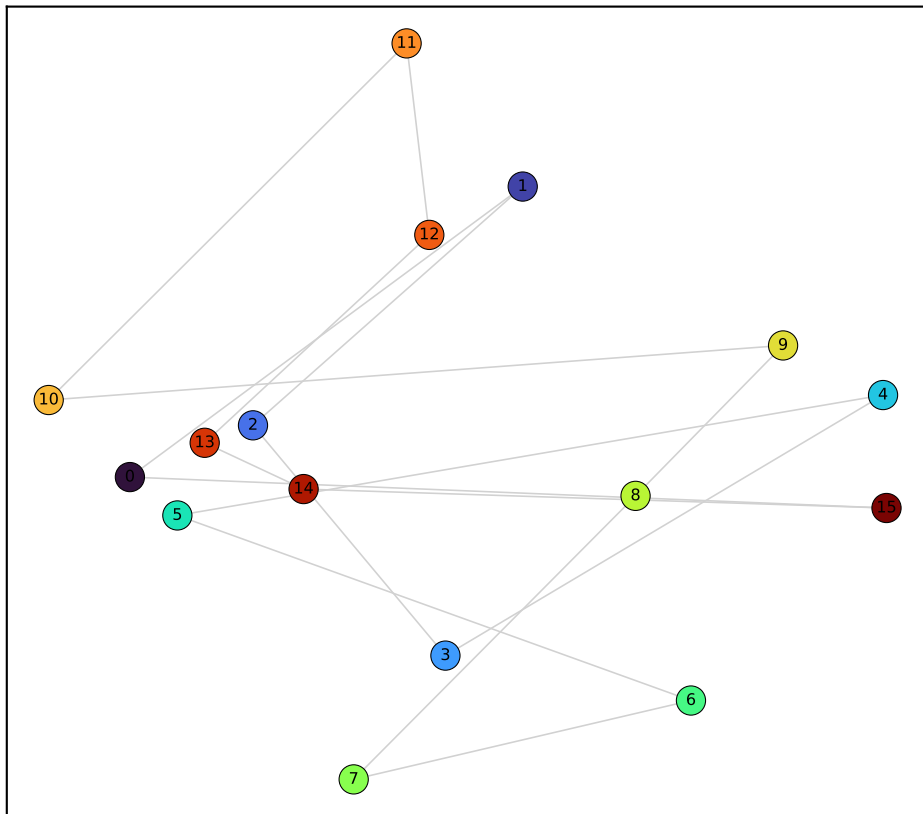


Qwen layer 15/35 (PC var 11·10%)

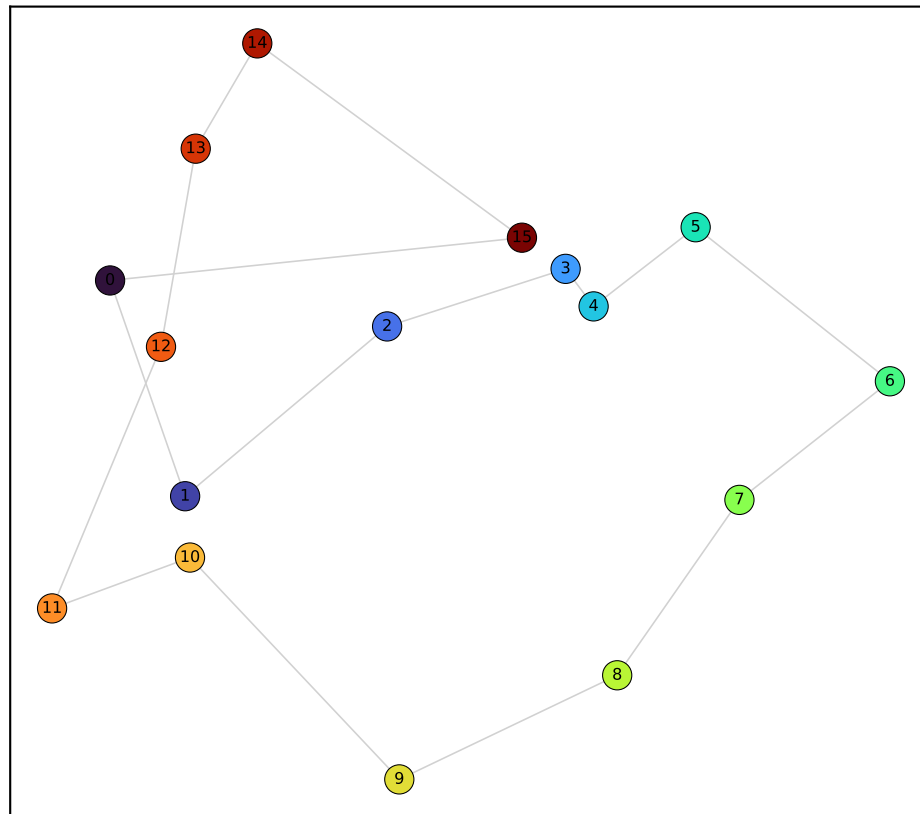


ring — node-mean PCA, depth 44% (edges = graph adjacency)

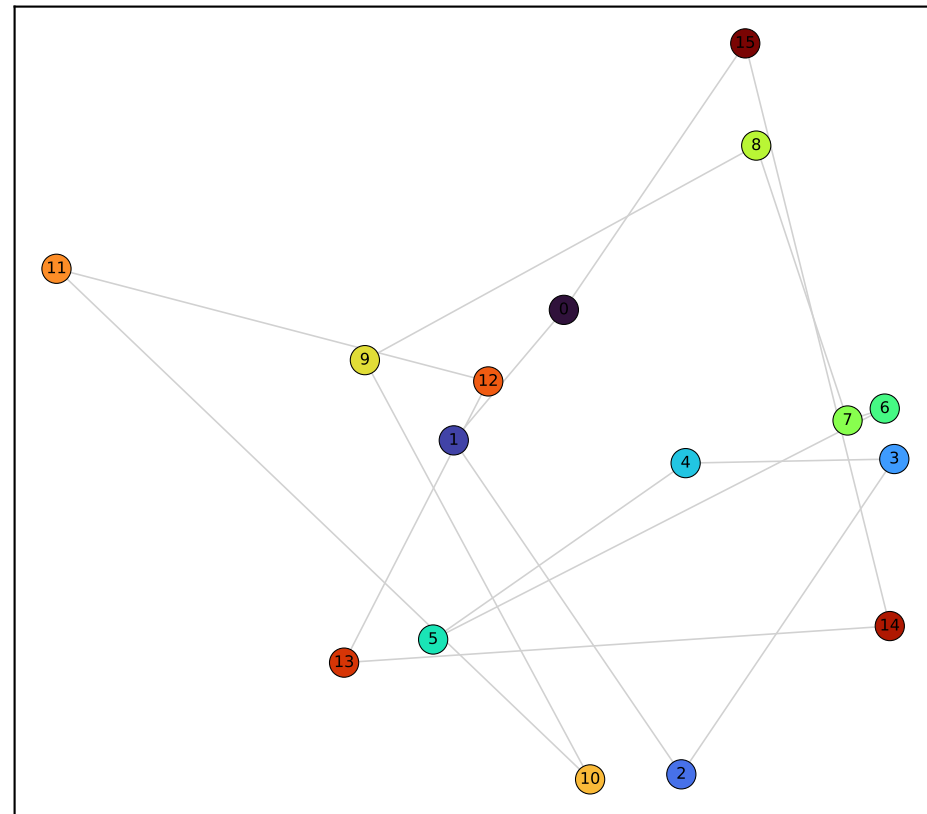
Llama layer 14/31 (PC var 13·11%)



Gemma layer 18/41 (PC var 14·12%)

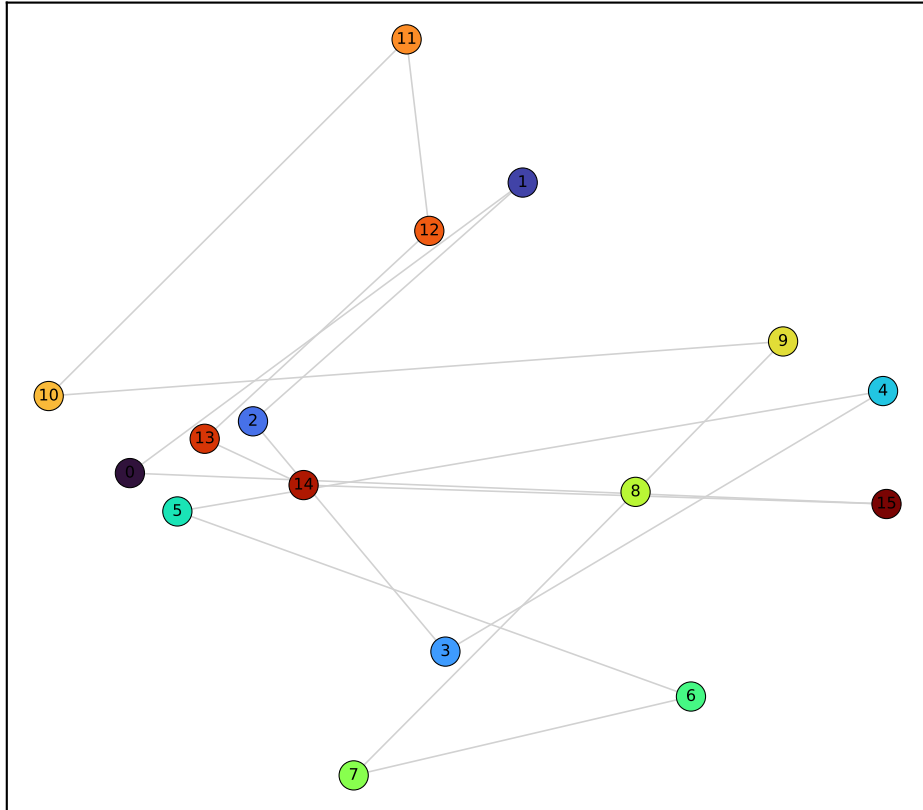


Qwen layer 15/35 (PC var 11·10%)

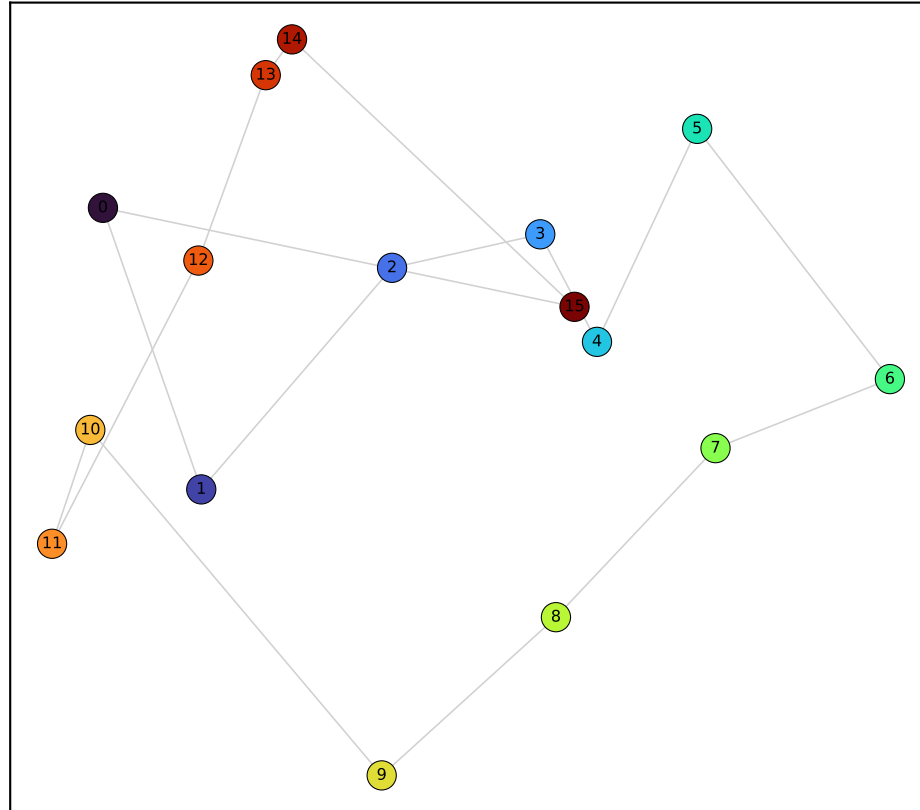


ring — node-mean PCA, depth 46% (edges = graph adjacency)

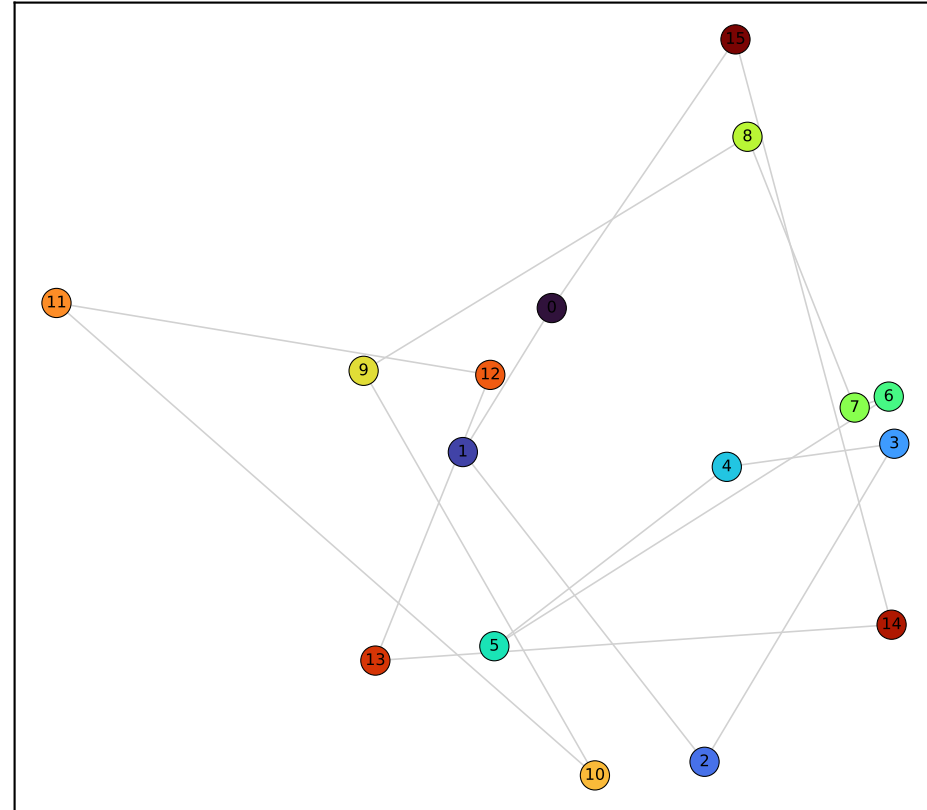
Llama layer 14/31 (PC var 13·11%)



Gemma layer 19/41 (PC var 13·11%)

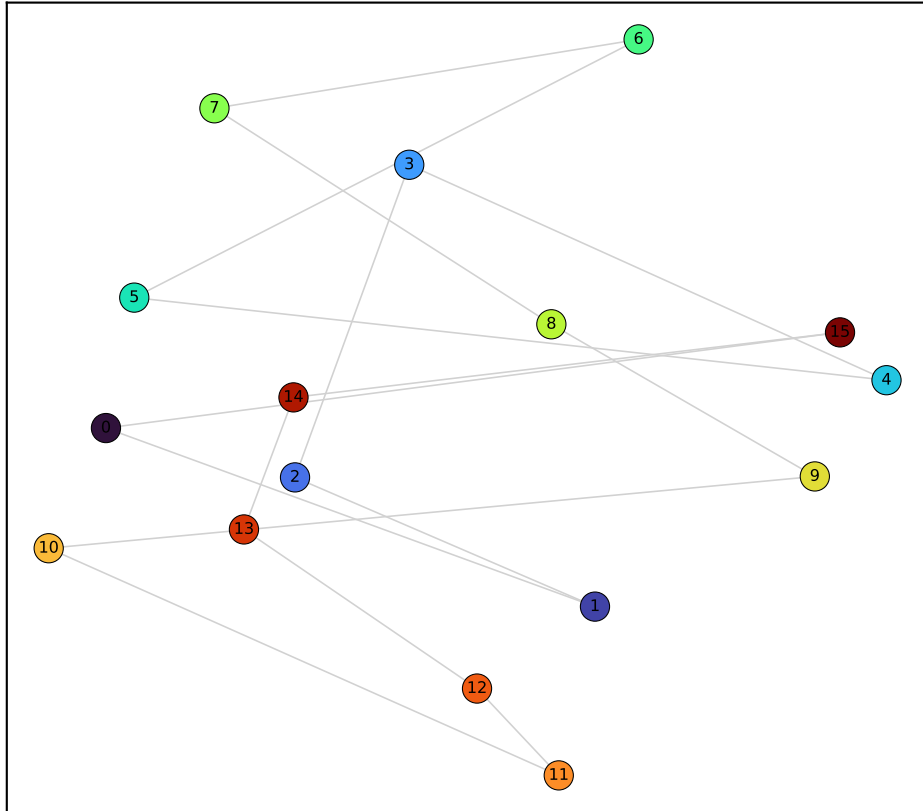


Qwen layer 16/35 (PC var 12·10%)

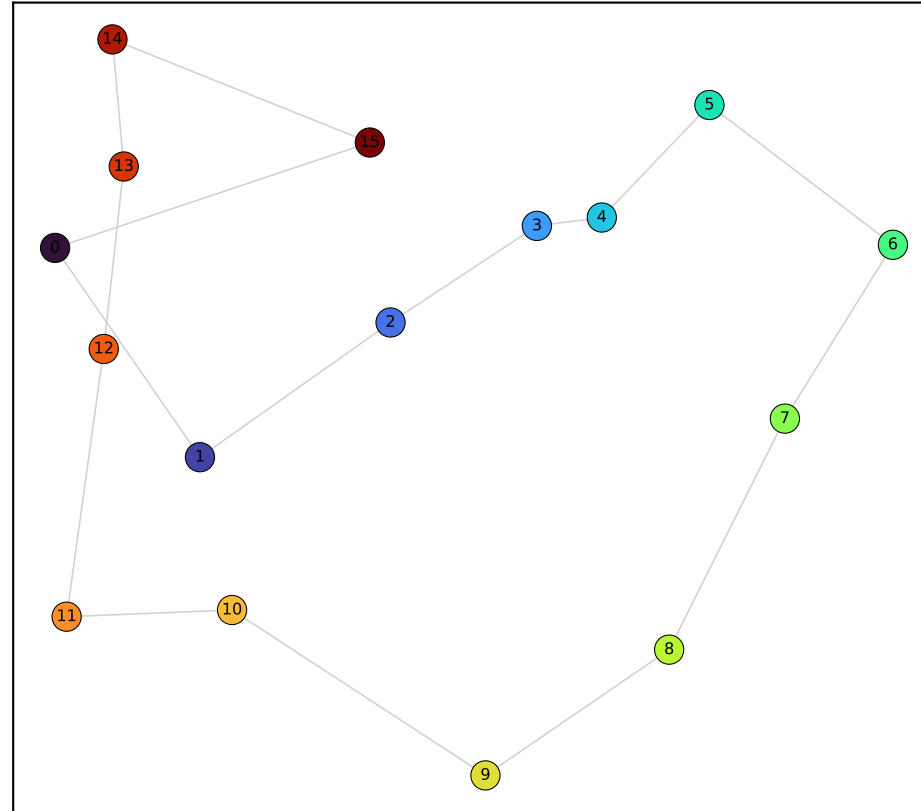


ring — node-mean PCA, depth 49% (edges = graph adjacency)

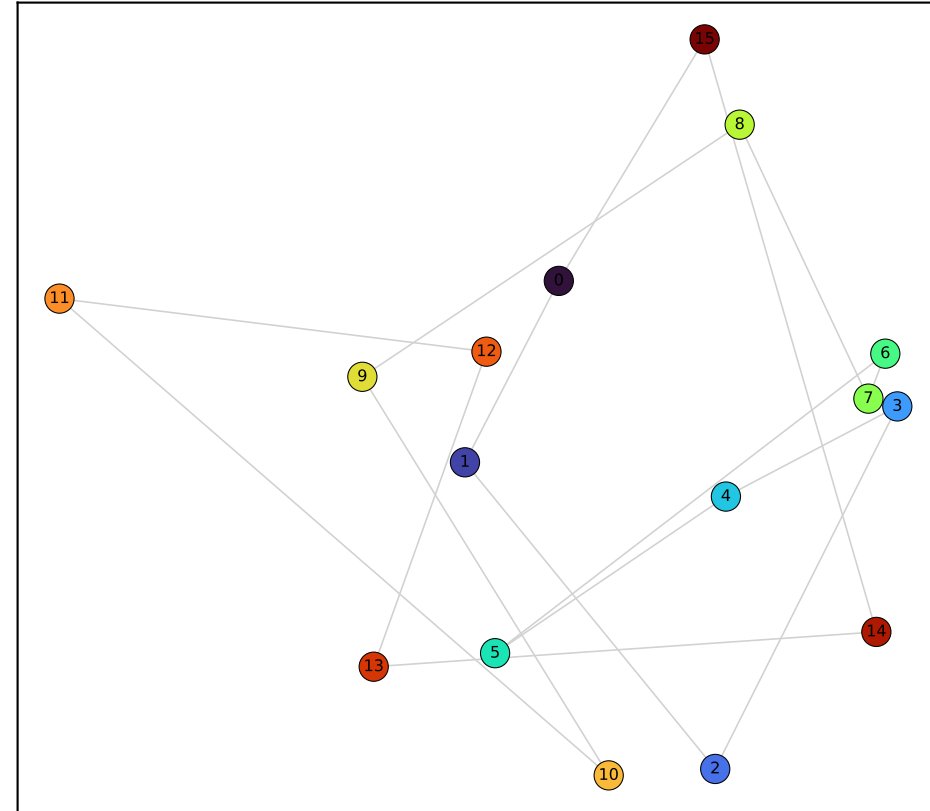
Llama layer 15/31 (PC var 13·11%)



Gemma layer 20/41 (PC var 15·12%)

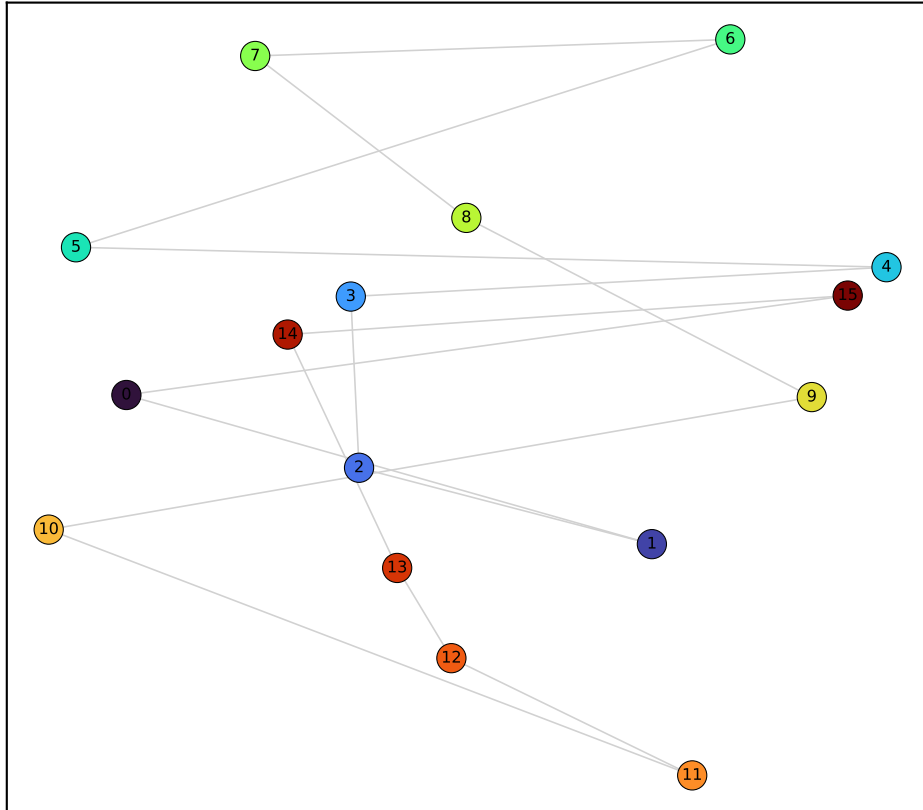


Qwen layer 17/35 (PC var 12·10%)

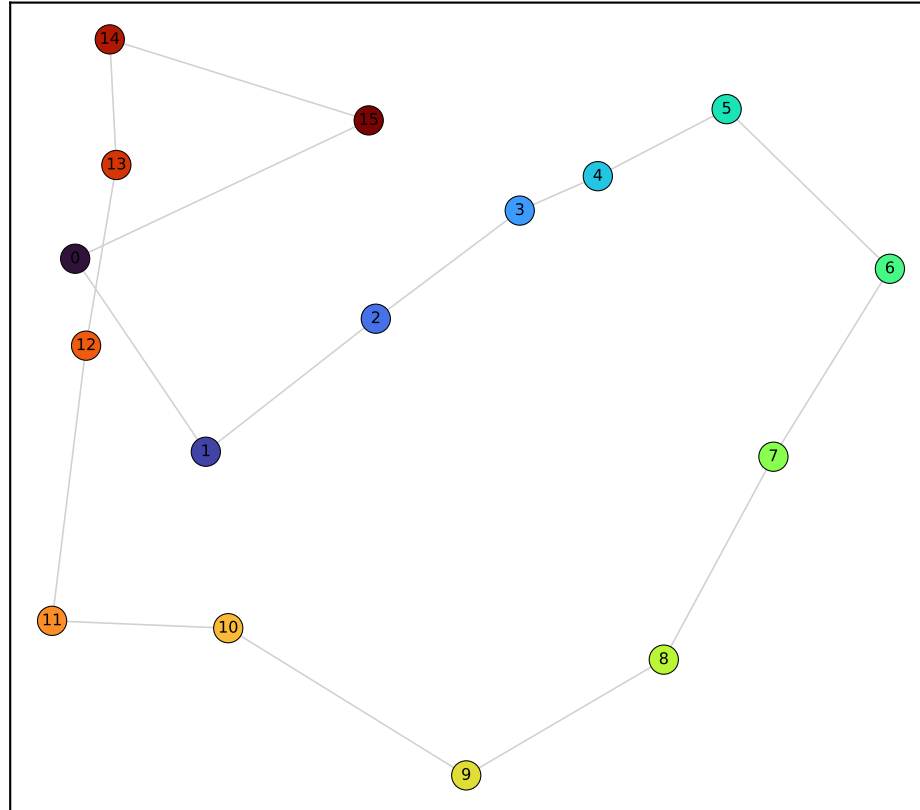


ring — node-mean PCA, depth 51% (edges = graph adjacency)

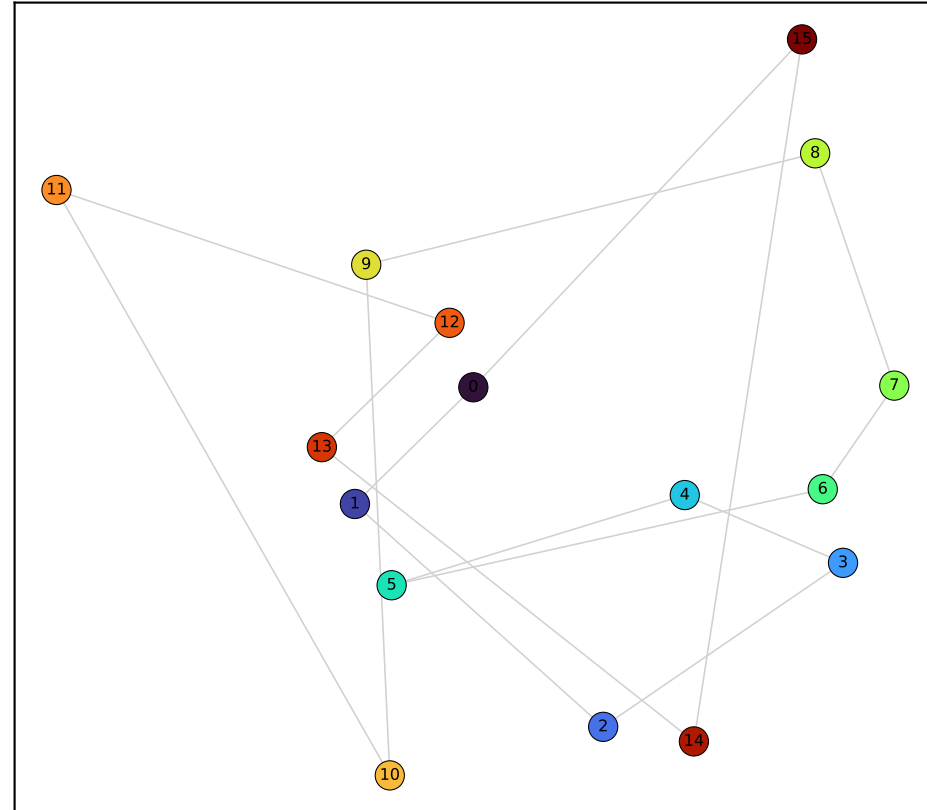
Llama layer 16/31 (PC var 13·11%)



Gemma layer 21/41 (PC var 16·13%)

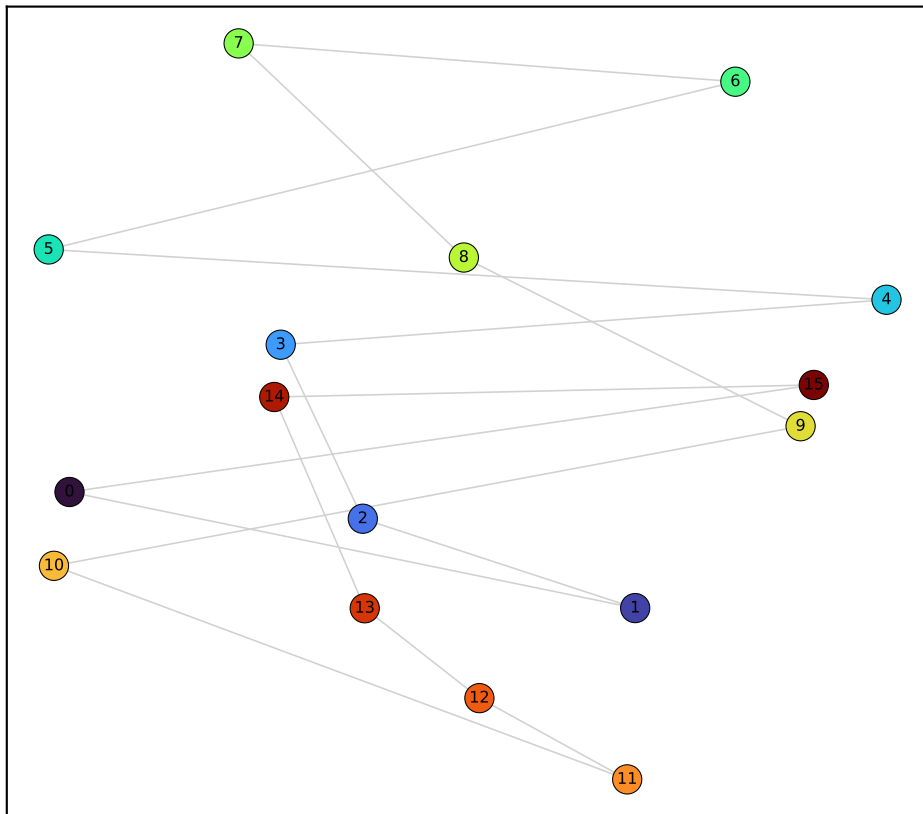


Qwen layer 18/35 (PC var 11·10%)

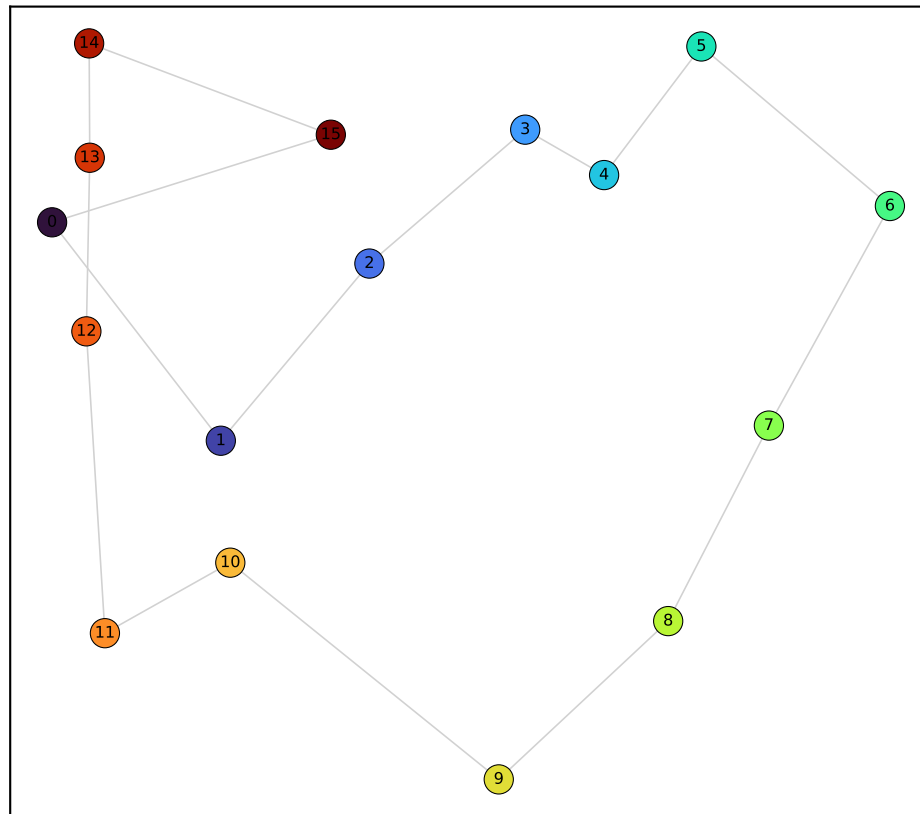


ring — node-mean PCA, depth 54% (edges = graph adjacency)

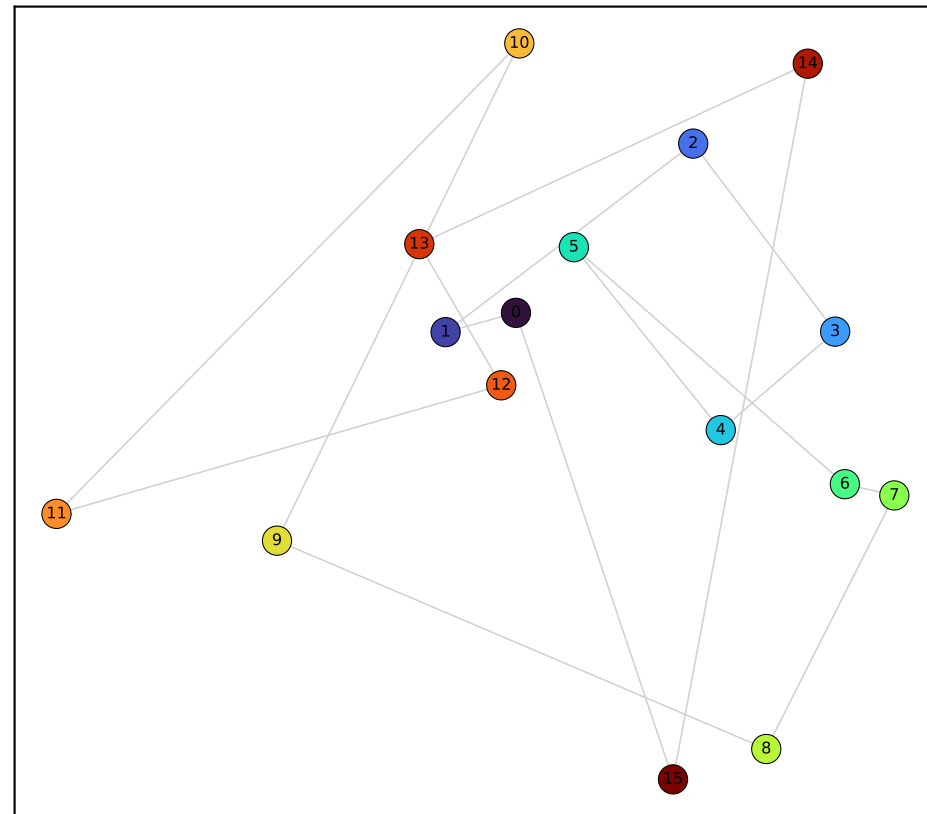
Llama layer 17/31 (PC var 13·12%)



Gemma layer 22/41 (PC var 16·12%)

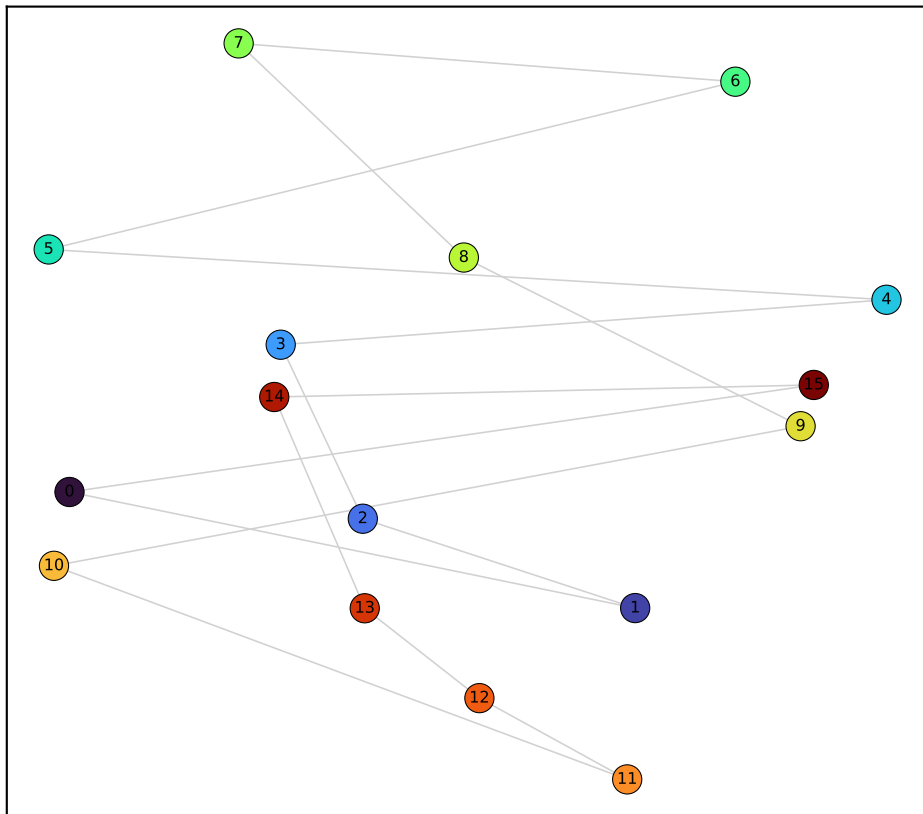


Qwen layer 19/35 (PC var 12·10%)

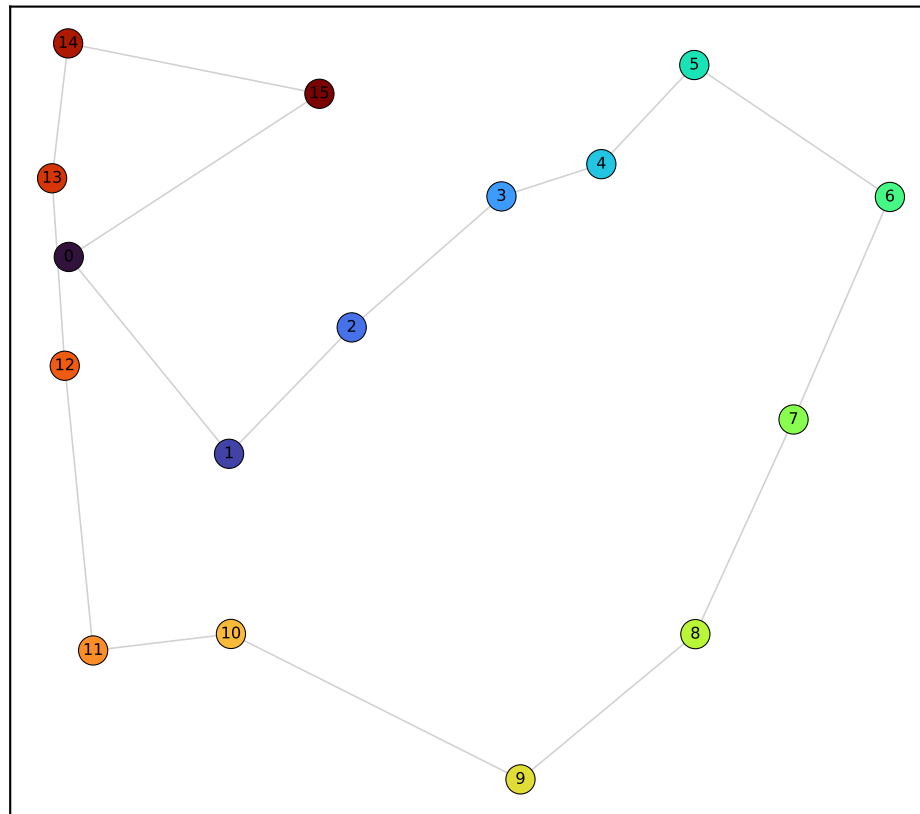


ring — node-mean PCA, depth 56% (edges = graph adjacency)

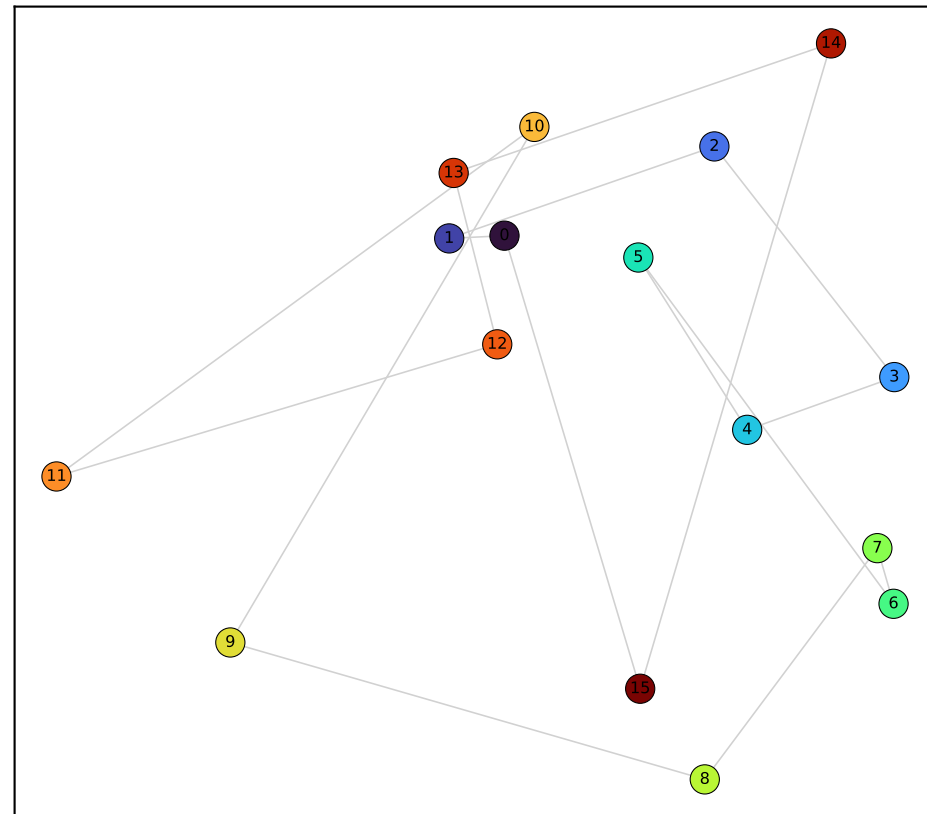
Llama layer 17/31 (PC var 13.12%)



Gemma layer 23/41 (PC var 16.12%)

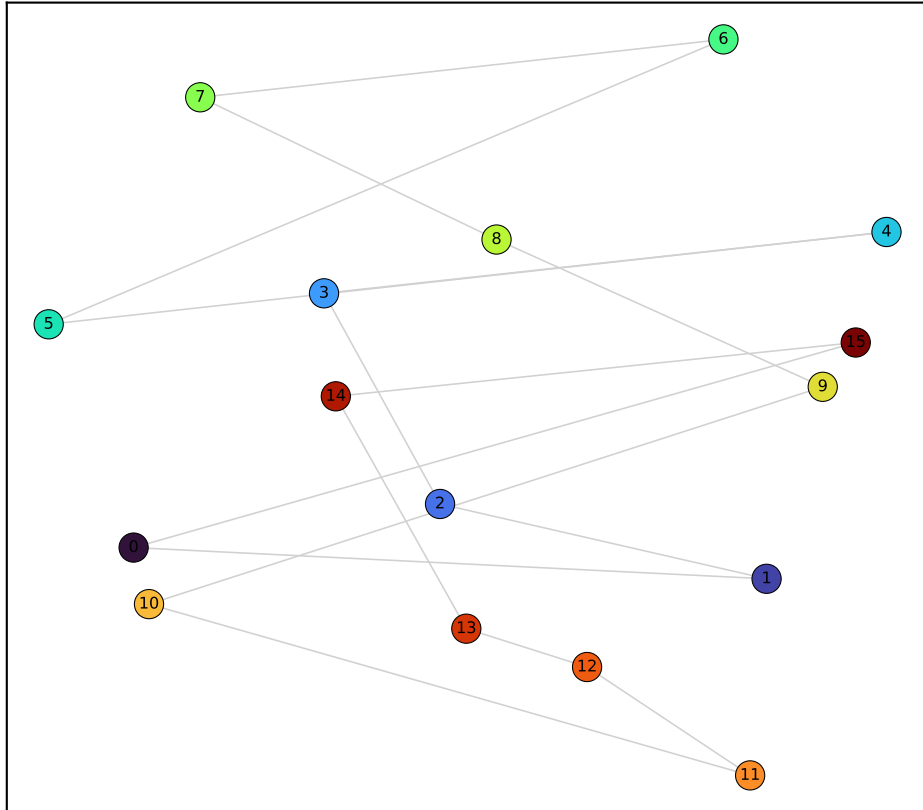


Qwen layer 20/35 (PC var 12.10%)

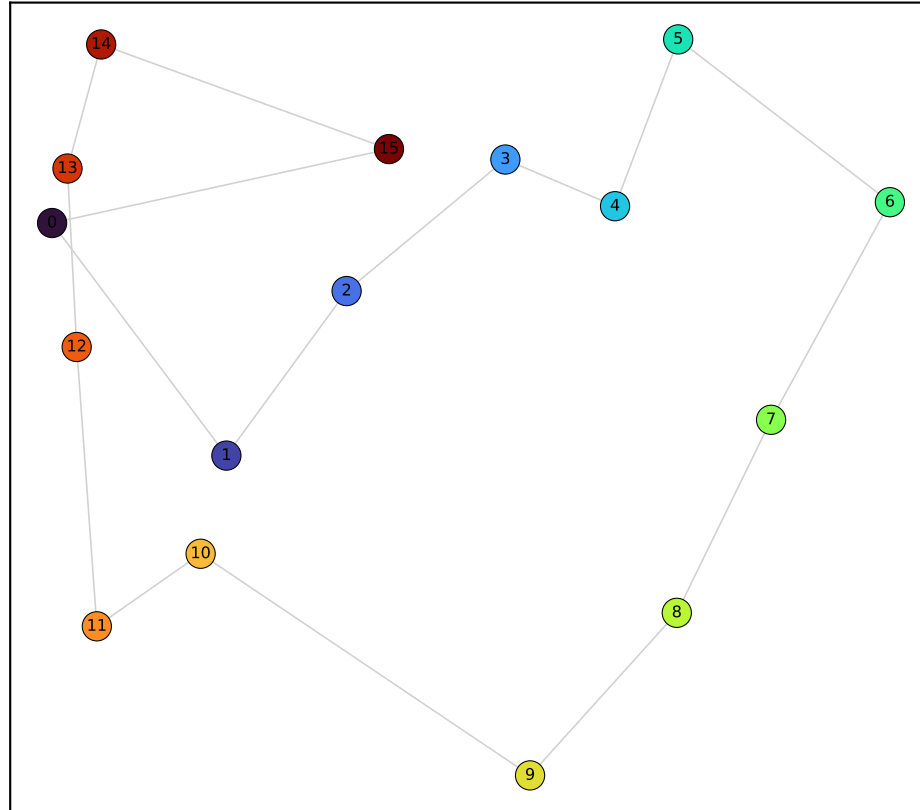


ring — node-mean PCA, depth 59% (edges = graph adjacency)

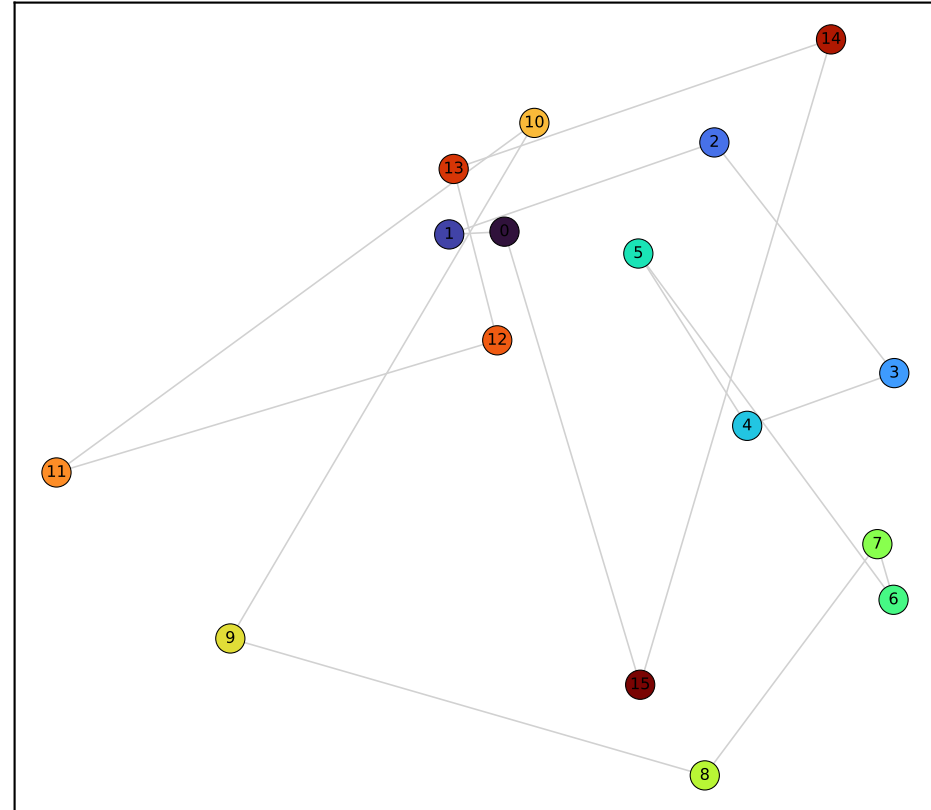
Llama layer 18/31 (PC var 14.11%)



Gemma layer 24/41 (PC var 15.12%)

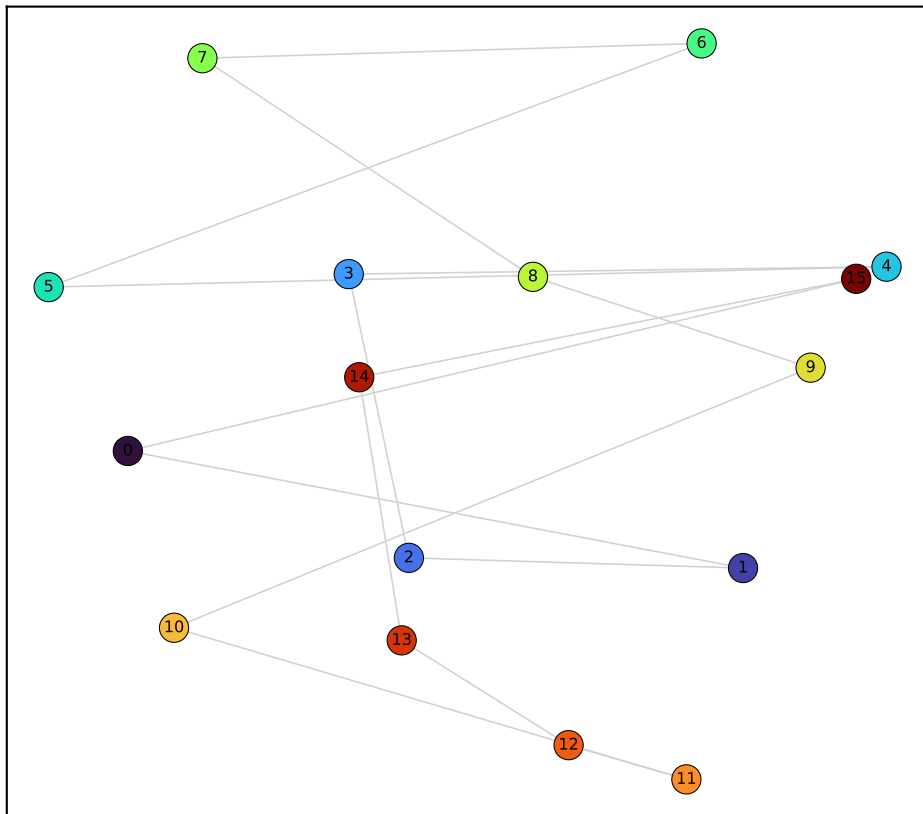


Qwen layer 20/35 (PC var 12.10%)

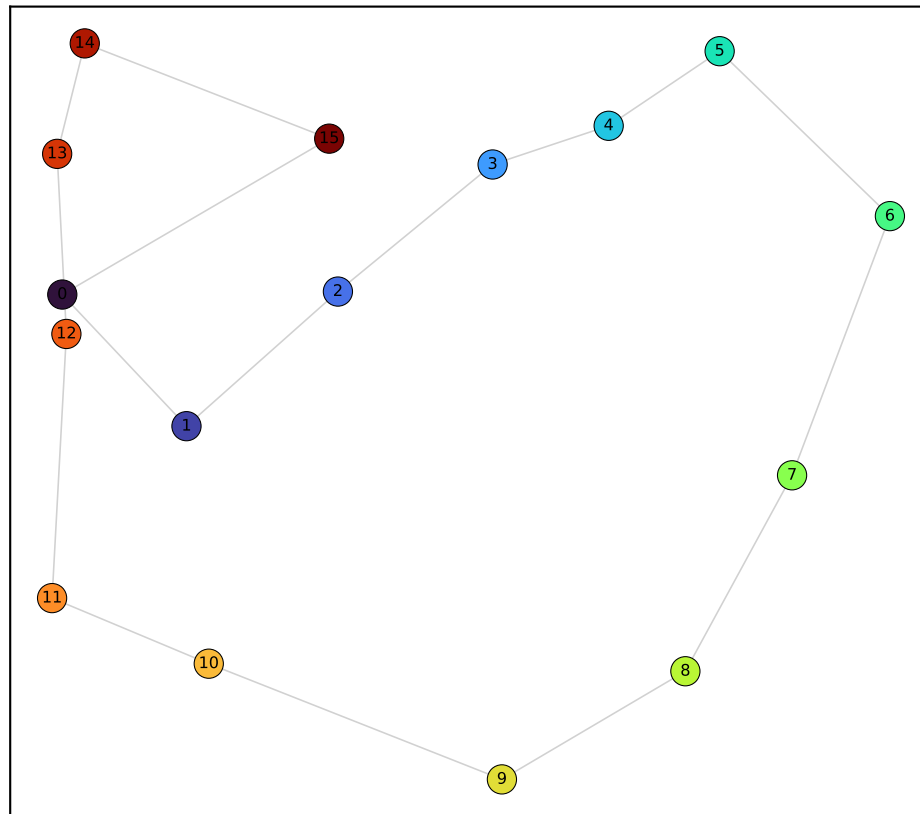


ring — node-mean PCA, depth 61% (edges = graph adjacency)

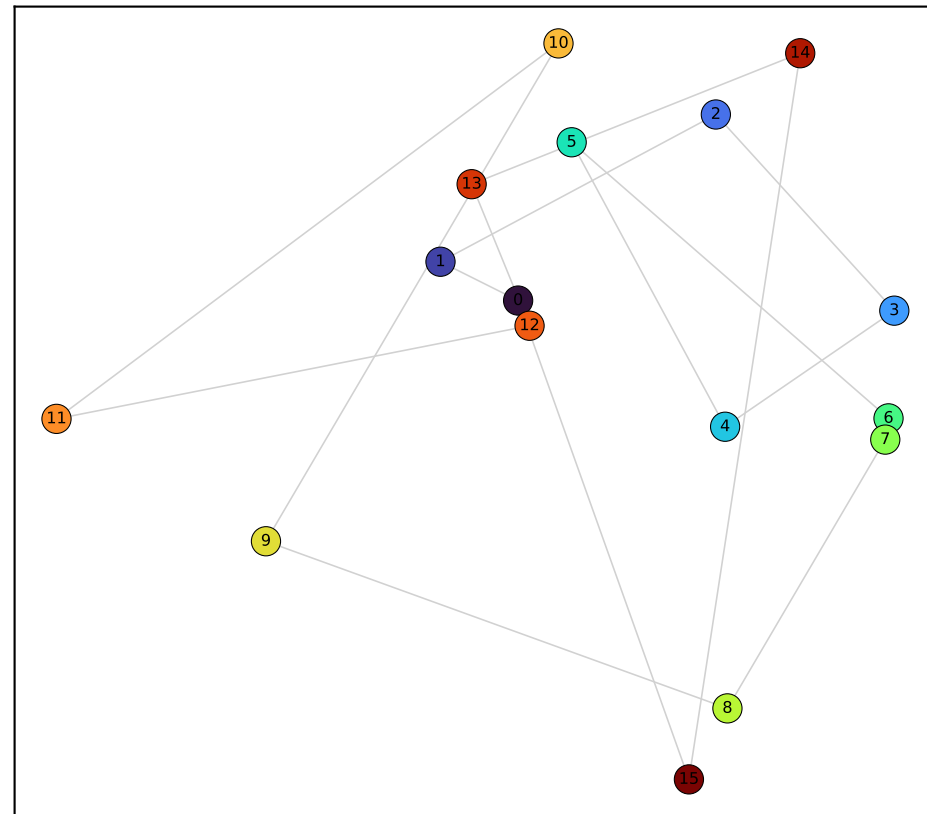
Llama layer 19/31 (PC var 14·11%)



Gemma layer 25/41 (PC var 16·13%)

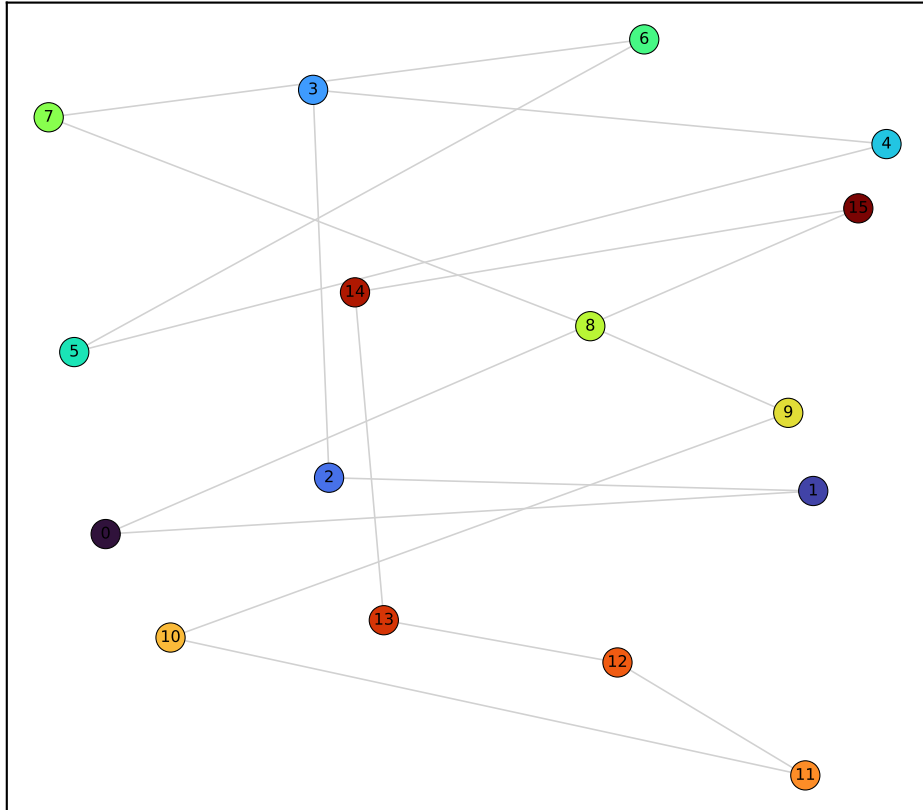


Qwen layer 21/35 (PC var 12·10%)

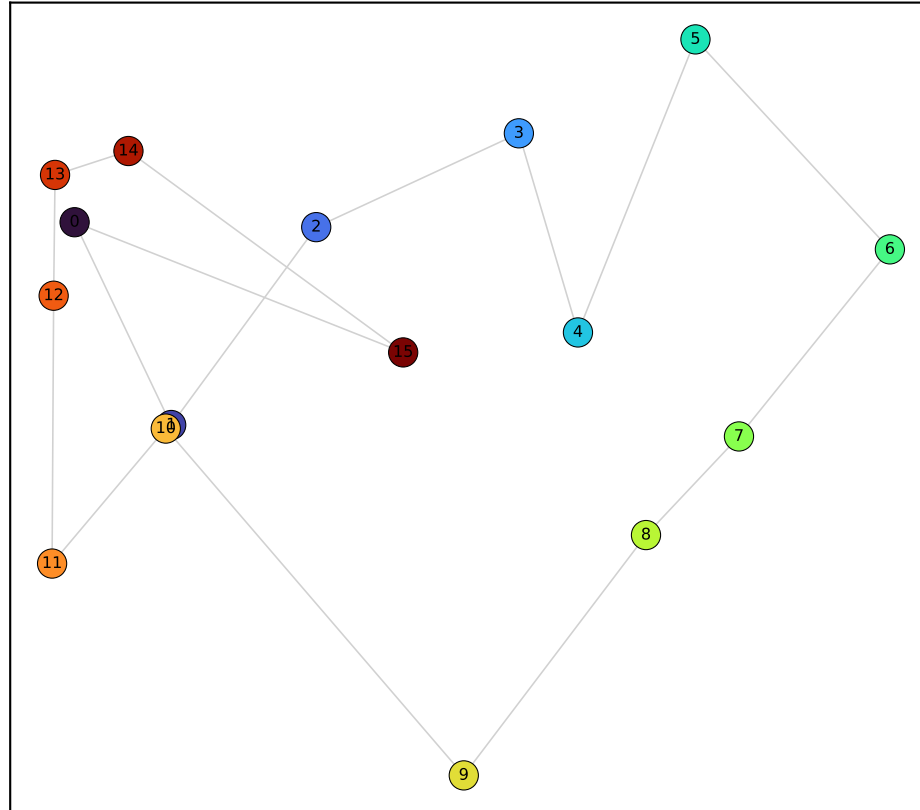


ring — node-mean PCA, depth 63% (edges = graph adjacency)

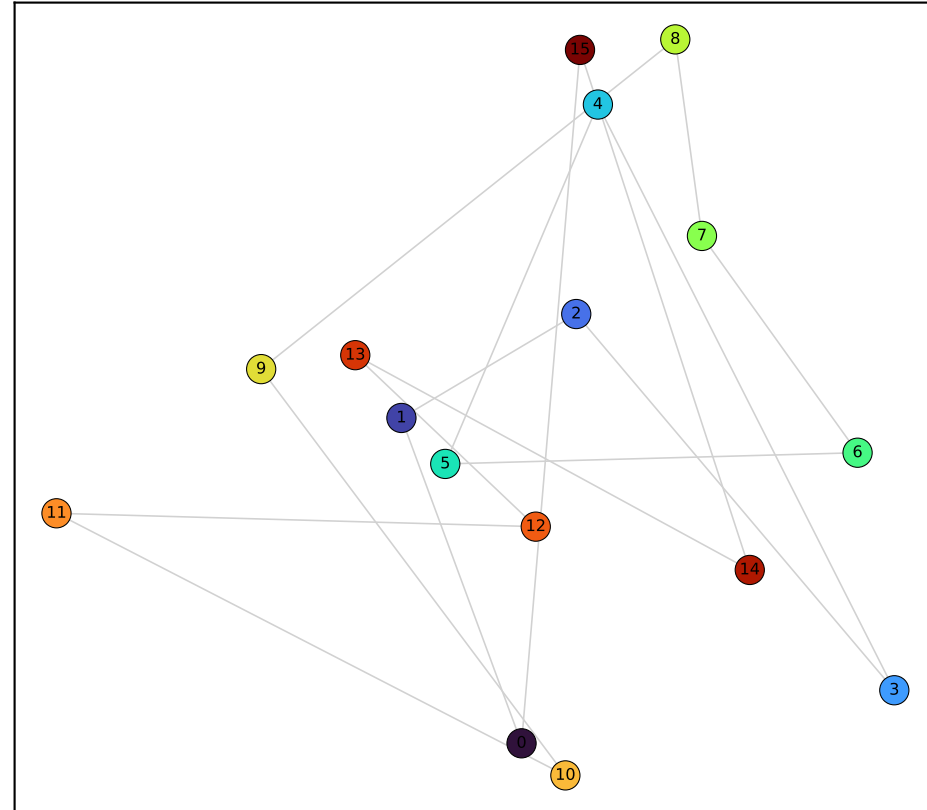
Llama layer 20/31 (PC var 13·10%)



Gemma layer 26/41 (PC var 15·11%)

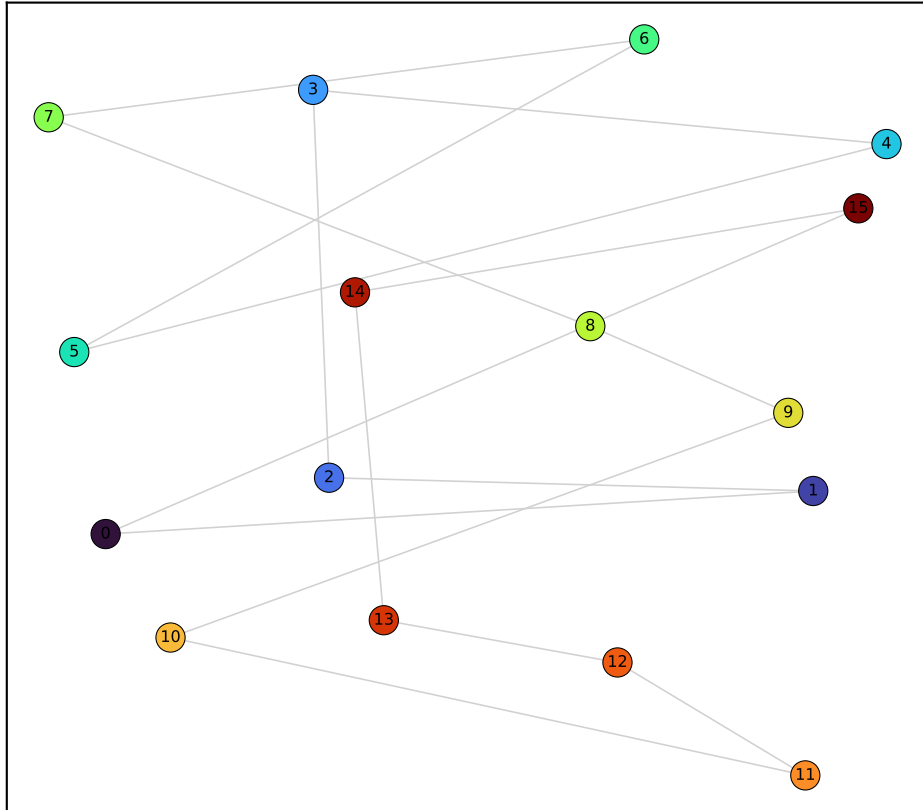


Qwen layer 22/35 (PC var 11·9%)

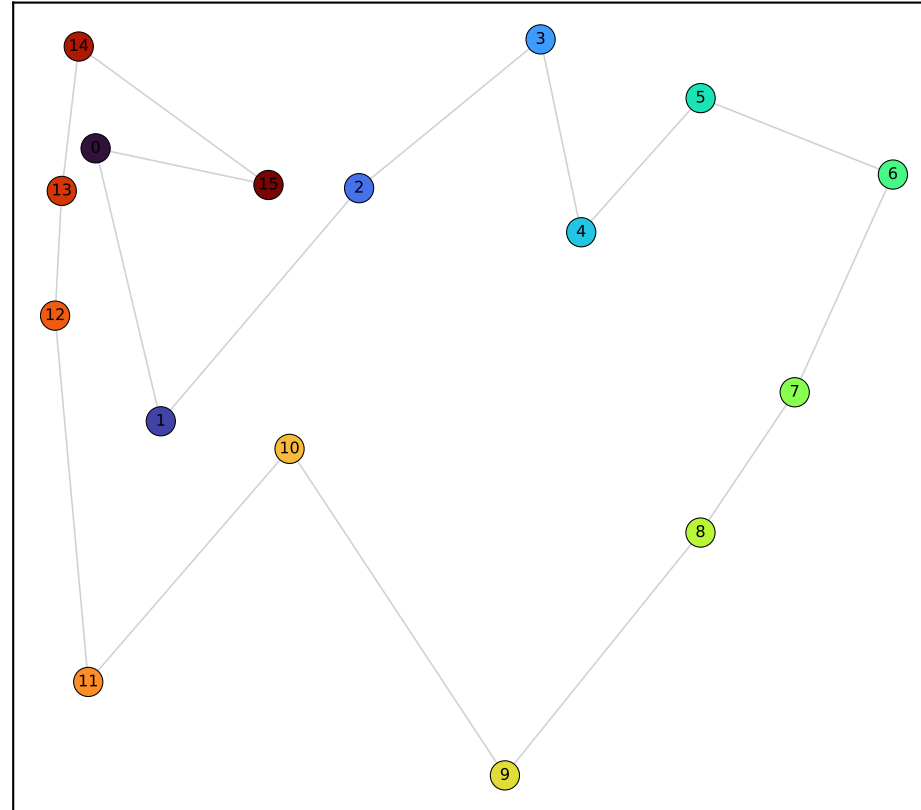


ring — node-mean PCA, depth 66% (edges = graph adjacency)

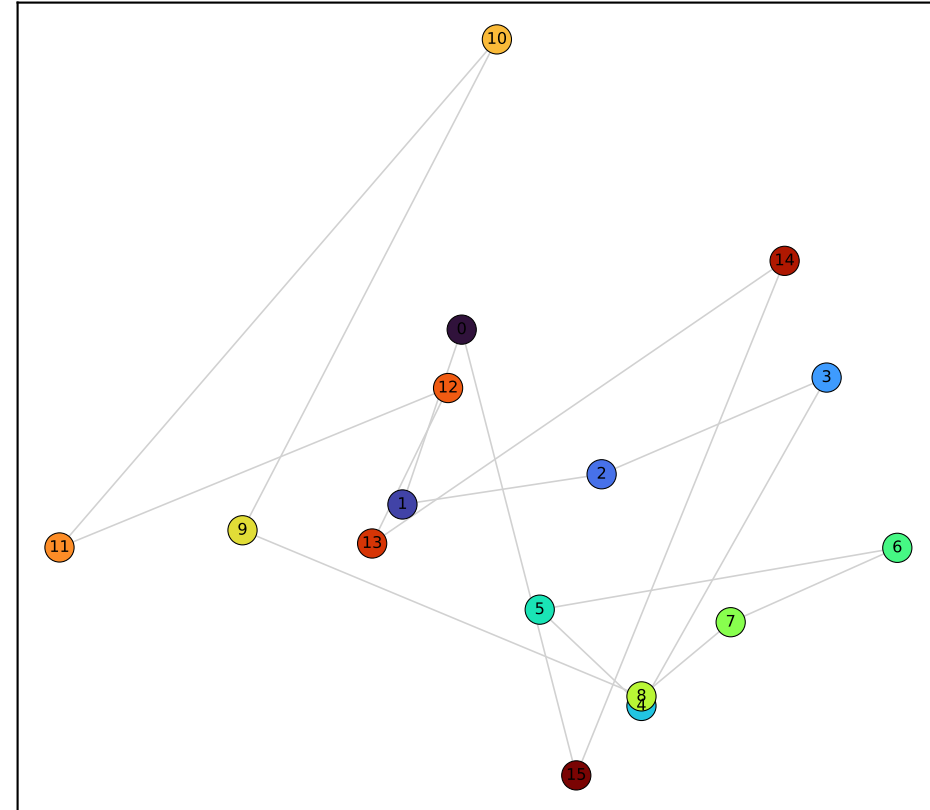
Llama layer 20/31 (PC var 13·10%)



Gemma layer 27/41 (PC var 15·11%)

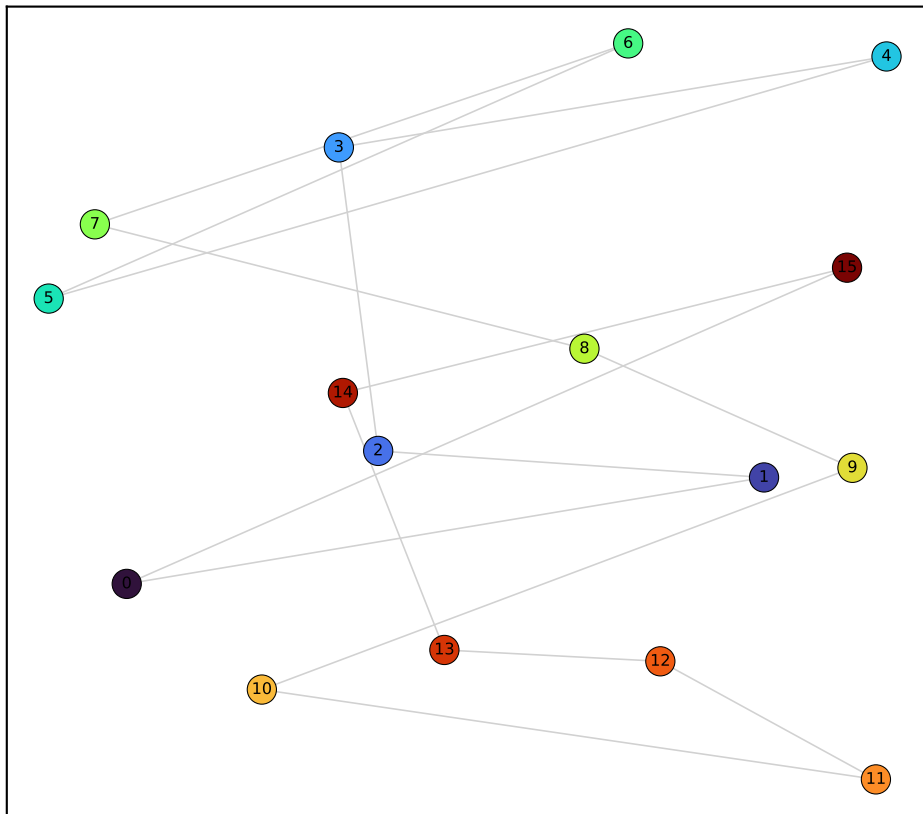


Qwen layer 23/35 (PC var 11·9%)

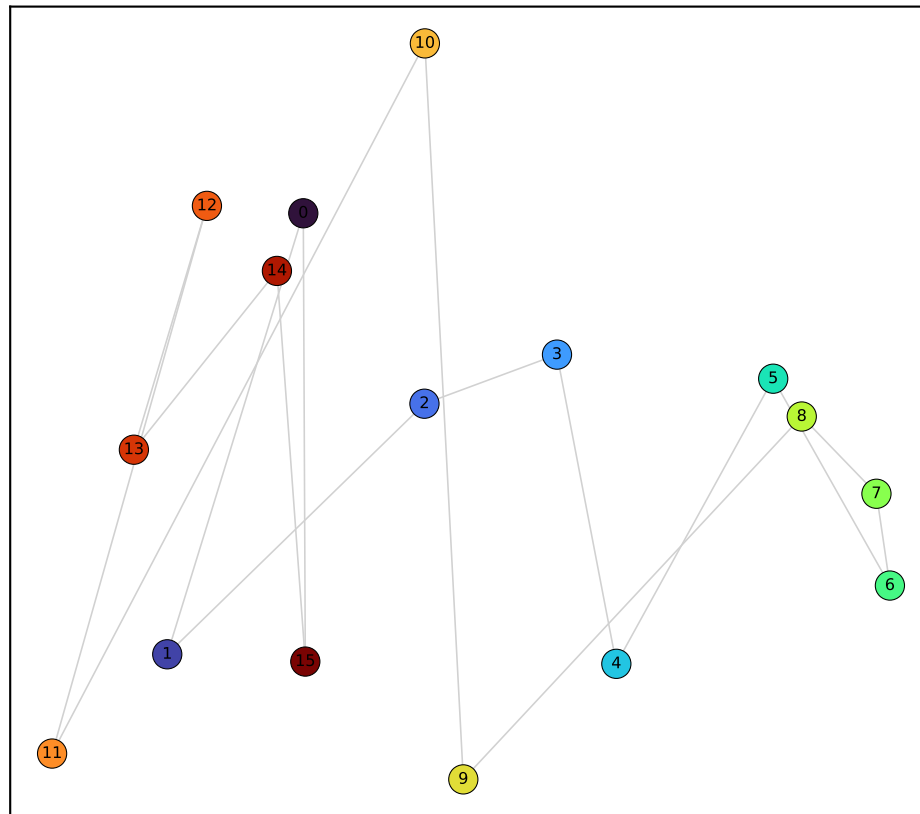


ring — node-mean PCA, depth 68% (edges = graph adjacency)

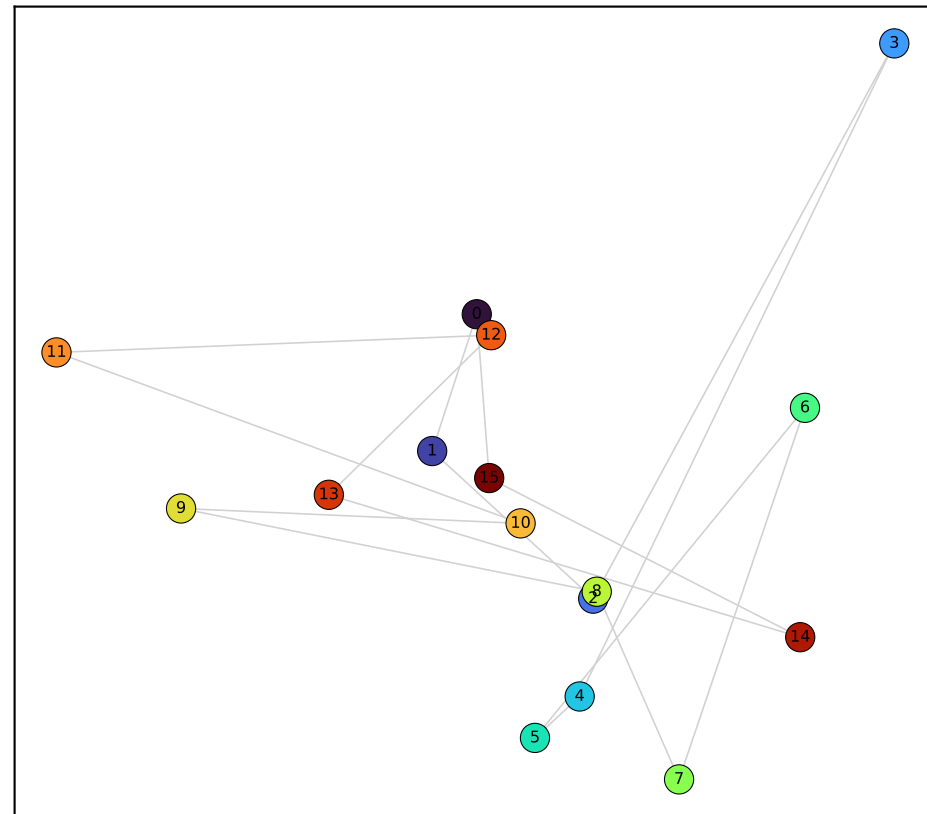
Llama layer 21/31 (PC var 12·11%)



Gemma layer 28/41 (PC var 13·11%)

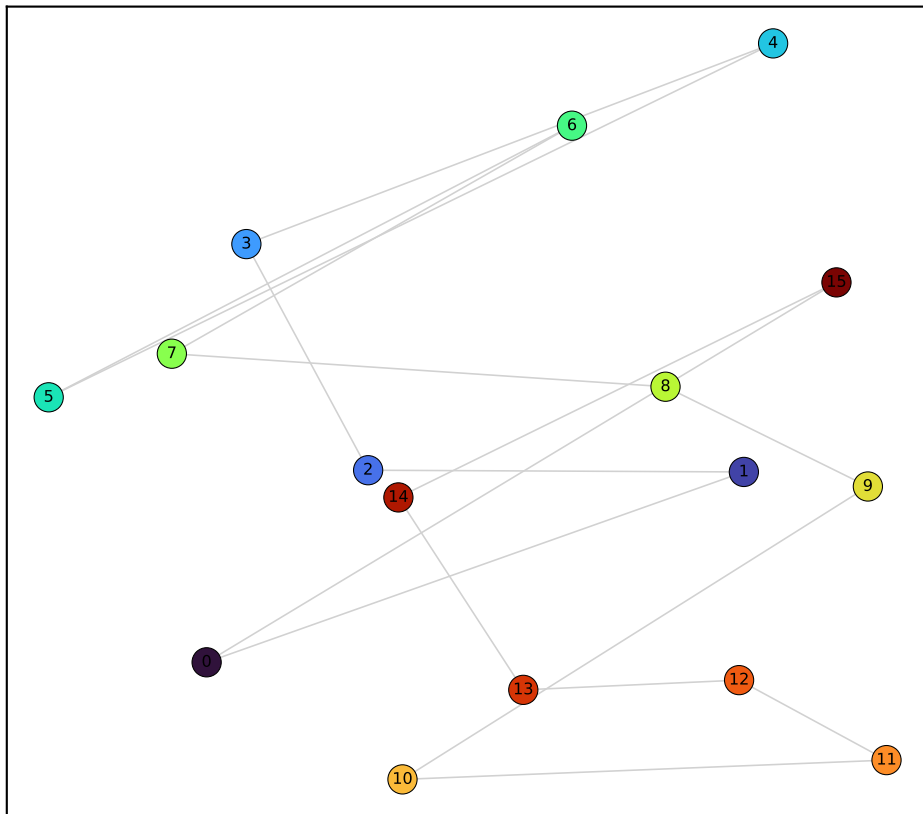


Qwen layer 24/35 (PC var 12·9%)

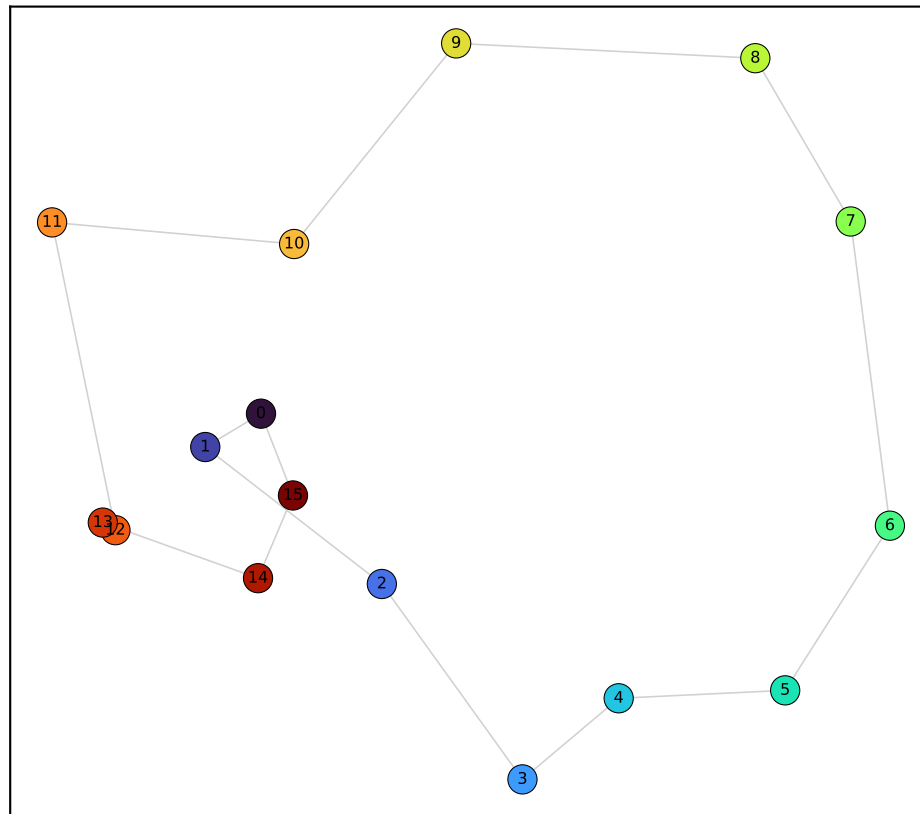


ring — node-mean PCA, depth 71% (edges = graph adjacency)

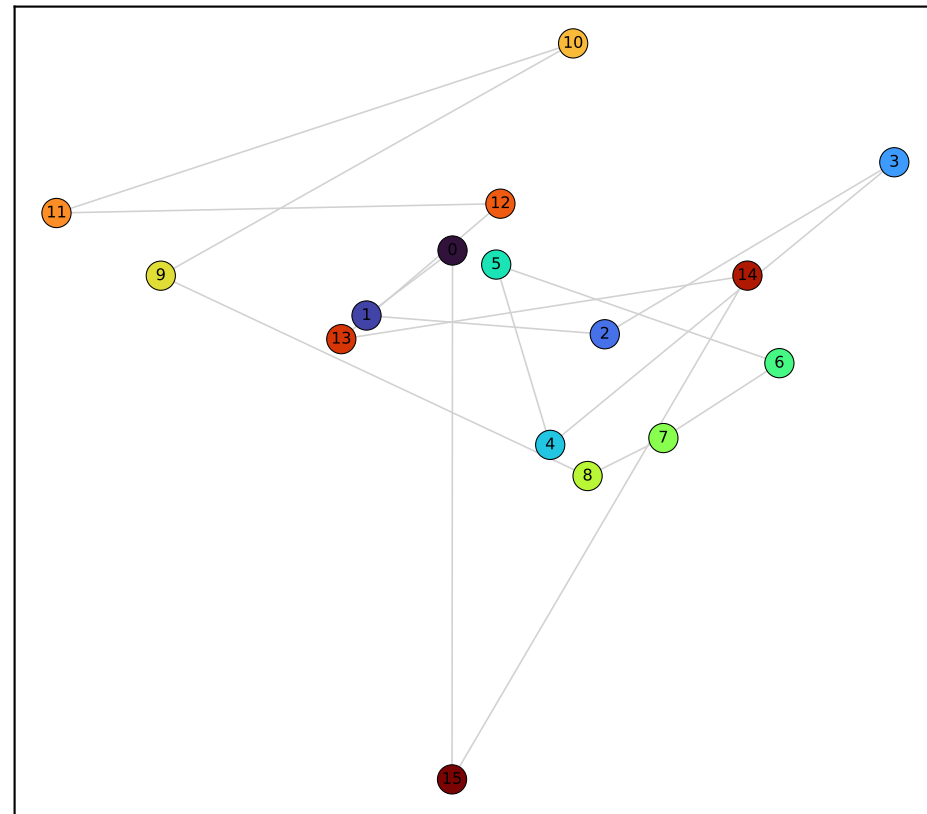
Llama layer 22/31 (PC var 12·11%)



Gemma layer 29/41 (PC var 15·11%)

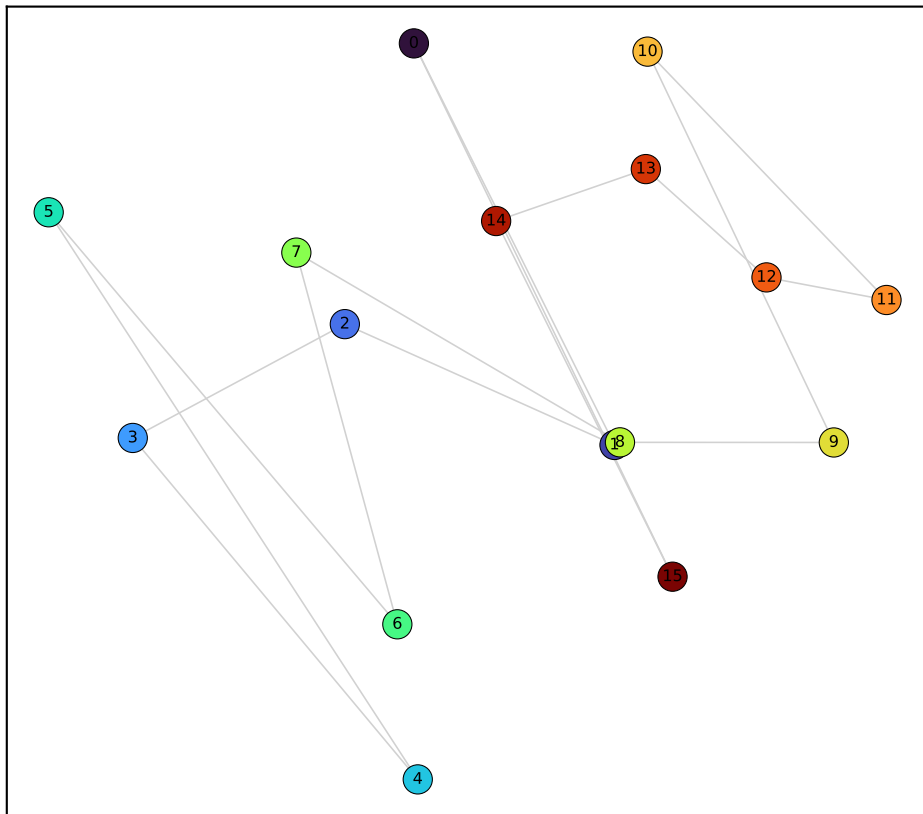


Qwen layer 25/35 (PC var 12·9%)

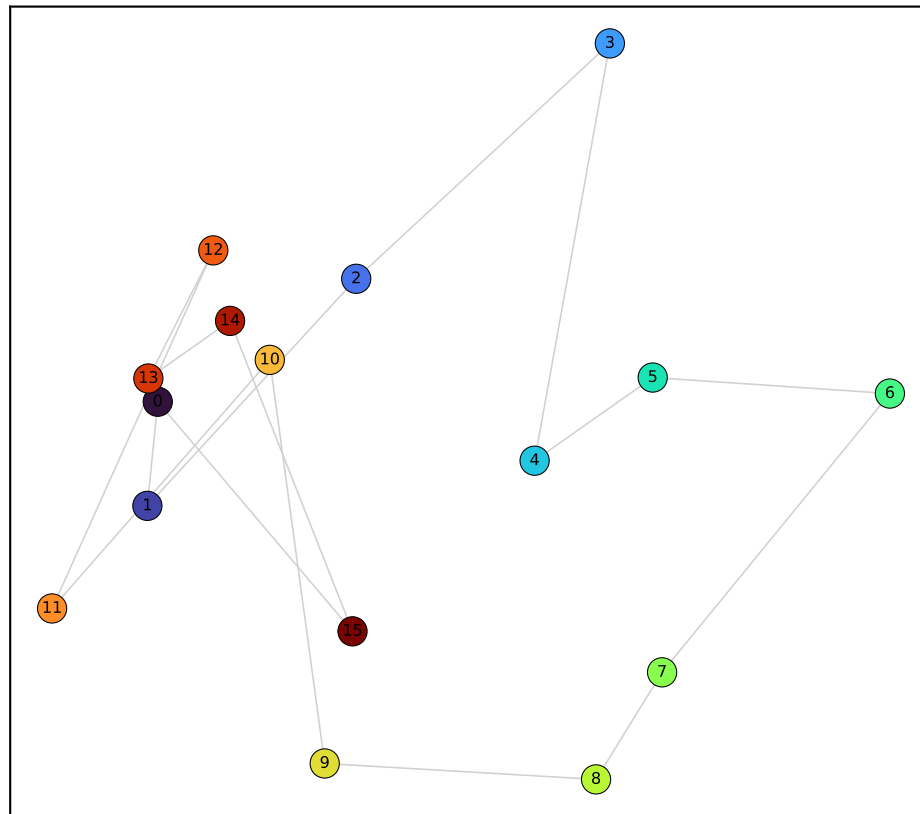


ring — node-mean PCA, depth 73% (edges = graph adjacency)

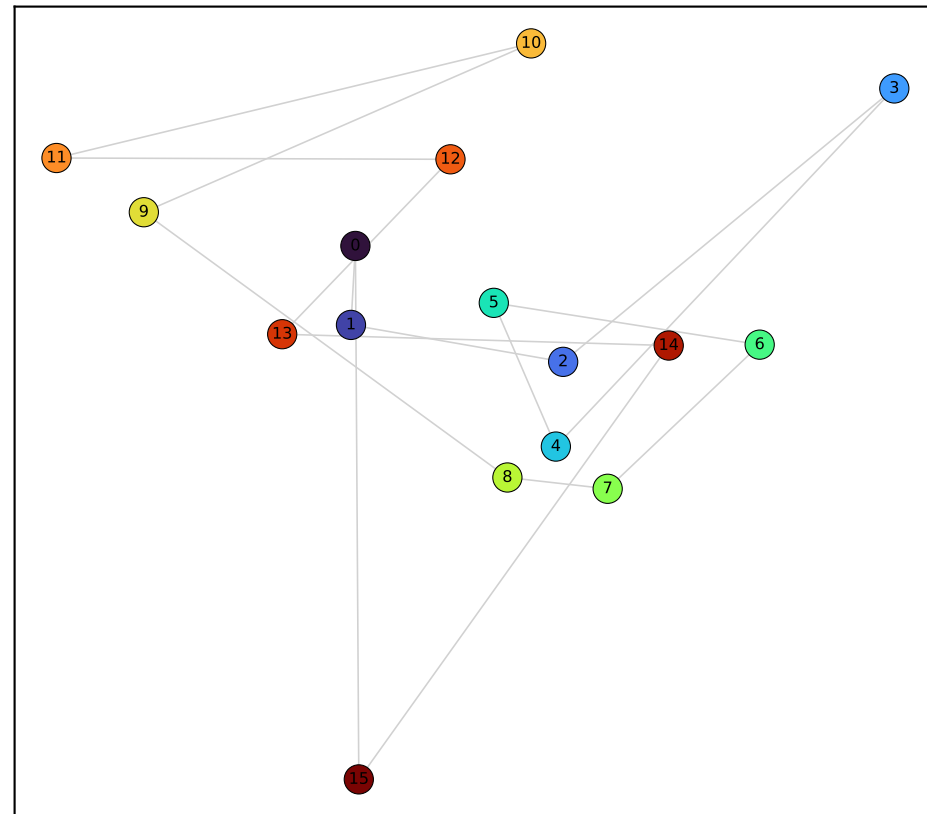
Llama layer 23/31 (PC var 13.11%)



Gemma layer 30/41 (PC var 11.9%)

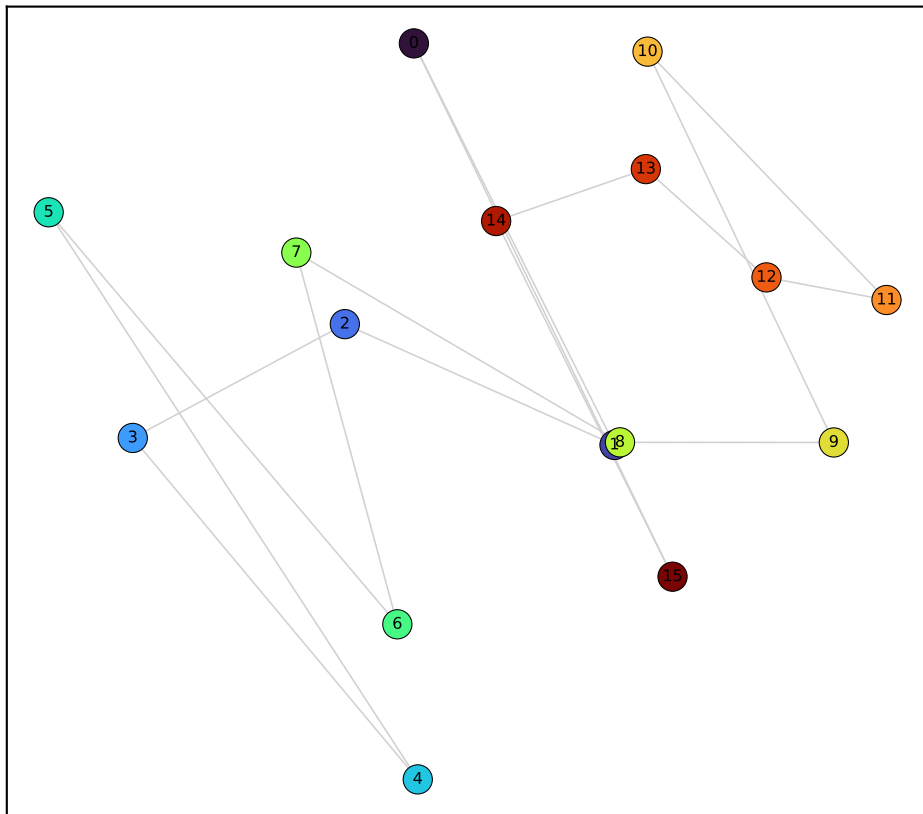


Qwen layer 26/35 (PC var 12.9%)

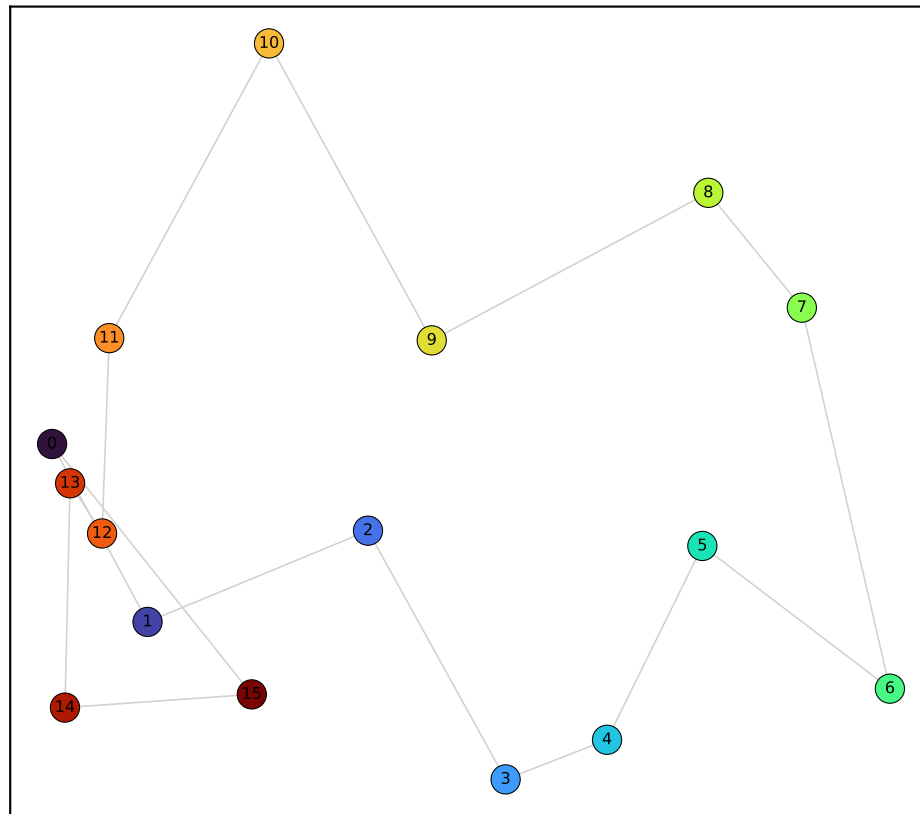


ring — node-mean PCA, depth 76% (edges = graph adjacency)

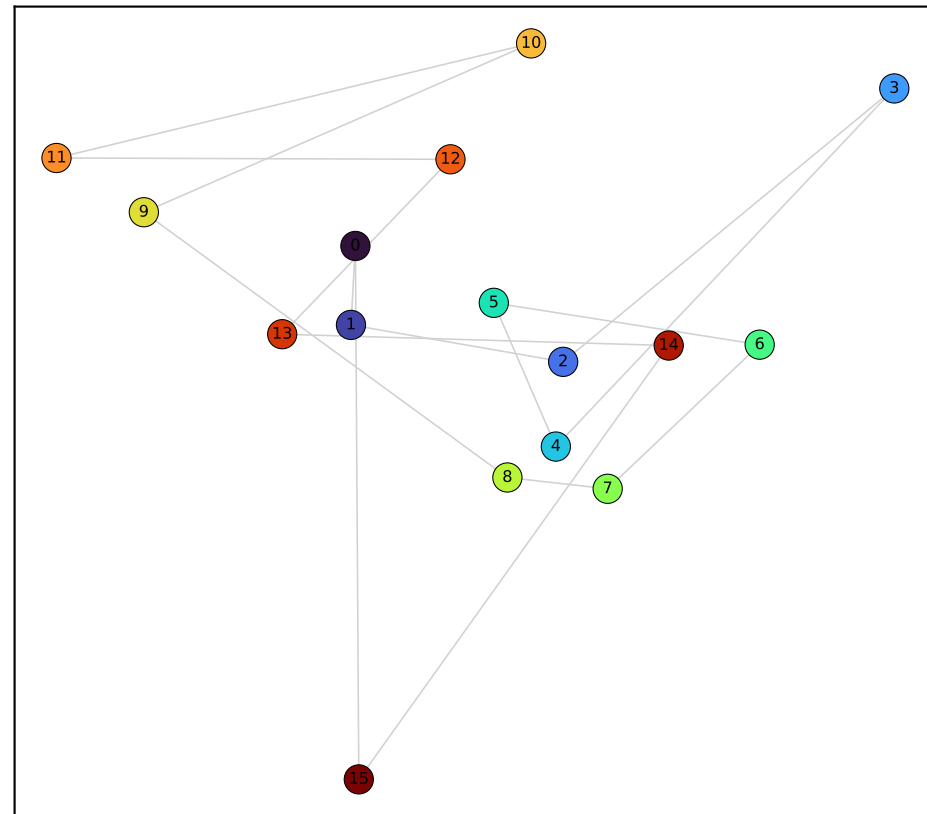
Llama layer 23/31 (PC var 13.11%)



Gemma layer 31/41 (PC var 12.10%)

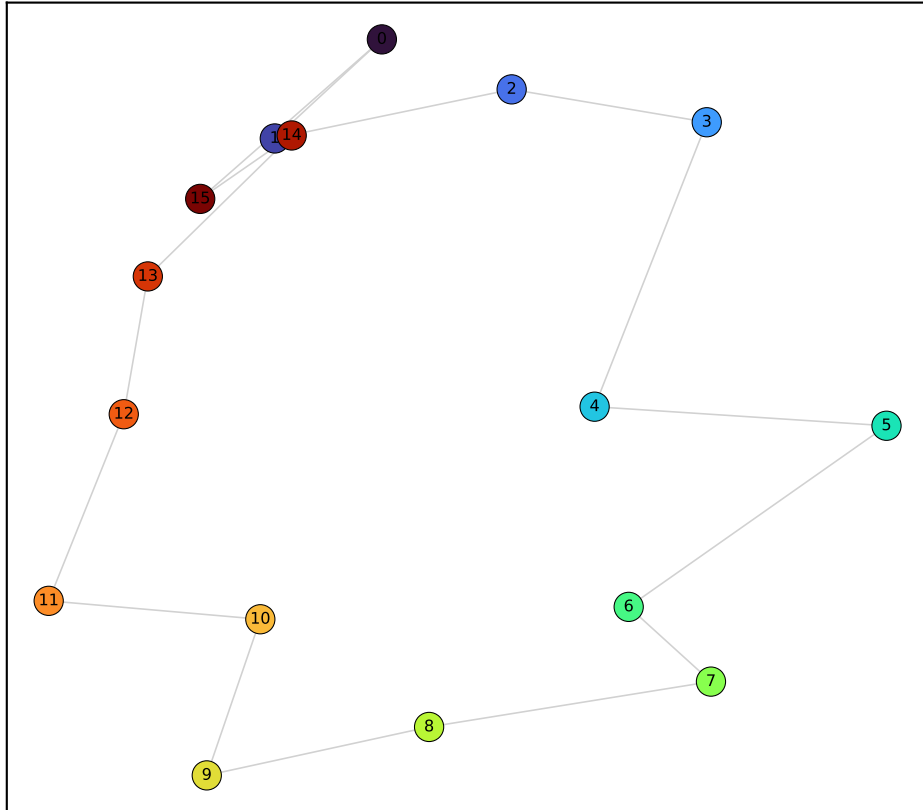


Qwen layer 26/35 (PC var 12.9%)

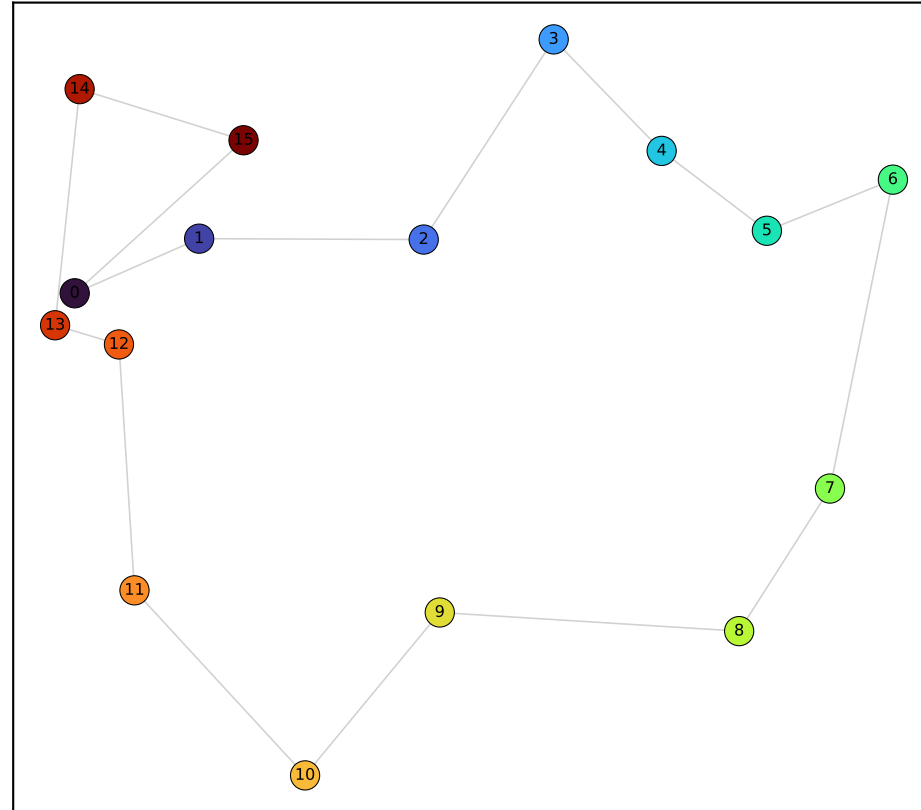


ring — node-mean PCA, depth 78% (edges = graph adjacency)

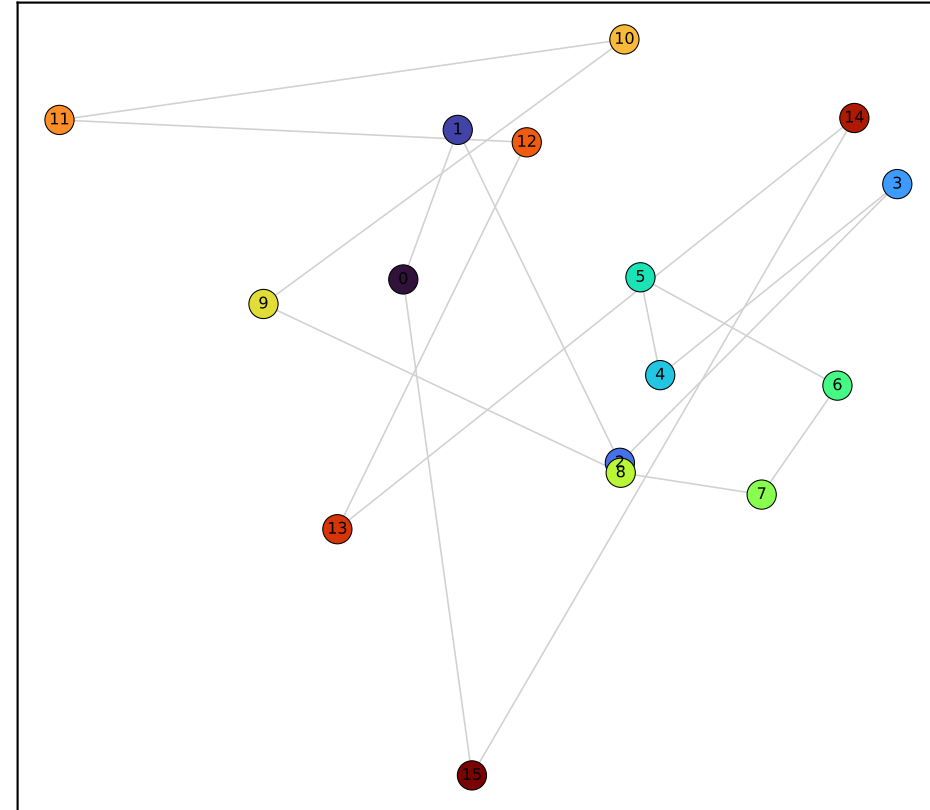
Llama layer 24/31 (PC var 14·12%)



Gemma layer 32/41 (PC var 12·10%)

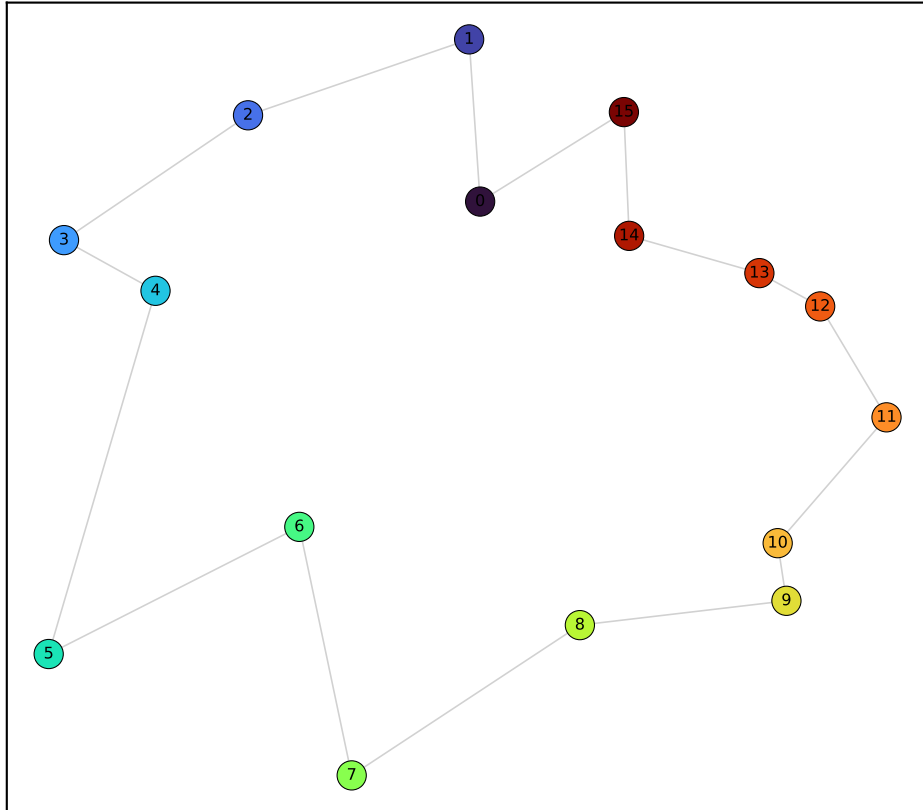


Qwen layer 27/35 (PC var 12·9%)

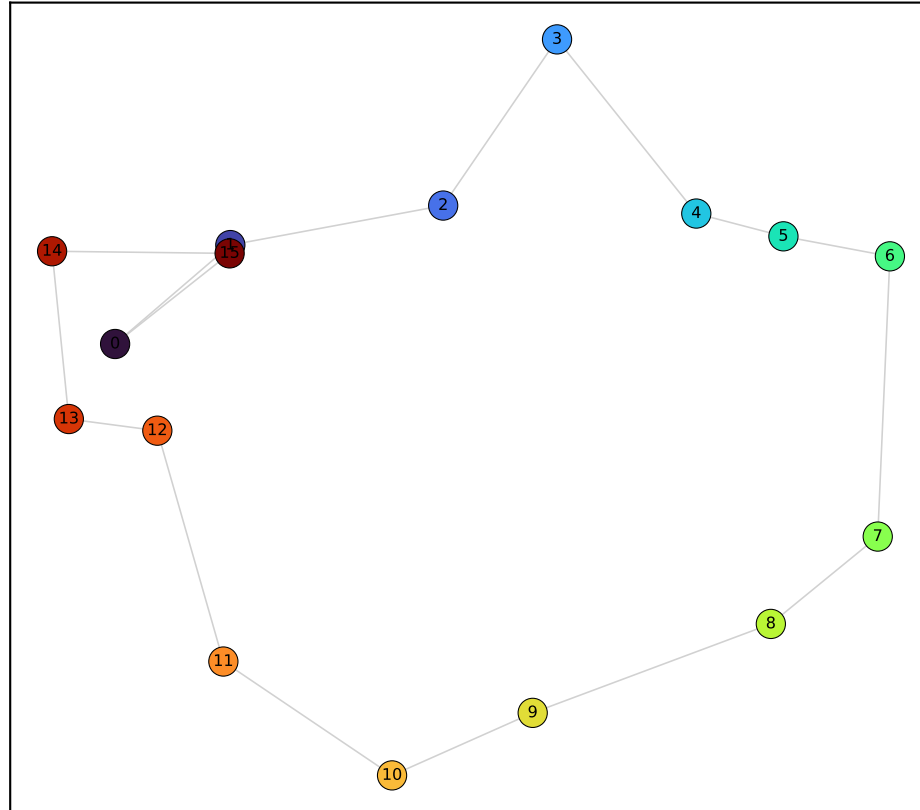


ring — node-mean PCA, depth 80% (edges = graph adjacency)

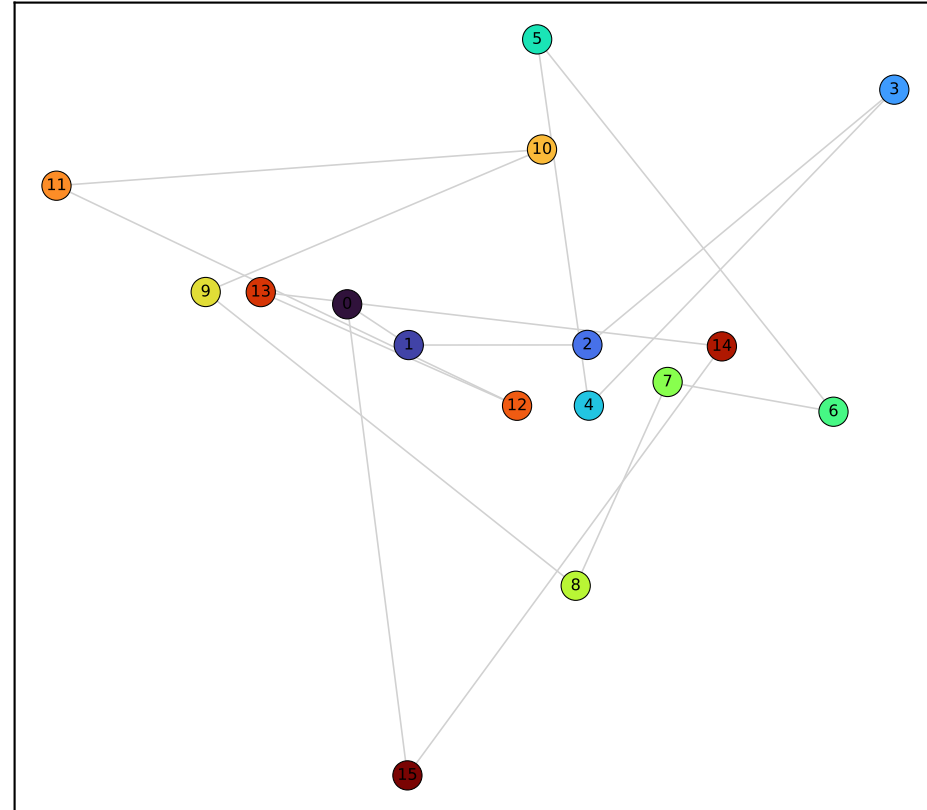
Llama layer 25/31 (PC var 15·13%)



Gemma layer 33/41 (PC var 13·10%)

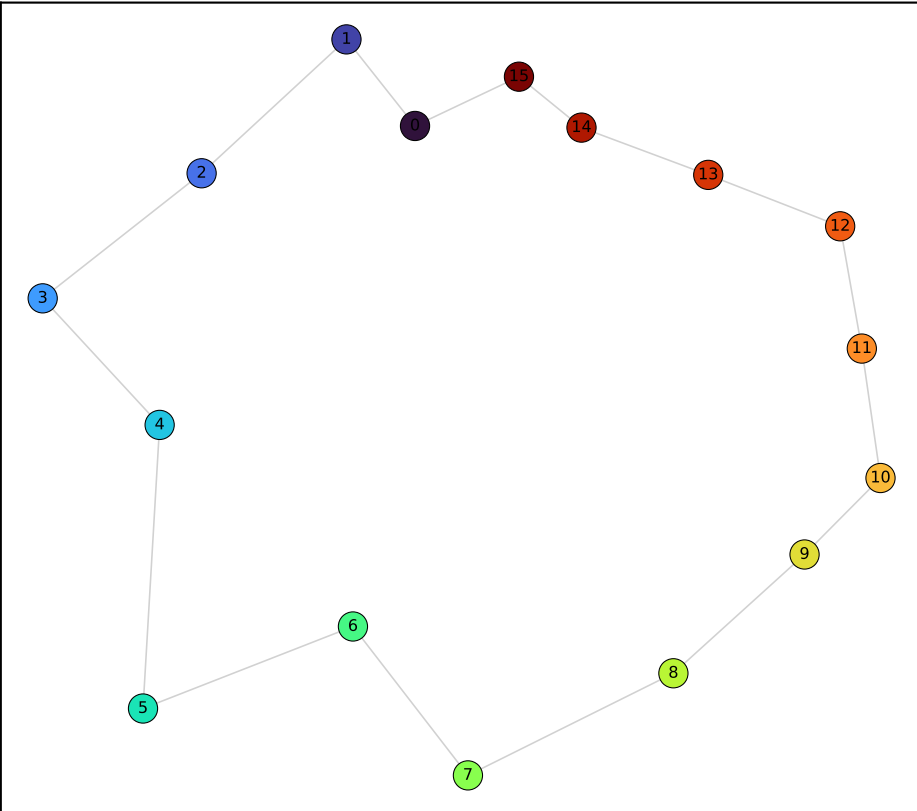


Qwen layer 28/35 (PC var 12·9%)

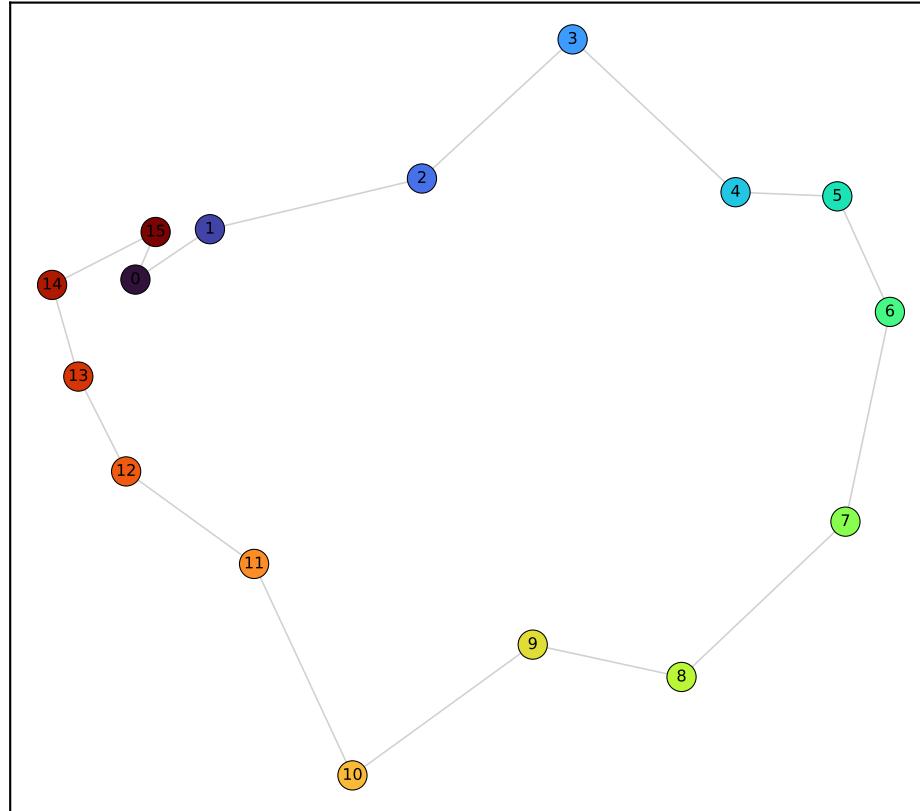


ring — node-mean PCA, depth 83% (edges = graph adjacency)

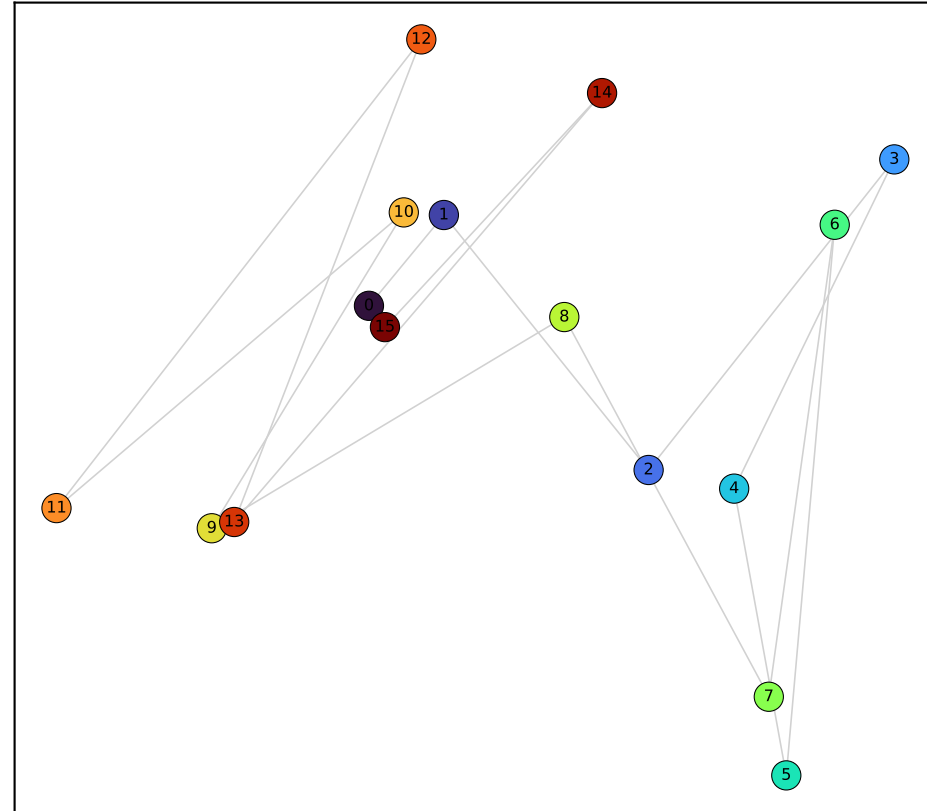
Llama layer 26/31 (PC var 17·14%)



Gemma layer 34/41 (PC var 14·11%)

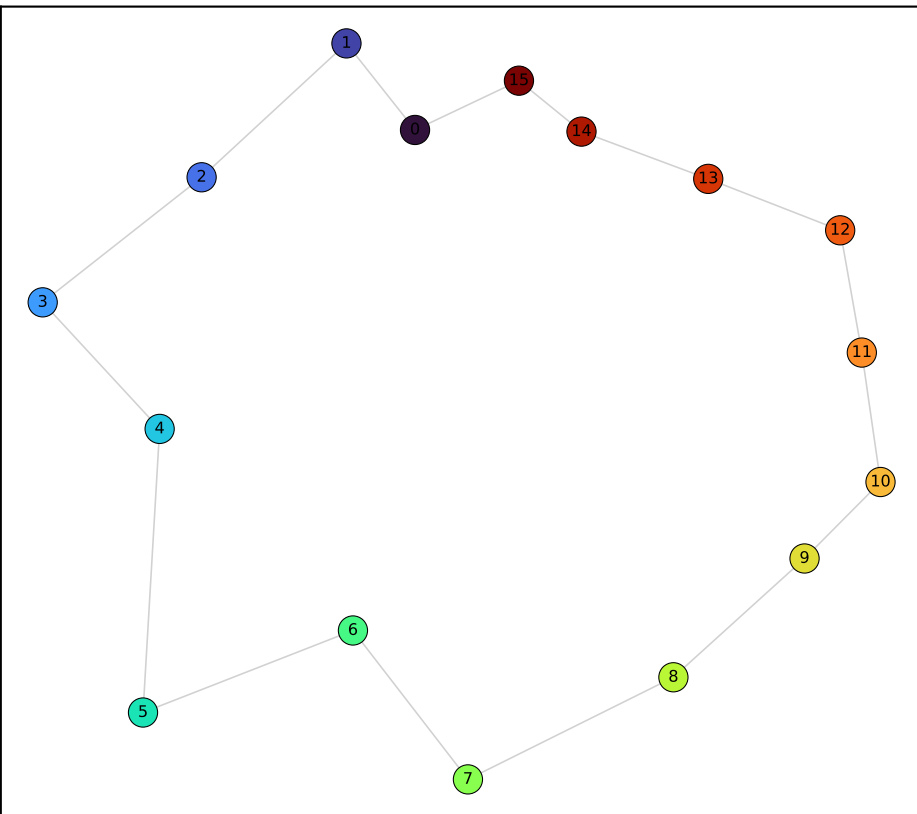


Qwen layer 29/35 (PC var 12·10%)

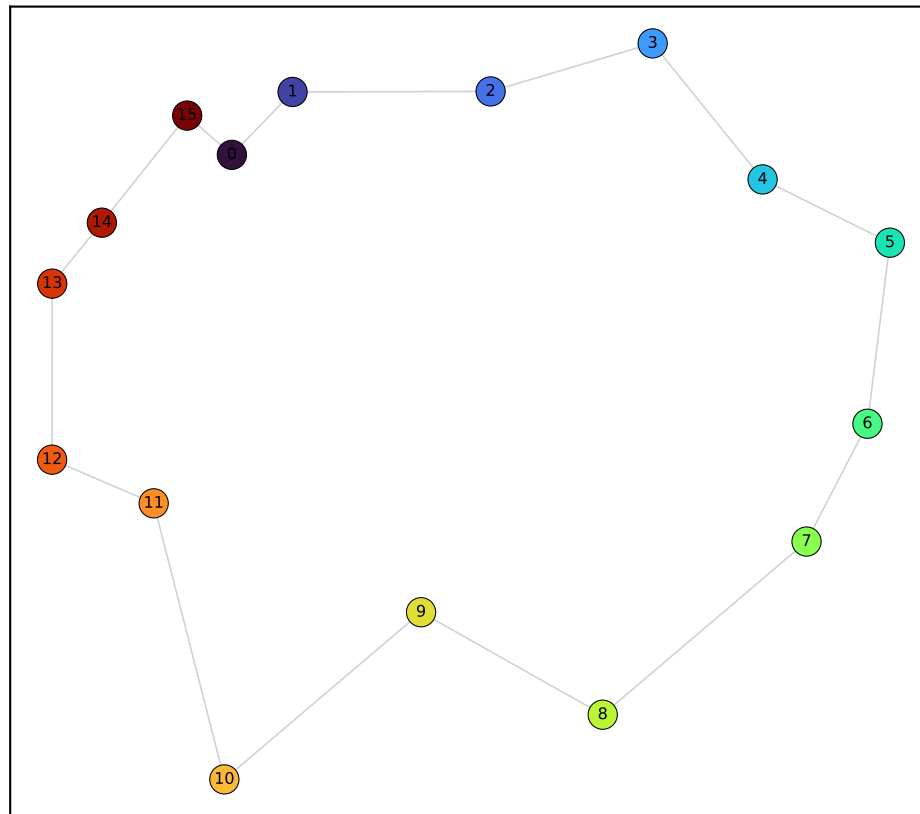


ring — node-mean PCA, depth 85% (edges = graph adjacency)

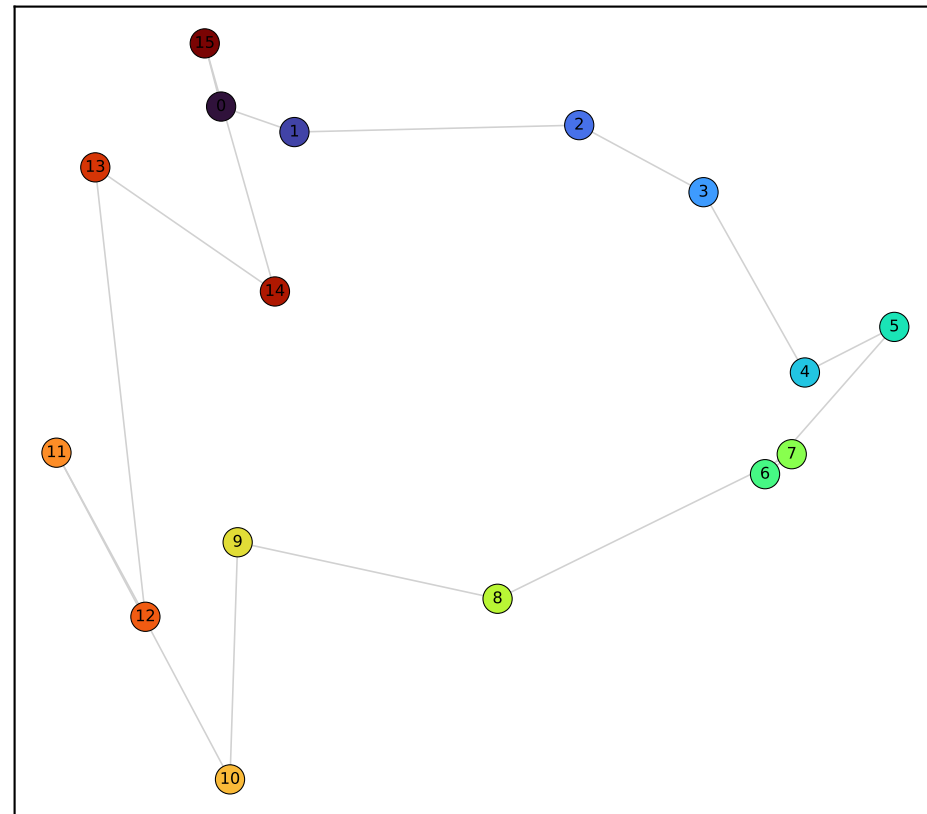
Llama layer 26/31 (PC var 17·14%)



Gemma layer 35/41 (PC var 17·14%)

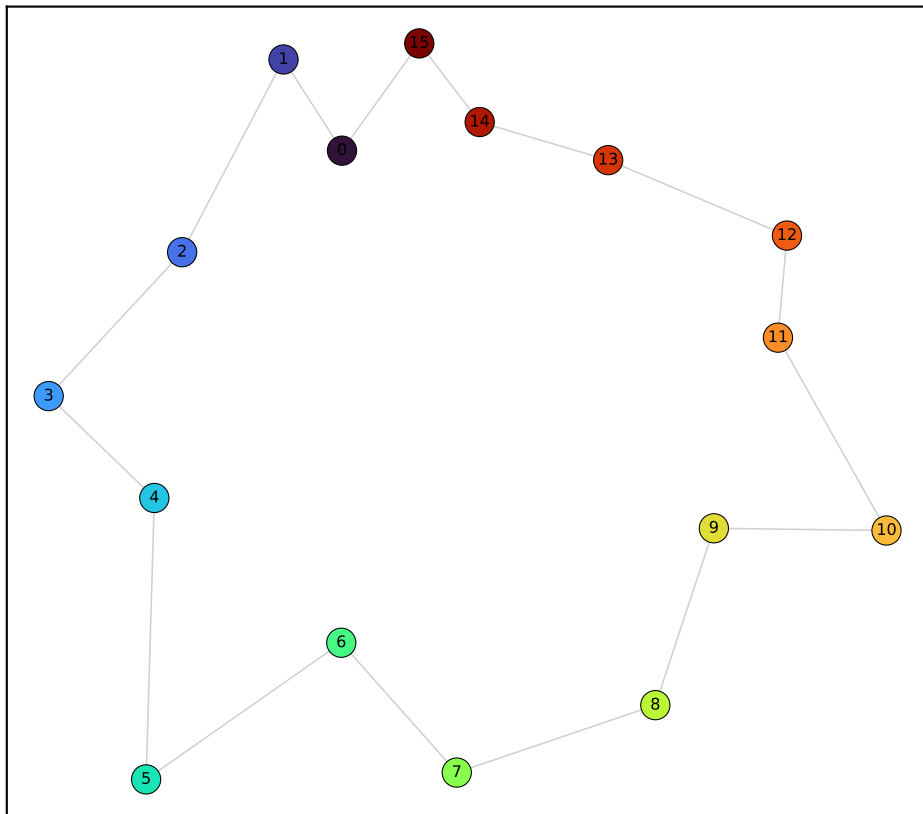


Qwen layer 30/35 (PC var 14·11%)

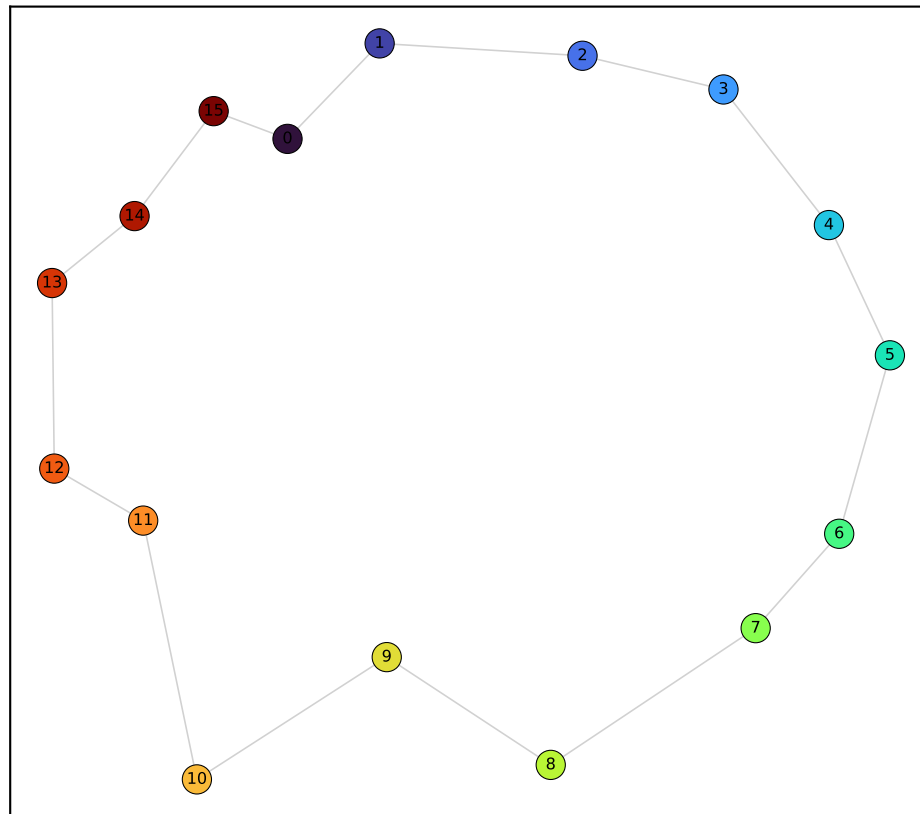


ring — node-mean PCA, depth 88% (edges = graph adjacency)

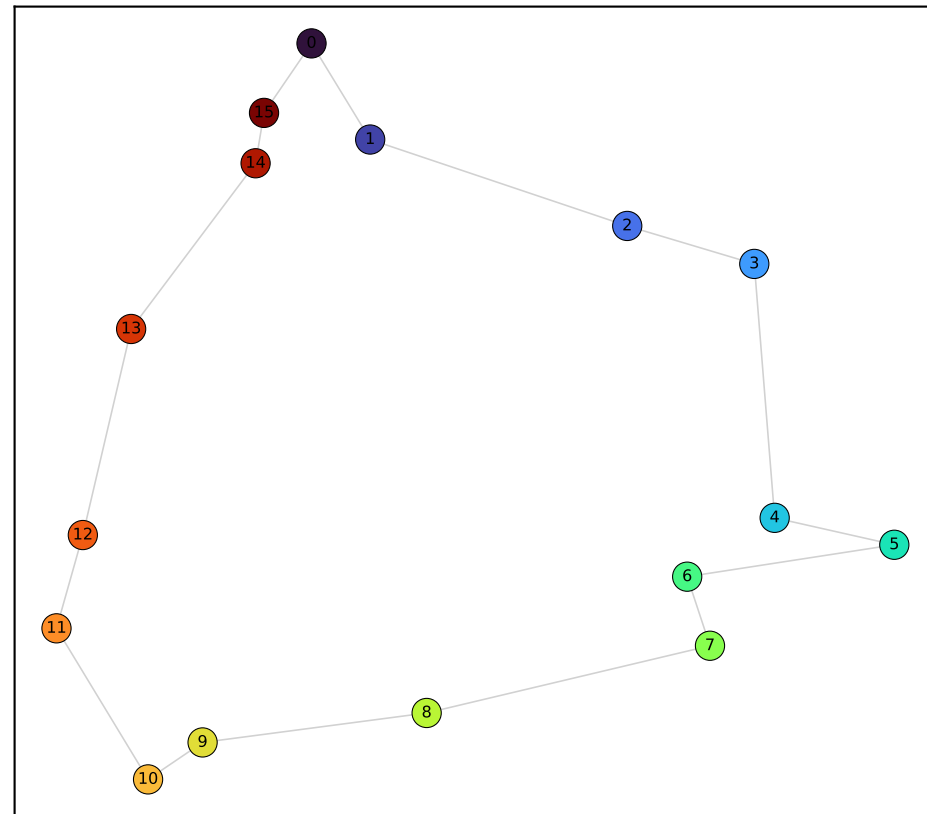
Llama layer 27/31 (PC var 18·14%)



Gemma layer 36/41 (PC var 19·15%)

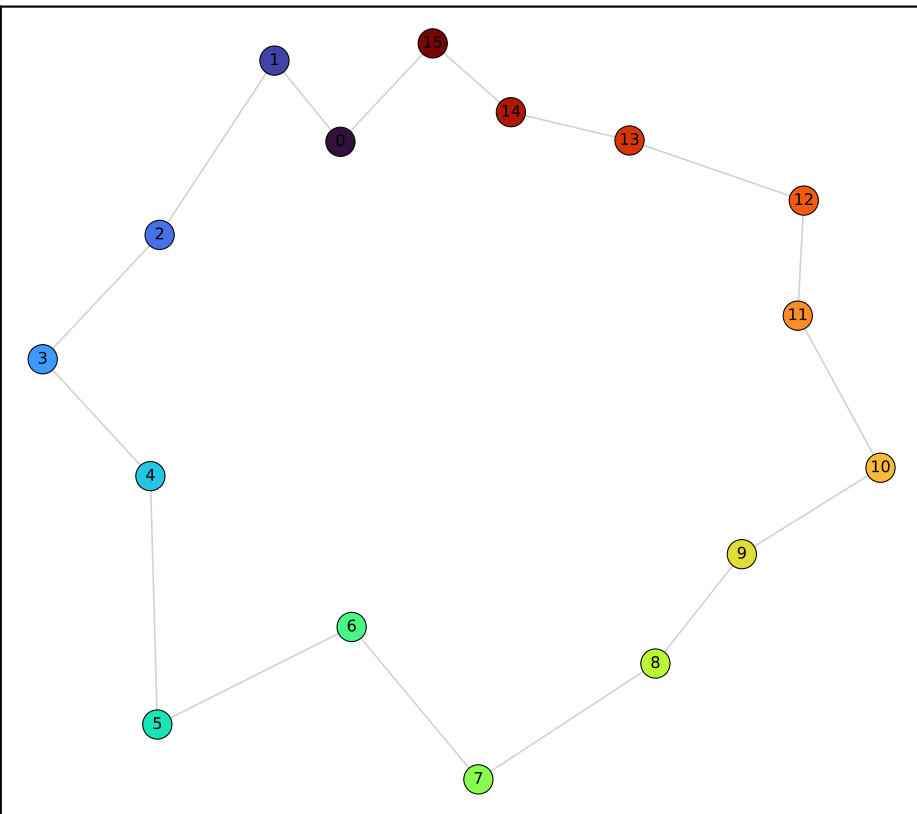


Qwen layer 31/35 (PC var 15·13%)

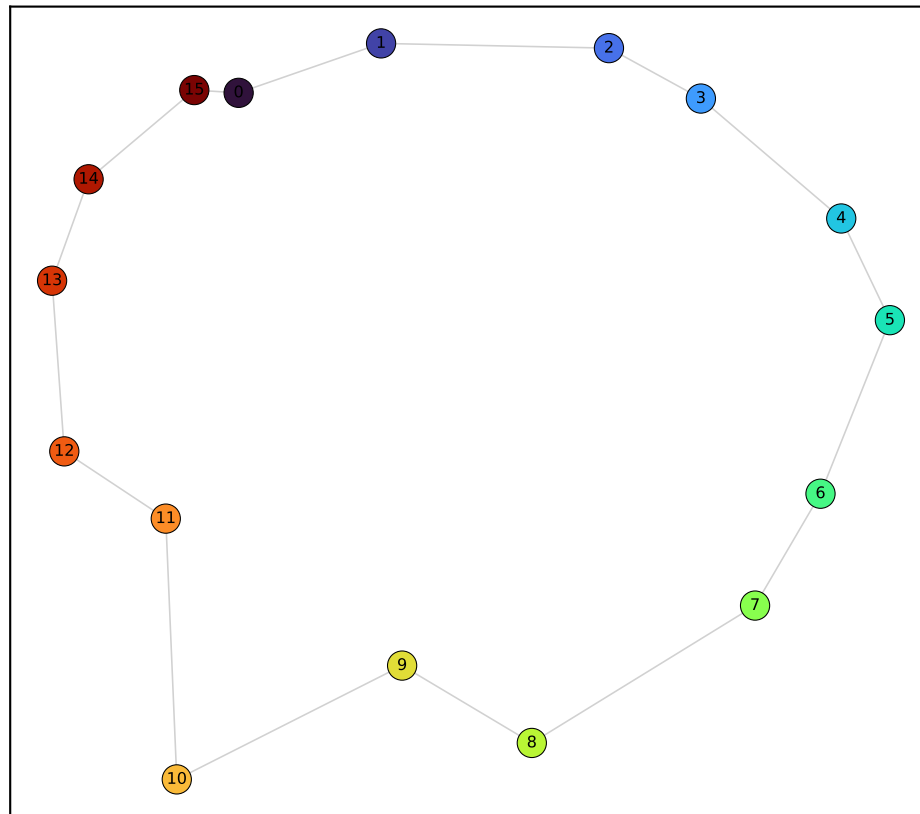


ring — node-mean PCA, depth 90% (edges = graph adjacency)

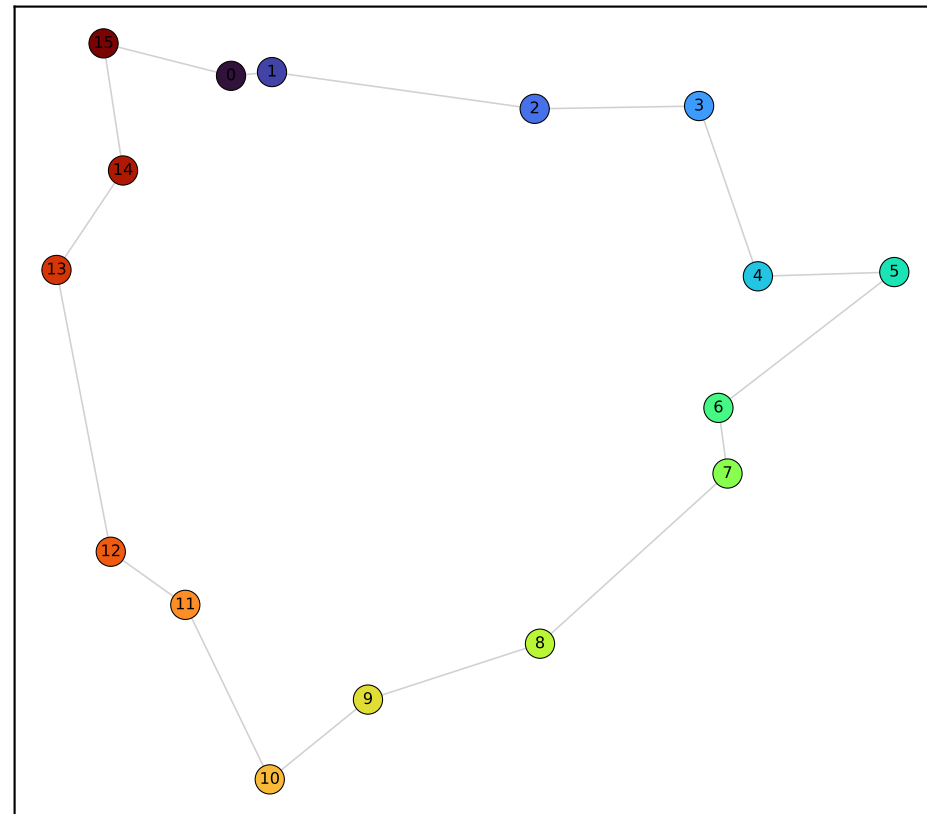
Llama layer 28/31 (PC var 19·16%)



Gemma layer 37/41 (PC var 20·17%)

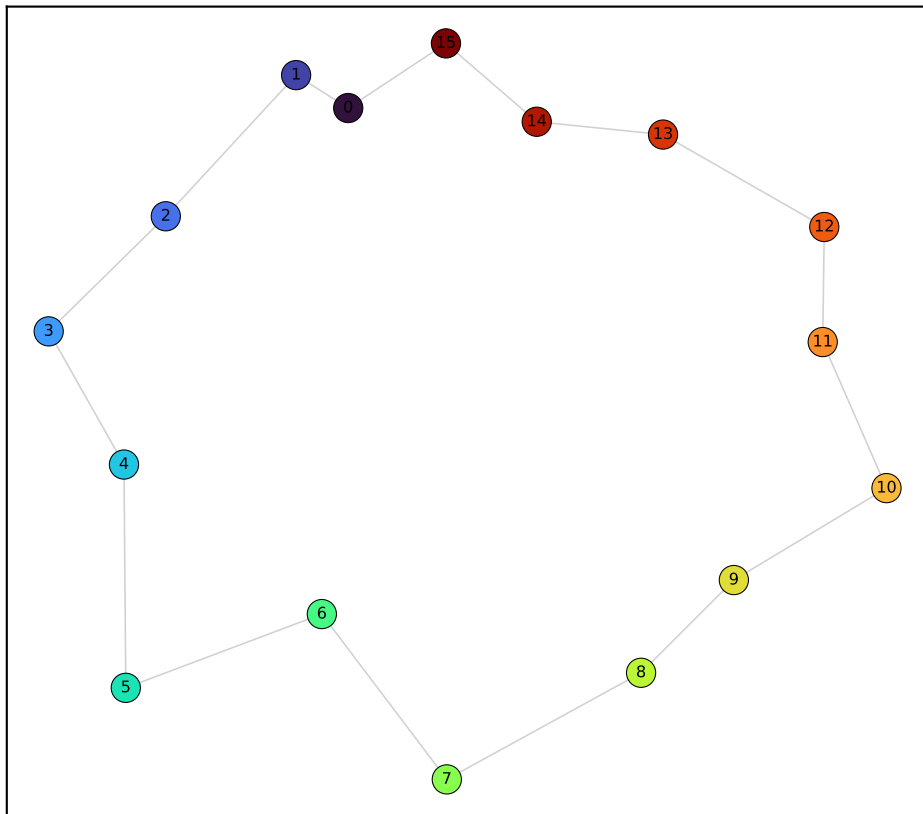


Qwen layer 32/35 (PC var 18·16%)

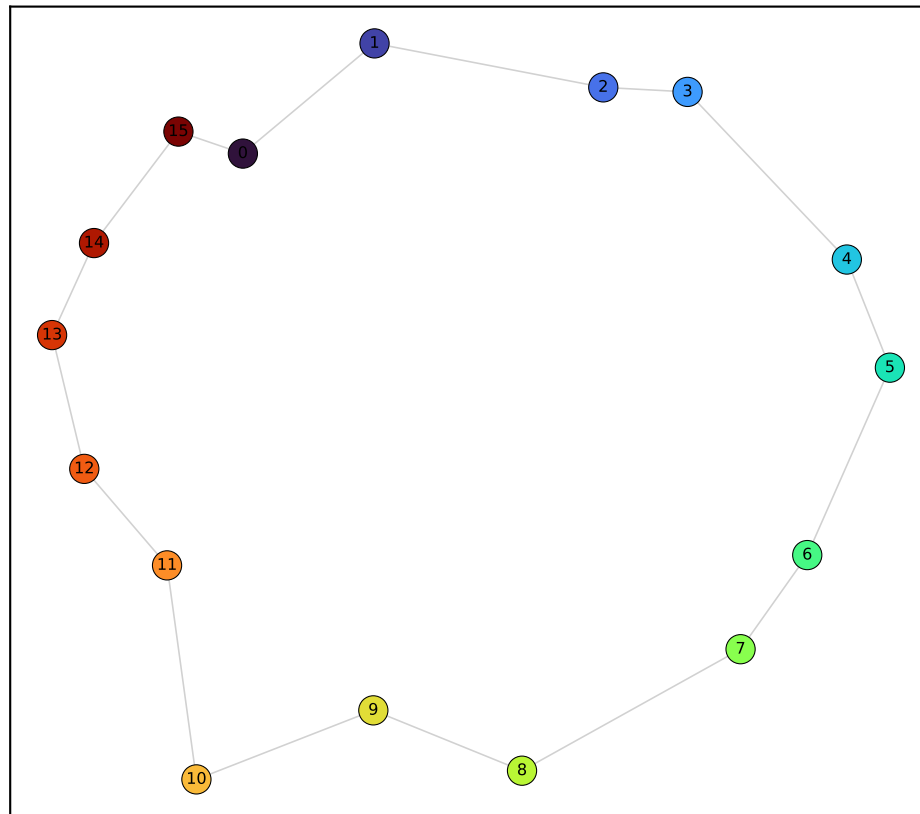


ring — node-mean PCA, depth 93% (edges = graph adjacency)

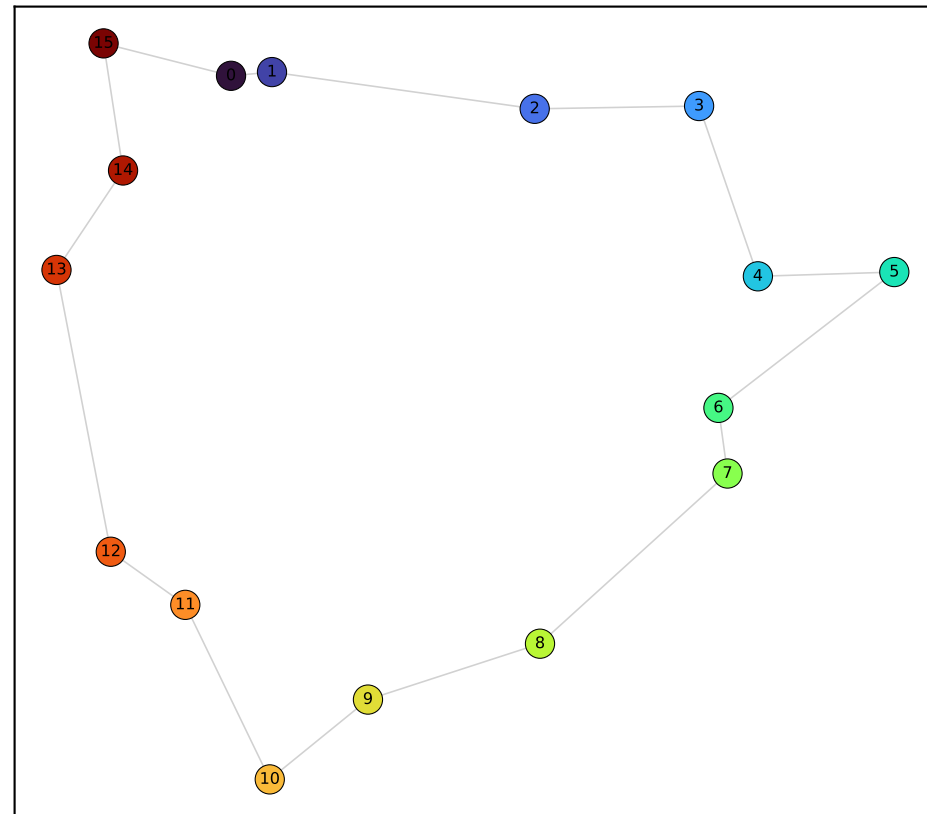
Llama layer 29/31 (PC var 20·16%)



Gemma layer 38/41 (PC var 20·17%)

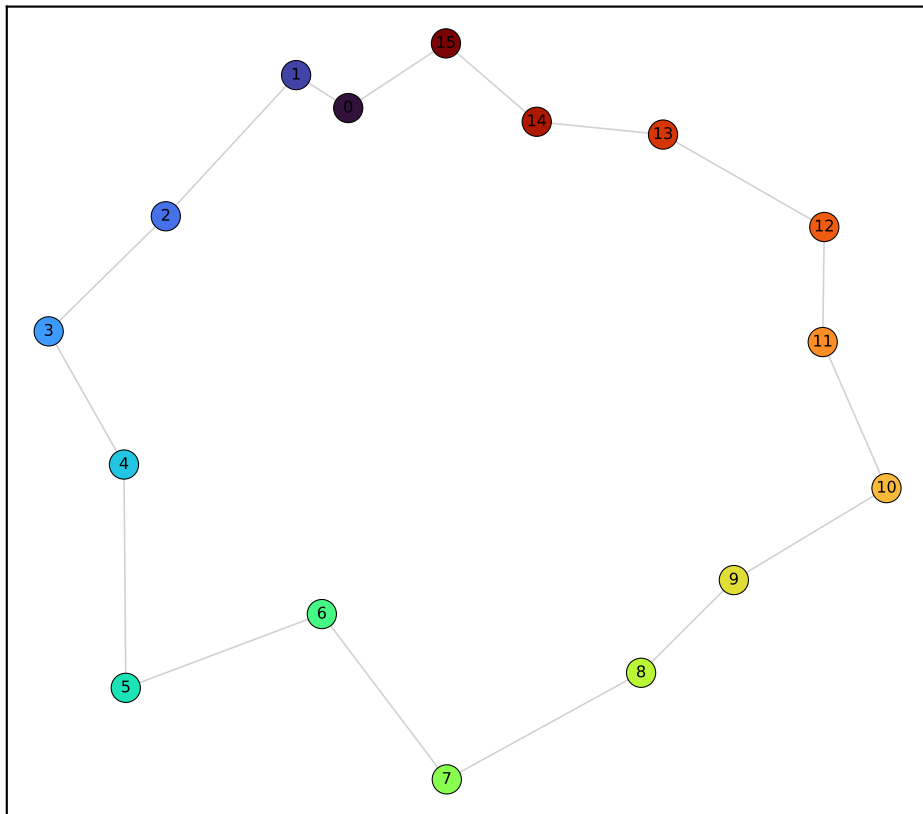


Qwen layer 32/35 (PC var 18·16%)

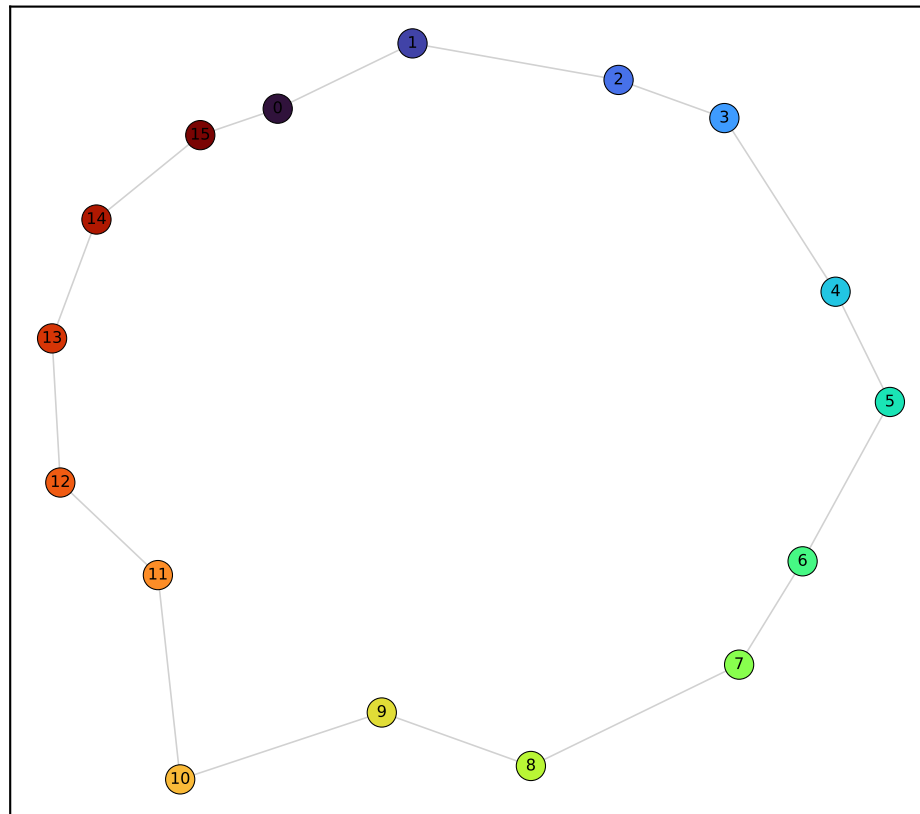


ring — node-mean PCA, depth 95% (edges = graph adjacency)

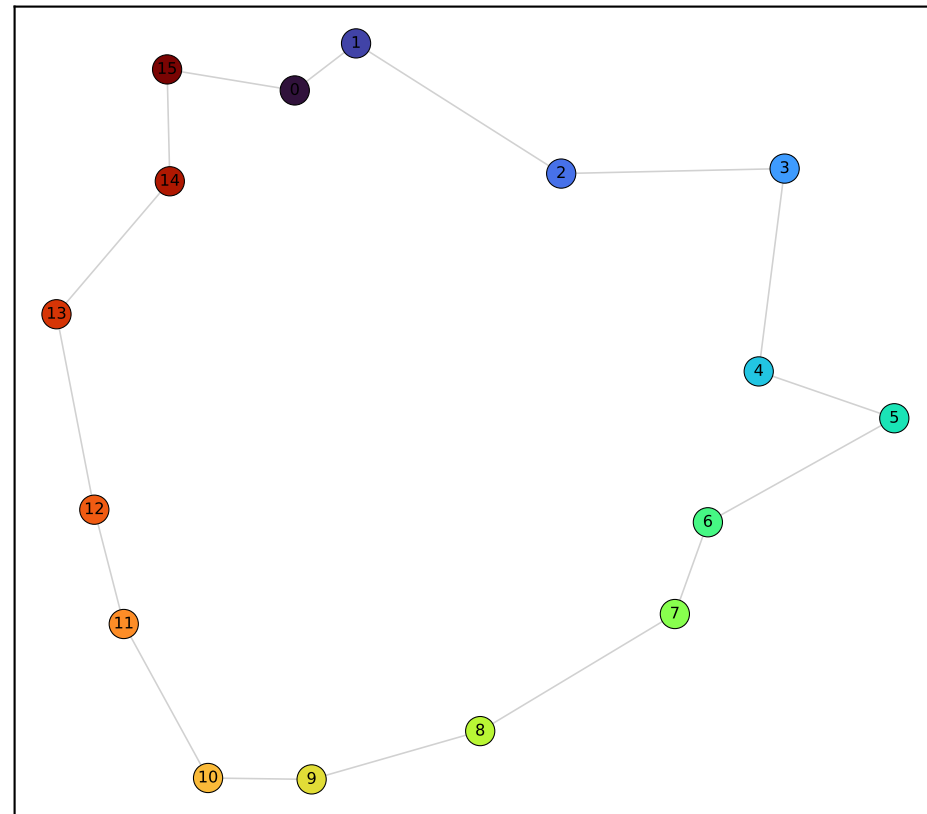
Llama layer 29/31 (PC var 20·16%)



Gemma layer 39/41 (PC var 24·19%)

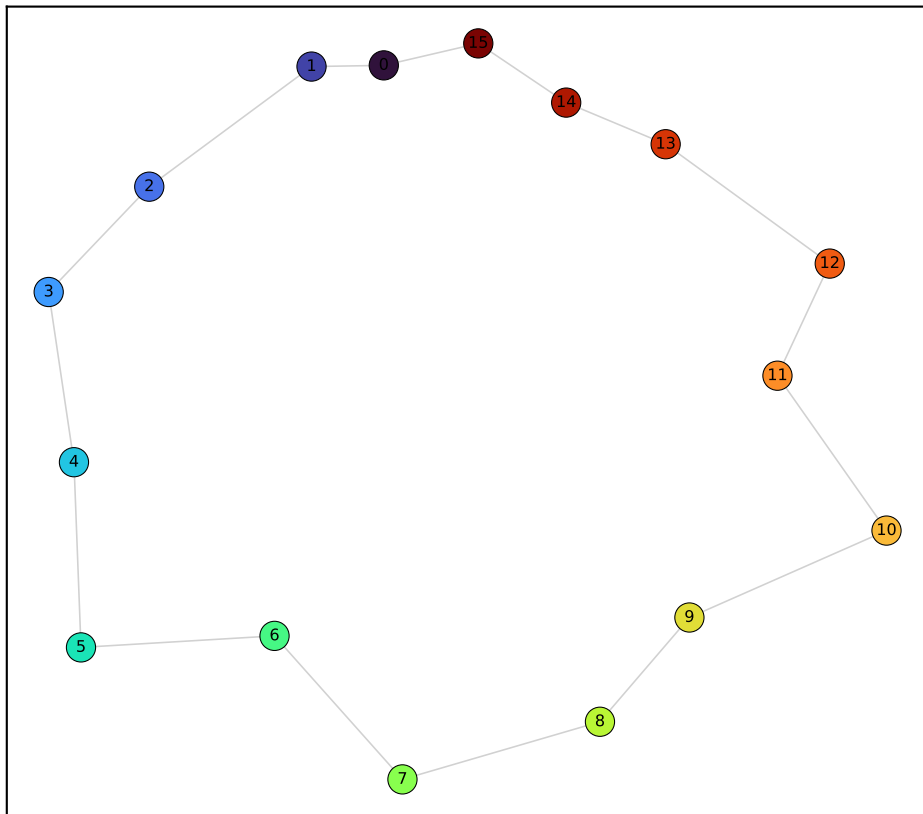


Qwen layer 33/35 (PC var 21·19%)

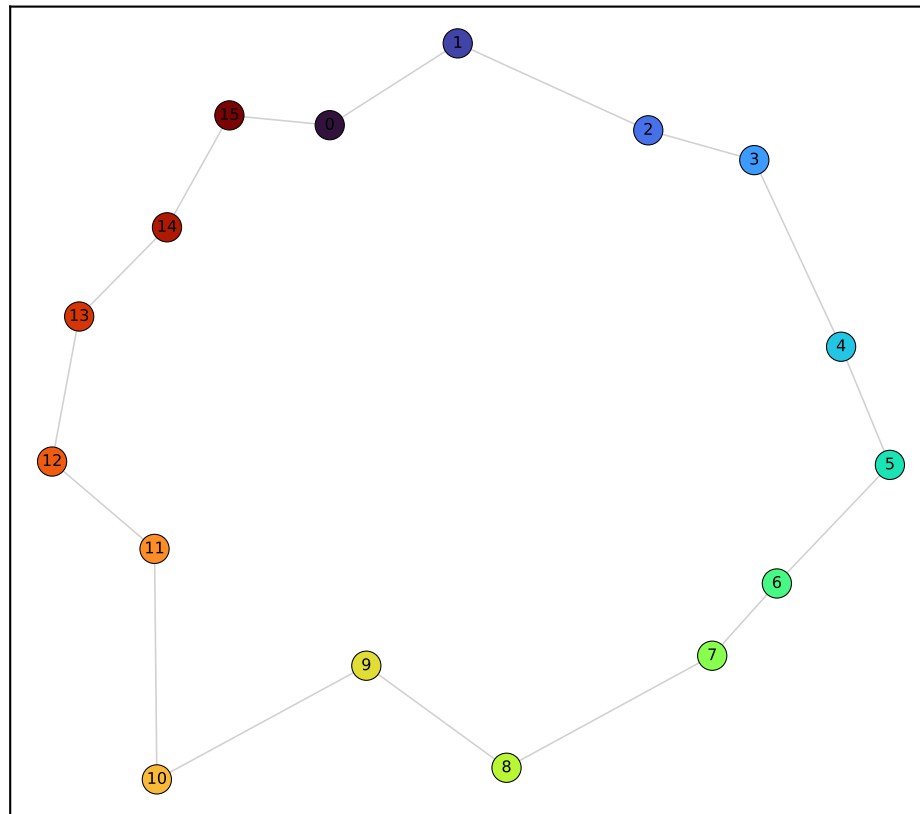


ring — node-mean PCA, depth 98% (edges = graph adjacency)

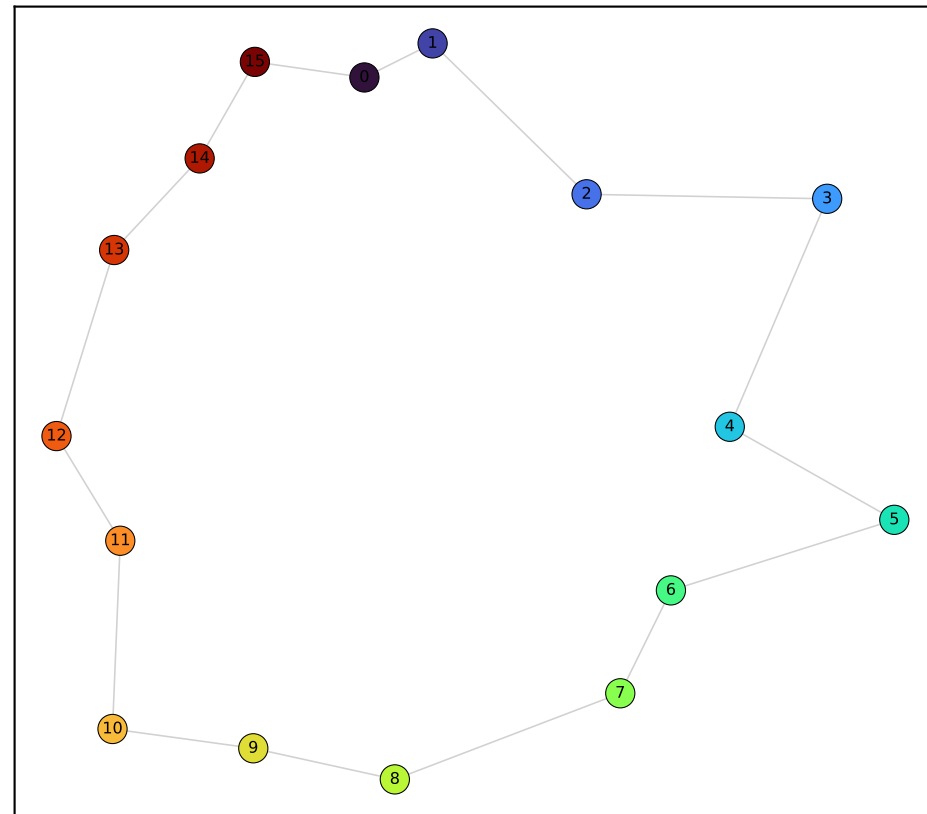
Llama layer 30/31 (PC var 22·18%)



Gemma layer 40/41 (PC var 25·20%)

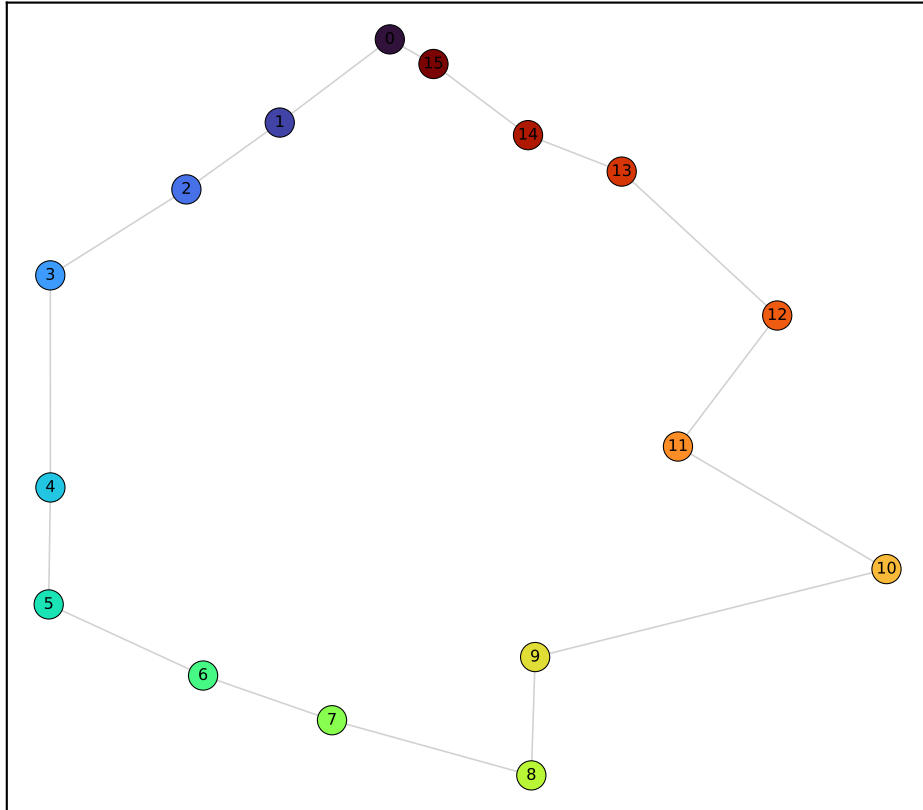


Qwen layer 34/35 (PC var 24·20%)

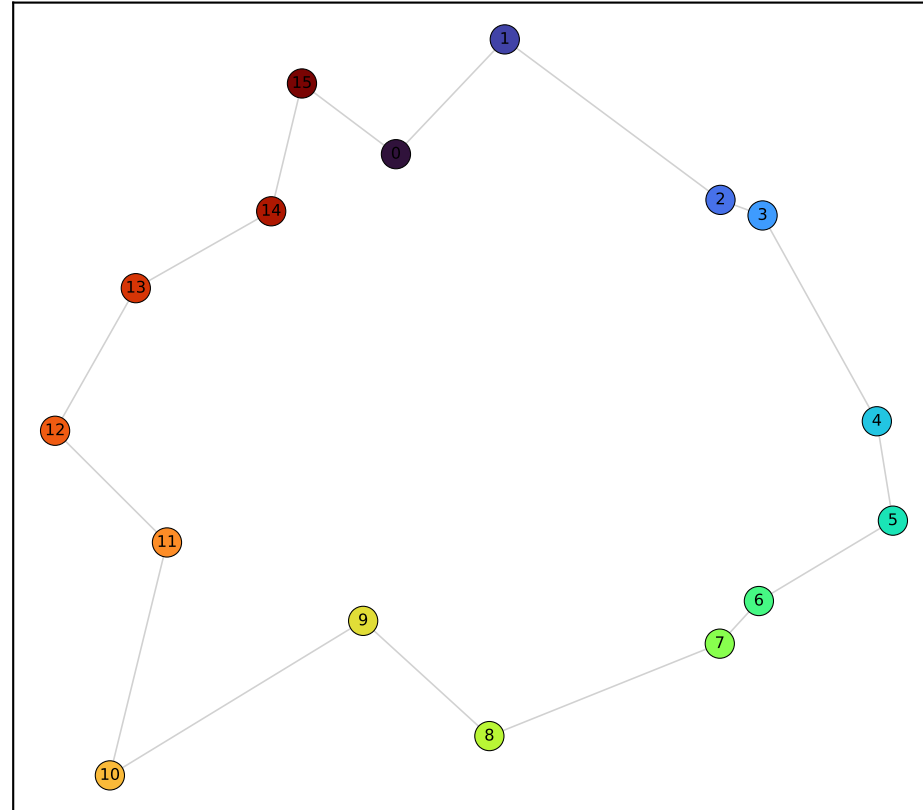


ring — node-mean PCA, depth 100% (edges = graph adjacency)

Llama layer 31/31 (PC var 29·21%)



Gemma layer 41/41 (PC var 24·19%)



Qwen layer 35/35 (PC var 32·19%)

